Internet Shutdowns and Software Development:

Evidence from Bangladesh

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Abstract

This paper studies how Bangladesh's nationwide Internet shutdown of July 2024 affected observable software-development activity. Using public GitHub events for 59,158 pre-active accounts associated with Bangladesh and fourteen comparison countries, I measure the daily share of eligible users with productive GitHub activity. During the July 18–23 shutdown, this share averaged 2.02 percent in Bangladesh, compared with 15.28 percent for a synthetic control: an estimated effect of −13.26 percentage points (−86.8 percent). An international difference-in-differences comparison yields −13.09 percentage points. Bangladesh is the most extreme of fifteen units in an in-space placebo test, although with fourteen placebo countries the exact rank p-value cannot fall below 0.0667. Git metadata show that 829 of the 2,932 commits authored during July 18–23 (28.3 percent) were first pushed after restoration. In levels, however, the backlog is small. For the cohort authored after the nationwide cutoff at about 15:00 UTC on July 18, creation of code observed publicly by August 10 fell 96.35 percent relative to duration-matched pre-shutdown windows, and this cohort's delayed commits equal 2.50 percent of normal volume; code pushed later or never published is unobserved. Against the synthetic series, the shutdown deficit is 8,145 active user-days and is only partly offset by September 4. An international extension estimates separate synthetic controls for fixed pre-treatment repository-domain cohorts, with explicit classification coverage, sample-support rules and placebo diagnostics. Forty eligible accounts publish during the interior shutdown days; an individual case audit finds direct automation evidence in 113 of their 180 pushes, qualifying an interpretation of residual activity as continued human coding. Within-Bangladesh recovery patterns are retained as descriptive appendix evidence. All outcomes measure public GitHub activity, not total software productivity. Because the shutdown coincided with severe domestic unrest, the estimates are interpreted as effects of the shutdown episode, assuming that no other Bangladesh-specific shock produced the same abrupt break.

Keywords: Internet shutdowns; software production; GitHub; synthetic control; difference-in-differences; digital economy.

1 Introduction

Internet connectivity is an input into an increasingly large set of production processes. For software developers, the dependence is especially direct: source code can be written locally, but modern development also relies on remote repositories, package registries, code review, issue tracking, continuous integration, documentation, and communication with collaborators. A nationwide loss of connectivity therefore interrupts both the transmission of code and many complementary tasks required to coordinate and ship software.

This paper studies a sharp episode of lost connectivity: Bangladesh's nationwide Internet shutdown in July 2024. Cloudflare reported that mobile networks began disconnecting early on July 18 and that national traffic fell to near zero later that day. Fixed broadband began to return on July 23, initially for selected sectors, while mobile connectivity recovered later, on July 28 (Belson, 2024a). Internet Society Pulse records the national disruption as beginning on July 18 and lasting roughly five and a half days (Internet Society, 2024a). The event offers an unusual setting in which a digitally intensive form of work can be observed at high frequency before, during, and after a near-complete national loss of Internet connectivity.

The paper makes four contributions. First, an explicit international synthetic counterfactual measures the immediate collapse in observable public activity. Second, separate country-domain controls assess heterogeneity among users grouped by their most active pre-shutdown repository; the labels describe project applications, not verified employment industries. Third, Git metadata distinguish timestamped authorship from publication, revealing a small observed backlog alongside a large fall in the author-time creation of code eventually observed by August 10. A case review of residual authors and publishers documents selection on prior activity and a substantial automated component of continued publication. Fourth, cumulative national gaps and the separate within-Bangladesh recovery analysis describe persistence, without interpreting post-restoration baseline crossings as causal gains.

The empirical challenge is that GitHub activity is volatile and varies widely across countries and developers. I therefore construct a multi-country donor pool and use synthetic control as the primary design. The treated unit is Bangladesh, and the donor pool contains fourteen countries of different sample sizes, including 1,373 eligible Ghana-associated accounts and 446 Jordan-associated accounts; rates for the smaller donors are correspondingly noisier. The synthetic-control weights are chosen to match the pre-shutdown path of the daily active-user rate. The resulting counterfactual assigns 47.6 percent weight to Pakistan, 32.5 percent to Ghana, and 19.9 percent to Kenya. During July 18–23, the share of eligible Bangladeshi users with productive GitHub activity averaged only 2.02 percent, compared with 15.28 percent in the synthetic control. The estimated effect is -13.26 percentage points, or -86.8 percent relative to the synthetic counterfactual.

Several complementary designs support the same conclusion. An international difference-in-differences comparison yields an effect of −13.09 percentage points, and excluding national holidays changes the estimate only modestly to −13.38 percentage points. Holding the primary donor weights fixed, productive activity falls by 86.6 percent, shipping activity by 93.3 percent, and collaborative activity by 93.5 percent; re-optimizing the weights for each outcome gives similar declines of 86.7, 90.5, and 91.9 percent. Regression discontinuity in time is retained as an appendix sensitivity check.

The mechanism analysis separates observable online activity from underlying code creation. A Git commit can be created without Internet access, but pushing it to GitHub requires connectivity. For a large sample of public commits, I recover Git author and committer timestamps and match each commit to the first public PushEvent in which it appears. In the strict sample, in which the commit author is the same GitHub account that performs the push, 829 of the 2,932 commits authored during July 18–23 (28.3 percent) first appear in a public push on or after July 24; the committer timestamp gives 27.2 percent. Long publication delays are otherwise rare: 29.7 percent of shutdown-authored commits were pushed more than 24 hours after authorship, compared with an average of 1.1 percent across six pre-shutdown placebo windows of six days each. In levels, however, the backlog is small. For the cohort authored after the nationwide cutoff at approximately 15:00 UTC on July 18, observable author-time commit creation is 96–97 percent below comparable pre-shutdown windows (96.35 percent against duration-matched windows), and that cohort's delayed commits amount to 2.50 percent of normal volume (Section 7.1). These counts cover only code pushed publicly by August 10. Later pushes, unresolved metadata, and code never published are unobserved, so the data cannot separate work that was not done from work published after August 10 or never published.

Residual publication is especially informative about measurement. Only 40 eligible accounts record a primary event on July 19–22. Individual examination identifies automated heads in 113 of their 180 pushes, including dependency updates, profile graphics, metrics and deployment output. Removing those pushes alone reduces daily residual activity from 0.212 to 0.117 percent, while preserving the original national specification. These accounts overlap only partly with the 250 strict onset authors; neither continued posting nor matching Git author metadata establishes that a person kept coding inside Bangladesh.

A separate appendix analysis turns from the shutdown window to the restoration period. Using a fixed sample of 20,000 Bangladesh-associated accounts selected on pre-shutdown activity, and a text classifier that assigns repositories to NAICS sectors from historical README files, I trace observable push activity by the economic sector associated with the repositories in which it occurs. Restoration of connectivity did not imply an immediate or uniform recovery. Activity in Finance- and Information-related repositories began a two-week run at or above 80 percent of its

pre-shutdown level by the third week after the shutdown; Education and Health recovered more gradually; and Accommodation/Food remained well below its baseline throughout the observed window. Activity in pre-existing repositories recovered far less than sector-wide activity, consistent with part of the recovery occurring through new projects. Because this analysis compares each sector with its own pre-shutdown level rather than with an untreated control, it describes heterogeneity in recovery dynamics and does not estimate sector-specific causal effects.

The findings relate to three literatures. First, they add micro-founded evidence to work on the economic costs of Internet shutdowns and, more broadly, of infrastructure outages. Existing studies document broad effects on output, employment, investment, and economic activity (Tagat, Phokeer, and Kreitem, 2024; Gunsilius and Van Dijcke, 2023), and Allcott et al. (2016) show how firms cope when electricity becomes unavailable. Second, the paper contributes to research on Internet access as a productive input. Hjort and Poulsen (2019), for example, find that the arrival of fast Internet increased employment and firm performance in Africa, and Akerman et al. (2015) show that broadband complements skilled work. Third, the paper joins a growing literature that uses public GitHub activity as a high-frequency measure of software work. Kreitmeir and Raschky (2024) exploit Italy's temporary ChatGPT ban and construct daily GitHub outcomes; Hu et al. (2023) use open-source contributions to study lockdowns; and Wachs et al. (2022) demonstrate the usefulness of GitHub data for measuring the geography of software production.

The interpretation is deliberately narrower than "total software productivity." Public GitHub events omit private repositories, other hosting services, meetings, local coding that is never pushed publicly, and non-coding work. User location is self-reported rather than physically verified. Most importantly, the shutdown was implemented amid severe domestic unrest, so the design cannot separate every contemporaneous Bangladesh-specific shock from the loss of connectivity. The paper therefore estimates the collapse in observable GitHub activity during the shutdown episode, and uses commit timestamps to assess how much of that collapse reflects delayed publication rather than forgone code creation.

The paper proceeds as follows. Section 2 sets out the shutdown chronology and explains why GitHub data separate code creation from publication. Section 3 reviews related work, and Sections 4 and 5 describe the sample, outcomes, and estimators. Section 6 reports the national, secondary-outcome and international repository-domain synthetic controls alongside the difference-in-differences comparison; discontinuity and descriptive recovery details are in the appendix. Section 7 turns to the commit-timestamp evidence, Section 8 examines diagnostics and identification threats, including the concurrent unrest, and Sections 9 and 10 interpret the findings.

2 Institutional Background

2.1 The July 2024 shutdown

The shutdown followed student protests against quotas in government jobs (Belson, 2024a). The political context matters for identification because the decision to restrict connectivity was not randomly assigned. A separate university-teacher strike began on July 1 over the pension scheme (United News of Bangladesh, 2024), and Amnesty International documented violence against quota protesters before the nationwide cutoff (Amnesty International, 2024). Reuters reported partial Internet restoration alongside relaxed curfew hours and office reopening on July 24 (Paul, 2024). These concurrent changes can affect both development and education-related activity. The empirical design therefore relies on the timing and severity of the network disruption, together with cross-country and within-Bangladesh comparisons, rather than assuming that the broader domestic environment was unchanged.

Available network measurements nevertheless provide a precise chronology. Cloudflare reports outages at major mobile operators beginning during the early hours of July 18, followed by a near-complete national loss of Internet traffic around 21:00 local time. Fixed broadband began to recover on July 23, initially as a limited restoration that prioritized sectors including banking, technology, exporters, outsourcing providers, and media. Some fixed-broadband providers returned later, while major mobile providers did not show substantial recovery until July 28 (Belson, 2024a). Internet Society Pulse also records a national shutdown beginning July 18 and notes the gradual return of connectivity on July 23 (Internet Society, 2024a).

The main treatment window in this paper is July 18–23 inclusive. This choice captures the six daily observations most clearly associated with the near-total national disruption while treating July 24 onward as the start of the restoration period in daily data. The distinction is necessarily approximate because restoration began during July 23 and differed across networks. For that reason, the main estimates average over the full July 18–23 window, the weekly event study reports the restoration weeks separately, and the regression discontinuity in time is used only as a robustness check.

2.2 Why GitHub activity is informative about software production

Distributed version control separates code creation from code publication. Developers can create commits locally, but a push to a remote GitHub repository requires network connectivity. Other productive actions—opening or merging pull requests, reviewing code, closing issues, publishing releases, and commenting on collaborative work—are even more directly online. This distinction makes software development useful for studying an Internet shutdown: public GitHub activity mea-

Table 1: Institutional timeline

Date	Connectivity status	Empirical treatment
July 16–17	Selective social-media/service blocking reported before the nationwide outage (Internet Society, 2024a)	Pre-treatment warning period; July 17 Ashura holiday handled in robustness (The Daily Star, 2024)
July 18	Mobile networks disconnect; traffic falls toward zero nationally	Treatment begins
July 18–23	Near-total nationwide Internet disruption	Main total-shutdown window
July 23–27	Partial fixed-broadband restoration	Restoration / post-treatment transition
July 28 onward	Mobile connectivity begins to recover	Post-treatment

Sources: Cloudflare Radar reporting in Belson (2024a) and Internet Society Pulse (Internet Society, 2024a); the Ashura holiday is documented by The Daily Star (2024). Dates are operationalized at daily frequency for the empirical analysis.

sures an observable online margin, while commit metadata can reveal whether some code was created locally and uploaded later.

GH Archive records the public GitHub event stream and makes it available through hourly archives and Google BigQuery (GH Archive, n.d.). The data include events such as PushEvent, PullRequestEvent, PullRequestReviewEvent, IssuesEvent, IssueCommentEvent, and ReleaseEvent. For historical PushEvents, the event payload contains the number of commits and a list of commit objects, subject to a 20-commit timeline-event cap documented by GitHub. I account for this limitation in the mechanism analysis by defining a clean sample of fully observed pushes and by reporting robustness to progressively smaller push sizes (GitHub, n.d.).

3 Related Literature

3.1 Connectivity, complementary inputs and adjustment

The economic premise is that connectivity complements some productive tasks while others can be performed locally. Hjort and Poulsen (2019) study the arrival of submarine Internet cables and terrestrial network access in twelve African countries using difference-in-differences. They find higher employment, with firm evidence linking the expansion to entry, productivity and exports. Their setting concerns the adoption of a better technology over time; it does not imply that suddenly removing access has an equal and opposite effect. Existing users may have accumulated complementary skills and organizations, and a short outage leaves little time to reorganize production. Bangladesh's episode therefore provides evidence on the short-run dependence of an established digital activity on connectivity, rather than the long-run return to expanding broadband.

Akerman et al. (2015) use a public broadband rollout in Norway as a source of variation in firms' adoption. Their evidence supports complementarity with skilled workers performing non-routine abstract tasks and substitution for unskilled routine work. The relevant implication here is task heterogeneity: local editing may remain feasible without Internet access, while publishing, reviewing and coordinating changes depend directly on online services. It would be inappropriate to infer workers' skill or occupation from a repository label alone. Instead, I distinguish observable tasks and study whether users associated with different repository domains exhibit different public-activity gaps.

A related literature asks how firms adapt when an essential infrastructure input fails. Allcott et al. (2016) instrument Indian manufacturers' electricity shortages with hydroelectric supply variation. Their estimates distinguish revenue and producer-surplus losses from smaller productivity losses, in part because inputs can be stored; access to generators also changes exposure. The parallel for software is the possibility of storing local work and uploading it later. The analogy is a mechanism hypothesis, not an assumption that software firms and factories adjust identically. I examine it by separating commit-author timestamps from publication timestamps and reporting delayed output both as a share of observed commits and relative to normal output levels.

3.2 Restrictions, networks and public software activity

Connectivity restrictions may affect people who retain access through their collaborators. Zhang and Zhu (2011) exploit the blocking of Chinese Wikipedia and document reduced contributions among users who were not themselves blocked. This evidence on participation incentives makes cross-border spillovers relevant to the present design: a donor-country developer working with Bangladeshi collaborators need not be wholly untreated. A national synthetic control is credible only to the extent that such spillovers and donor-specific disturbances do not explain the counterfactual path. Aceto et al. (2018) survey the measurement and causes of Internet outages, while Schachenhofer et al. (2023) analyze organizational information flows. These studies motivate the distinction between connectivity, coordination and final output, but do not by themselves identify the effect of Bangladesh's July episode. Cross-country shutdown-cost models such as Tagat, Phokeer, and Kreitem (2024) and the regional application in Gunsilius and Van Dijcke (2023) address broader economic activity; this paper's contribution is a directly observed, high-frequency public software margin with a transparent denominator.

GitHub traces are useful because they record actions and timing, but their economic interpretation depends on the measure. McDermott and Hansen (2021) use activity from millions of accounts to study labor reallocation during COVID-19, including changes toward weekends and nonstandard working times. Their approach illustrates why an aggregate count can change because of timing

rather than a proportional change in labor supplied. Wachs et al. (2022) map the geography of open-source software and Birkholz and Gomtsyan (2026) study international software-production networks. These applications motivate country comparisons while also making self-reported locations and cross-border collaboration central measurement concerns. Nagle (2019) relates firms' use of open-source software to productivity; the economic relevance of open source does not, however, provide a conversion factor from an individual GitHub event to value added. Likewise, Hu et al. (2023) study pandemic lockdowns in China, a disruption whose effects on time allocation and online work need not resemble loss of connectivity.

The closest measurement precedent is Kreitmeir and Raschky (2024), who combine Italy's temporary ChatGPT ban with daily GitHub output for Italian and other European users in a difference-in-differences design. They report heterogeneous responses by experience and task type. Their conversion of sparse daily actions into indicators motivates my extensive-margin outcome, but the treatments differ: restricting one tool leaves Internet access available, whereas a network outage impedes publication and collaboration themselves. Accordingly, I separately define public activity, shipping and coordination, rather than treating them as interchangeable measures of total productivity. The commit-timing analysis further asks whether apparently missing publication is observed later. It cannot recover work that never becomes public or validate the physical time and location of authorship.

3.3 Comparative designs and the contribution of this study

Abadie and Gardeazabal (2003) introduce a synthetic comparison in their study of conflict in the Basque Country. Abadie, Diamond, and Hainmueller (2010) formalize the method, and Abadie et al. (2015); Abadie (2021) emphasize the relationship between donor choice, predictor balance and the credibility of extrapolating the pre-treatment relationship. This framework is appropriate to a country-level episode, but it does not remove concurrent unrest or compensate automatically for poor fit. Ferman and Pinto (2021) show why imperfect pre-treatment fit deserves explicit attention. I therefore report the optimization problem, all candidate donors, lag specifications, holdout fit, placebo comparisons and donor exclusions. The fitted weights are empirical solutions, not claims that the positive-weight countries are structurally identical to Bangladesh.

Alternative estimators answer related questions under additional choices. Synthetic difference-in-differences adds time reweighting (Arkhangelsky et al., 2021); here that reweighting emphasizes the unusually active final pre-shutdown weeks, so its larger estimate is presented as sensitivity evidence. Regression discontinuity in time requires care about serial correlation, functional form and coincident shocks (Hausman and Rapson, 2018); its details are relegated to the appendix. The central contribution is the combination of a transparent national counterfactual, task-specific public

outcomes, fixed pre-treatment repository-domain heterogeneity, and a separately defined residual-author cohort. These components distinguish disruption from observed adaptation while preserving the limits of what public traces can establish.

4 Data and Measurement

4.1 What GitHub records

GitHub hosts software projects and provides tools for developers to coordinate work. A *repository* is a project's collection of files and their version history. Git, the version-control system used by GitHub, allows a developer to save a *commit*—a recorded set of file changes—on a local computer without an Internet connection. A *push* uploads one or more commits to a remote repository and requires connectivity. A push can therefore occur later than the work represented by its commits. Neither a commit nor a push measures hours worked or the economic value of the code.

A *pull request* proposes changes for review; closing it need not mean acceptance, whereas merging it incorporates the proposed changes. An *issue* records a task, question or bug. Reviews and comments record coordination. A *release* publishes a named software version. Repository, branch or tag creation is also recorded as an event. GH Archive stores public GitHub event records and their publication times; it is not a record of all computer work, private development, or every local Git operation (GH Archive, n.d.; GitHub, n.d.).

4.2 GitHub users and country assignment

The analysis begins from GitHub accounts returned by country-location searches for Bangladesh and fourteen potential donor countries: the Philippines, Pakistan, Sri Lanka, Nepal, Indonesia, Vietnam, Malaysia, Thailand, Kenya, Nigeria, Ghana, Morocco, Jordan, and Tunisia. GitHub profile location is self-reported. I therefore interpret the sample as accounts associated with each country rather than verified physical residence at every point in time.

The country universe is the set for which the original project assembled location-search lists and a common historical event panel. It is a data-availability sample spanning South and Southeast Asia, Africa and the Middle East, not a random sample of countries or an automatically selected set of Bangladesh's closest economic peers. The archived materials do not establish an ex ante income, climate or electricity threshold that determined these fourteen countries, so I do not retrospectively claim one. All fourteen enter the optimization before weights are estimated; only Pakistan, Ghana and Kenya receive positive primary weights. Table A1 reports every country's sample size, pre-period activity and fitted weight. Small samples, country-specific disturbances and cross-border

spillovers are evaluated as limitations rather than screened using the size of the treatment-period gap.

For each account, I use GH Archive to recover public daily GitHub events from May 1 through September 4, 2024; all event dates are in UTC. The activity requirement is that an account records any of the nine event types retained by the extraction query, including issue comments and commit comments not counted in `productive_events`, on at least four days between May 1 and July 17. This requirement is necessary but not sufficient for inclusion: the further exclusions described below also apply. The final eligible sample contains 59,158 users, including 10,241 accounts associated with Bangladesh. This restriction focuses the design on developers with observable pre-shutdown activity and reduces the influence of dormant accounts. It also means that the estimand is an effect on pre-active public GitHub users, not on every person who has a GitHub account.

The sample also excludes suspected automation accounts, but the original login-pattern list and behavioral thresholds used for that step are unavailable. Applying the activity requirement alone yields 60,729 candidates, compared with 59,158 retained accounts; the 1,571 additional exclusions cannot be attributed to specific rules, because the original exclusion rules remain unresolved. Both stored bot flags are false for every retained account, which confirms their status but does not recover the exclusion rule. The stored behavioral metrics—maximum daily commits between May 1 and July 17, the number of dates with positive commits, and that count divided by 78 calendar days—can be reproduced for all retained accounts, but their exclusion thresholds remain unknown.

4.3 Outcomes

The primary outcome is the daily active-user rate: the share of eligible users in a country who record at least one productive GitHub action on day t. Operationally the indicator is $1\{\texttt{productive_events} > 0\}$. I prefer this extensive-margin measure for two reasons. First, GitHub activity is sparse at the user-day level; Kreitmeir and Raschky (2024) similarly convert daily GitHub actions into binary indicators because such actions are relatively rare at the user level. Second, raw commit counts in this sample contain extreme outliers. The original extraction query, recovered from the project's query history, defines `productive_events` as the count of seven event types: pushes, pull-request events, pull-request reviews, review comments, issue events, releases, and creation of a repository, branch or tag. Pull-request and issue events count across actions, not only openings or closures. Ordinary issue comments and commit comments do not enter this primary count, although they enter the collaboration measure. An independent re-extraction exactly reproduces the primary counts on all 799,417 overlapping eligible user-day records through July 23. The earlier component list omitted creation and several actions, explaining why a positive primary indicator need not coincide with those narrower components. The primary label describes

this operational basket and should not be read as a judgement that every event produces useful software.

Let $\mathcal{U}_c$ be the fixed set of eligible accounts in country c, $N_c = |\mathcal{U}_c|$, and P_{it} the count of these seven event types published by user i on UTC date t. The outcome is

$$Y_{ct} = 100 \frac{\sum_{i \in \mathcal{U}_c} \mathbf{1}\{P_{it} > 0\}}{N_c}. \tag{1}$$

An eligible user with no event on a date contributes zero; the denominator is not the number of users observed that day. Several qualifying events still count as one active user-day. A change from 15 to 2 percent is a decline of 13 percentage points, while the relative change is $100(2 - 15)/15$ percent. Event publication dates and Git author timestamps are different measurements throughout the paper.

I construct secondary outcomes along the software-production pipeline. The secondary productive-output indicator indicates a day with commits, issue closure, pull-request closure, or a release. Shipping activity indicates a day with a merged pull request or a release. Collaboration activity indicates a day with code review or comment activity. I also construct count outcomes per 1,000 eligible users: collaboration events and shipping output. These measures separate recorded output from downstream delivery and coordination. The primary outcome and the secondary productive-output indicator are distinct measures: their Bangladesh pre-period means are 17.25 and 15.38 percent, respectively (Table 2).

A small number of accounts generate implausibly large daily or weekly commit counts, so the primary country-day synthetic-control analysis does not rely on a raw commit-count headline. National holidays are coded at the country-day level and used in robustness checks.

4.4 Commit-level mechanism data

To distinguish coding from publication, I build a separate commit-level dataset for Bangladesh. I extract public PushEvents by eligible Bangladeshi users, retain PushEvents for which the historical payload reveals the complete set of commit SHAs, and keep the earliest observed push for each repository-SHA pair. I then query GitHub commit metadata to recover `authoredDate`, `committedDate`, and GitHub-linked author and committer logins. The metadata-query target contains 86,065 repository-SHA pairs, rather than the full pre-period universe: only 17,000 of 134,479 pairs first pushed before July 18 (12.64 percent) were queried, while all 69,065 pairs first pushed on or after July 18 were targeted; 70,576 commit objects are successfully resolved, an 82.0 percent resolution rate.

The strict mechanism sample requires (i) a non-missing author timestamp, (ii) a nonnegative

Table 2: Data and outcome construction

Component	Definition / sample	Scale
Primary outcome	At least one of the seven public event types defined in the text	Country-day percentage (effects in pp)
Secondary productive output	Any commits, issues closed, pull requests closed, or releases	User-day indicator / country-day rate
Shipping activity	Any merged pull request or release	User-day indicator / country-day rate
Collaboration activity	Any PR review, issue comment, PR review comment, or commit comment	User-day indicator / country-day rate
Collaboration events	Total review/comment events	Per 1,000 eligible users
Shipping output	Merged pull requests plus releases	Per 1,000 eligible users
Mechanism sample	Public repo-SHA pairs from fully observed PushEvents, matched to commit metadata	Commit level

Notes: Public GitHub events do not measure private repository activity, work on other platforms, meetings, or local code that is never pushed publicly.

author-to-push delay, and (iii) equality between the GitHub login linked to the commit author and the PushEvent actor. This leaves 59,920 commits. The equality condition avoids attributing a commit written by another person to the Bangladeshi user who happened to push it. Git author and committer timestamps are embedded metadata rather than independently verified work-time measurements, so they are interpreted as evidence consistent with offline production rather than as an infallible clock of when coding occurred.

4.5 International repository-domain cohorts

To construct international comparisons within repository domains, I retrieve the repository associated with each of the nine retained public event types for the unchanged 59,158 eligible accounts during May 1–July 23. For each account, I select its repository with the largest number of pre-shutdown days containing a primary event. Ties are resolved by the number of those events and then the repository name in lexical order. The rule uses May 1–July 17 only and selects a repository for 59,127 accounts; the remaining 31 have no qualifying primary event in that period. The selected repositories comprise 58,122 distinct projects. This is a mutually exclusive assignment, unlike the multi-sector activity measures used in the older Bangladesh recovery analysis.

I apply the same published multilingual NAICS classifier across all countries, using repository name and a root README from the latest retrievable default-branch commit no later than July 17, 2024. The model revision is pinned and the training project's text-formatting function is used. Historical source text is retrieved through GitHub using the same cutoff and procedure for Bangladesh and every donor country. Of the 58,122 selected repositories, 35,160 yield a nonempty historical

README; 420 have empty README files, 11,804 have no root README, 9,949 are unavailable, 788 have no retrievable commit at the cutoff, and one has an unreadable text object. Current descriptions, topics and profile text are excluded. The main international assignment requires both an available historical README and a classifier score of at least 0.80. Accounts that fail these requirements remain in an explicit unclassified group. Sensitivities use 0.70 and 0.90 and, separately, permit name-only classifications at 0.80. Coverage is reported for every country in Table A7.

These groups measure a user's pre-treatment association with a repository application domain, not verified industry of employment. The most active repository need not represent all of a user's work. Retrieval follows the default branch available at query time, restricting its commit history to the pre-treatment cutoff; it cannot establish which branch was the default in July 2024. Histories may omit projects that became unavailable or whose history was rewritten, and the model's score is not a validated probability of correctness in this country sample. The reported test performance in the model card uses its own labelled data, not a hand-audited sample of these historical repositories (Quispe and Xu, 2026). The design therefore addresses heterogeneity in observable activity across inferred domains, conditional on classification and coverage, rather than estimating industrial value added.

4.6 Sector-recovery sample and repository classification

To study the period after connectivity returned, I construct a separate within-Bangladesh panel. It starts from the 23,866 Bangladesh-associated accounts with at least one day of selected public GitHub activity between January 18 and July 17, 2024, and draws a fixed sample of 20,000 of them. Eligibility depends only on pre-shutdown activity, so the sample composition cannot respond to the shutdown or to post-shutdown behavior. For these accounts, I recover all public GH Archive events from January 18 through November 23, 2024, aggregated to the user-repository-day level. This sample differs from the 59,158-account synthetic-control sample in its eligibility rule (one rather than four pre-treatment active days), its observation window, and its restriction to Bangladesh, so its activity levels are not directly comparable with the country-day rates in Section 6.

Each repository is assigned to a two-digit sector of the North American Industry Classification System (NAICS) by a publicly released multilingual transformer classifier that predicts a NAICS sector from repository text (`aquiro1994/naics-github-classifier-multilingual`; Quispe and Xu, 2026). The classifier input is the repository name and a historical snapshot of the root README file. For repositories observed on or before July 17, 2024, the snapshot is taken as of July 17; for repositories first observed after the shutdown, it is taken as of their first observed activity date. Current repository descriptions and topics are deliberately excluded, because they may have been written after the study period. Of the 168,875 distinct repositories in the panel, a

historical README is recovered for 92,979 (55.1 percent); the remainder lack a root README at the target date or could not be retrieved. The main specification assigns a sector only when the classifier's confidence score is at least 0.80, which holds for 32,491 repositories across all sectors; repositories below this threshold do not contribute to any sector. Thresholds of 0.70 and 0.90 are used as robustness checks.

The sector measures the economic sector associated with the repositories in which observable GitHub activity occurs. It is not the developer's employer or industry of employment: the same developer may contribute to repositories classified in several sectors, and a repository's sector describes the application domain inferred from its text rather than the organization that funds the work.

5 Empirical Strategy

5.1 Synthetic control

The headline design treats Bangladesh as the single treated country and the fourteen other countries as the donor pool. Let Y_{ct} denote the daily active-user rate for country c on date t. Synthetic-control weights are nonnegative, sum to one, and are chosen to minimize the mean squared difference between Bangladesh and the weighted donor combination over the pre-treatment period May 1–July 17. The estimated daily treatment effect is the gap between Bangladesh and the synthetic control during July 18–23.

The method suits the setting because treatment occurs at the country level and the number of aggregate comparison units is modest. Its main advantage here is transparency: the counterfactual is an explicit weighted combination of observed countries, and the credibility of the design can be evaluated through pre-treatment fit, donor weights, in-space placebos, and leave-one-out exercises (Abadie, Diamond, and Hainmueller, 2010; Abadie, 2021). Writing w_j for the weight on donor j, the estimated average treatment effect on the treated over the shutdown is

$$\mathrm{ATT}_{\mathrm{SC}} = \frac{1}{|\mathcal{T}_1|} \sum_{t \in \mathcal{T}_1} \Big[Y_{\mathrm{BD},t} - \sum_{j} w_j Y_{j,t} \Big], \tag{2}$$

where $\mathcal{T}_1$ denotes the shutdown days.

How the weights are obtained. Let $\mathcal{T}_0$ be the 78 dates from May 1 to July 17, and let $\Delta^{14} = \{w \in \mathbb{R}^{14} : w_j \geq 0, \sum_{j=1}^{14} w_j = 1\}$. The estimator solves

$$\widehat{w} = \arg\min_{w \in \Delta^{14}} \frac{1}{78} \sum_{t \in \mathcal{T}_0} \left(Y_{\mathrm{BD},t} - \sum_{j=1}^{14} w_j Y_{jt} \right)^2 . \tag{3}$$

Thus every pre-shutdown daily outcome is a lagged predictor. If X_1 is Bangladesh's 78×1 vector and X_0 the 78×14 donor matrix, the objective is $(X_1 - X_0 w)' V (X_1 - X_0 w)$ with $V = I_{78}/78$. No intercept or extrapolation beyond the donors' convex hull is permitted. The optimization uses only pre-treatment observations; the six shutdown observations do not select donors or tune weights. The three positive weights are solutions to this constrained least-squares problem, rather than subjective measures of economic similarity. A zero weight does not mean a country was dropped from the original donor pool. Numerically, nonnegative least squares uses a heavily weighted adding-up constraint, followed by normalization and verification that weights sum to one. The reported fit is evaluated in percentage points on the original daily series.

Covariates and identifying assumptions. The primary specification matches lagged outcomes only. It includes no temperature, precipitation, electricity-use or macroeconomic covariates, and it does not estimate a separate predictor-weighting matrix. Historical outcomes may summarize persistent determinants of activity, but this does not guarantee balance on every relevant factor. To assess dependence on the lag representation, I also report fits using weekly means, the last 28 or 14 daily observations, and weekday means; these are sensitivity exercises, not replacements selected by their treatment estimates. Pre-treatment holdout errors are reported alongside them. Electricity consumption measured during the shutdown could itself respond to the disruption and must not be added as an ordinary control for its total effect. Weather controls would require a geographically defensible measure of developers' exposure; assigning national developers the weather of one capital city would introduce an unvalidated proxy. Neither is silently included in the estimates.

For $\widehat{Y}^0_{\mathrm{BD},t} = \sum_j \widehat{w}_j Y_{jt}$ to approximate Bangladesh without the episode, the pre-period relationship must remain stable, donors must not be affected by Bangladesh's shutdown or their own coincident interventions, and no other Bangladesh-specific shock may generate the same discontinuity. International collaboration can transmit spillovers, while domestic unrest can violate the last assumption. Good pre-treatment fit is a diagnostic for these conditions, not proof of them. Placebos re-estimate the weights for each comparison country; the small donor pool limits the attainable rank probabilities. The main estimate is therefore conditional on these assumptions and concerns observable public activity during the shutdown episode.

5.2 Difference-in-differences and event study

As a complementary design, I compare the change in Bangladesh's mean outcome from the pre-period to the shutdown with the average corresponding change across the donor countries:

$$\mathrm{ATT}_{\mathrm{DiD}} = \left[\bar{Y}_{\mathrm{BD},1} - \bar{Y}_{\mathrm{BD},0}\right] - \frac{1}{14}\sum_{j=1}^{14}\left[\bar{Y}_{j,1} - \bar{Y}_{j,0}\right], \tag{4}$$

where $\bar{Y}_{c,0}$ is country c's mean daily active-user rate over May 1–July 17 and $\bar{Y}_{c,1}$ its mean over July 18–23. Unlike the synthetic control, which chooses donor weights to reproduce Bangladesh before the shutdown, this comparison gives equal weight to every donor. It is used primarily as a magnitude check against the synthetic-control estimate.

To trace the dynamics of the gap, I replace the shutdown window with weekly event periods k and compute

$$\hat{\theta}_k = \left[\bar{Y}_{\mathrm{BD},k} - \bar{Y}_{\mathrm{BD},0}\right] - \frac{1}{14}\sum_{j=1}^{14}\left[\bar{Y}_{j,k} - \bar{Y}_{j,0}\right], \tag{5}$$

where $\bar{Y}_{c,k}$ is country c's mean daily active-user rate in event period k; $k = -11, \ldots, -1$ are seven-day bins ending July 17 (bin -11 also contains May 1), $k = 0$ is the shutdown (July 18–23), and $k = +1, \ldots, +6$ are seven-day bins from July 24 to September 3. Equation (5) is computed directly from the stored country-day panel of the fifteen countries; it involves no regression and no user-level fixed effects. By construction, $\hat{\theta}_0 = \mathrm{ATT}_{\mathrm{DiD}}$, so the shutdown-week coefficient reproduces the aggregate DiD magnitude. Under parallel pre-shutdown paths, the coefficients for $k < 0$ would fluctuate narrowly around zero. As shown in Section 6.2, Bangladesh's pre-period is volatile and does not display a clean conventional parallel-trends pattern. The weekly coefficients are therefore not interpreted as an identified dynamic causal treatment path. They are diagnostic evidence on whether the largest break in the series is concentrated at the shutdown, and they complement, rather than replace, the synthetic control.

Because treatment varies only at the country level and there are fifteen countries, I do not rely on user-clustered standard errors. Uncertainty is instead summarized with in-space placebos: each donor is assigned the treatment in turn and compared with the equal-weighted mean of the remaining thirteen donors. I report the standard deviation of the fourteen placebo estimates, a donor-placebo reference interval equal to $\hat{\theta}_k \pm 1.96$ times that standard deviation, and the rank of Bangladesh's absolute estimate among the fifteen units. These reference intervals describe how far donor countries move under the same calculation; they are not conventional sampling confidence intervals, and because Bangladesh's pre-period is more volatile than any donor's, they are not fully calibrated measures of uncertainty for Bangladesh.

Temporal robustness. Regression discontinuity in time is retained only as a sensitivity check. Its cutoff predates the nationwide outage, and its estimates depend on bandwidth and sample; the equation and complete results are reported in Section A.4.

5.3 Inference and robustness

For synthetic control, I implement in-space placebos by assigning the treatment date to each donor country in turn and reconstructing its synthetic control using the remaining donors. The primary rank statistic compares Bangladesh's absolute ATT and post/pre RMSPE ratio with the placebo distribution. Abadie, Diamond, and Hainmueller (2015) illustrate the use of post/pre RMSPE ratios to scale post-intervention discrepancies by pre-intervention fit, alongside in-time placebos that reassign treatment to earlier dates. With fourteen placebo countries, the smallest possible finite-sample rank value is $1/(14 + 1) = 0.0667$; consequently, a value of 0.0667 means Bangladesh is the most extreme unit, not that the effect is weak.

I also re-estimate the primary synthetic control after removing each donor country one at a time. Holidays are excluded in a separate DiD specification. Finally, outcome-specific synthetic controls allow donor weights to vary by outcome, whereas the main secondary-outcome table deliberately reuses the active-rate synthetic-control weights. Reusing the primary weights reduces the risk that each new outcome is matched with a different donor combination chosen after observing results.

5.4 International synthetic controls by repository domain

Let G_i be the domain assigned before treatment and N_{cs} the fixed number of eligible users in country c assigned to group s. The country-domain outcome is

$$Y_{cst} = \frac{100}{N_{cs}} \sum_{i \in \mathcal{U}_c} \mathbf{1}\{G_i = s\}\mathbf{1}\{P_{it} > 0\}. \tag{6}$$

It counts the same primary public activity anywhere on GitHub among this fixed cohort; it does not condition on remaining in the original repository after the shutdown. Post-treatment switching therefore cannot change membership or the denominator. Each country-domain series is compared with the same domain in other countries, solving Equation (3) separately for s, on the same 78 pre-treatment dates. The shutdown ATT is the six-day mean of $Y_{\mathrm{BD},st} - \sum_j \widehat{w}_{js} Y_{jst}$.

The support rule requires at least 30 eligible users in the Bangladesh group and in each candidate donor group, and at least five eligible donor countries. This rule is based on sample support, not the observed effect. Every supported domain is reported, together with its denominator, donor count, pre-fit, placebo ranks and leave-one-out sensitivity. Unsupported domains are listed separately; they

are not assigned zero effects. The unclassified group is also estimated when supported. Confidence-threshold sensitivities change membership explicitly and never replace the original national sample. The July 1–17 holdout diagnostic refits weights through June 30; group definitions remain fixed at July 17, so this is a conditional fit diagnostic rather than a fully prospective prediction experiment.

A causal interpretation requires the same counterfactual stability and no-spillover conditions as national synthetic control, now within each group, as well as comparable measurement across countries. In particular, sector-specific unrest or differences in unobserved jobs may violate these conditions. Country placebos provide a reference distribution, with minimum rank value $1/(J_s+1)$ when J_s donors qualify; they do not justify conventional 5-percent significance claims or a causal ranking across many sectors. Different group effects need not be statistically distinguishable. The observed national rate can be reconstructed exactly as $Y_{ct} = \sum_s (N_{cs}/N_c) Y_{cst}$, including the unclassified group. The sector-specific synthetic counterfactual does not generally aggregate to the national synthetic counterfactual because weights and donor support differ across groups. A weighted sum of supported group effects is therefore reported only as a contribution for that covered sample, never substituted for the national headline.

6 Results

Table 3 summarizes the estimates for the primary outcome from the three designs. The synthetic control is the primary causal specification; the international difference-in-differences and the regression discontinuity in time are complementary analyses of the same shock. The subsections below present each design's main figure and table.

Table 3: Summary of estimates across designs: daily active-user rate

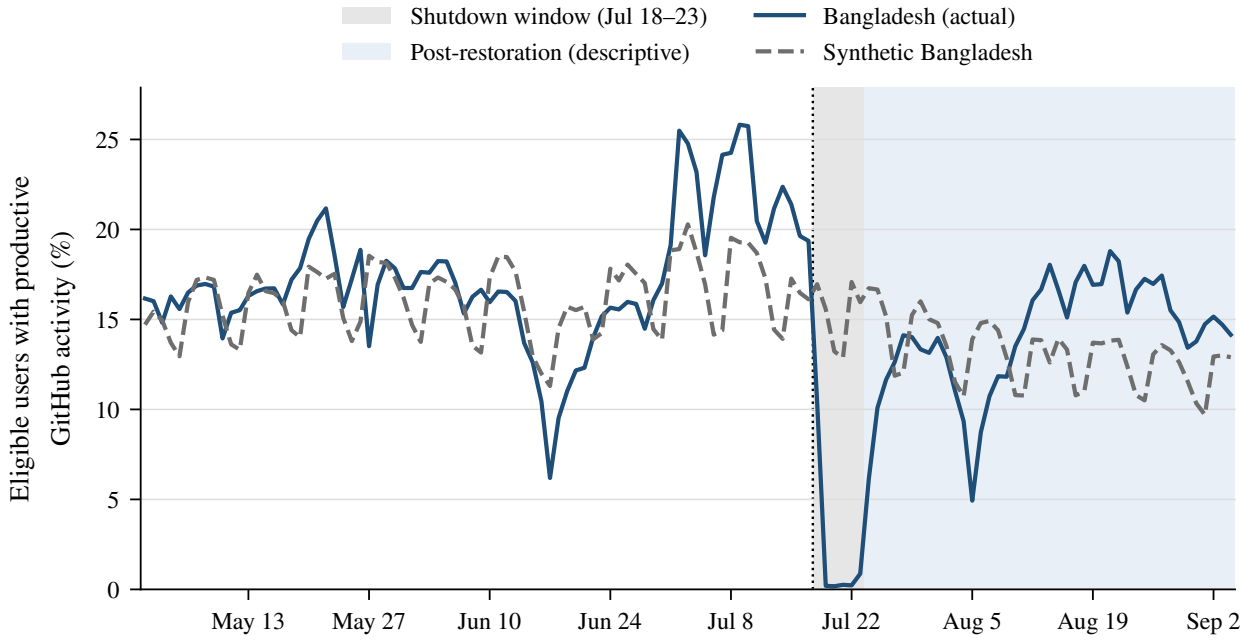
Design	Effect (pp)	Relative effect	Inference	Role
Synthetic control	−13.256	−86.8%	In-space placebo rank: 1/15 = 0.0667 (14 placebo countries)	Primary causal design
International DiD	−13.088	−86.6%	Donor-placebo rank: 1/15; placebo SD 1.17 pp	Magnitude check
DiD, holidays excluded	−13.379	−86.9%	Donor-placebo rank: 1/15; placebo SD 1.19 pp	Robustness
RDiT, local linear	−9.284	–	HAC p = 0.065 (SE 5.03 pp)	Temporal robustness

Notes: Outcome: daily percentage of eligible users with observable productive GitHub activity; SCM and DiD effects average July 18–23, 2024; the RDiT effect is a cutoff discontinuity. All effects are in percentage points. The inference column reports different objects that are not directly comparable. The rank values are the share of the 15 countries whose estimate is at least as extreme as Bangladesh's; 1/15 is the minimum attainable and means that Bangladesh is the most extreme unit, not $p < 0.05$. The RDiT p-value is a conventional time-series (HAC) p-value for the preferred specification (cutoff July 17, bandwidth 7 days, degree 1). Levels, fit statistics, and full inference are in Tables 4, 5 and A13.

6.1 Synthetic control: main effect on daily productive activity

The synthetic-control estimate shows a collapse in observable GitHub activity (Figure 1 and Table 4). Bangladesh's active-user rate averages 2.021 percent during July 18–23, compared with a synthetic counterfactual of 15.277 percent. The ATT is −13.256 percentage points, equivalent to an 86.8 percent reduction relative to the synthetic counterfactual. The pre-treatment RMSPE is 3.260 percentage points and the post/pre RMSPE ratio is 4.192.

Figure 1: Actual and synthetic Bangladesh: daily share of eligible users with observable productive GitHub activity



Notes: Daily percentage of eligible Bangladesh-associated accounts with observable productive GitHub activity (solid) and its synthetic control (dashed; Pakistan 47.6%, Ghana 32.5%, Kenya 19.9%, weights fit on May 1–July 17; Table 4). The shaded shutdown interval defines the primary treatment estimand. Post-restoration differences are shown descriptively and are not interpreted as additional causal treatment effects. The blue band (July 24–September 4) extends the synthetic series with the same weights; average gaps in weeks +4 and +5 (4.24 and 4.67 percentage points) remain within the range of pre-treatment weekly gaps (−2.86 to 5.88). June 17 coincides with a national holiday. August 4–5 includes a second Internet shutdown; August 5 also falls in an initially declared holiday period, so the dip cannot be attributed to a holiday alone (Belson, 2024c; Bangladesh Sangbad Sangstha, 2024). Pre-treatment fit is imperfect (RMSPE 3.26 percentage points), most visibly in early July. Source: Author's calculations using GH Archive.

Keeping the original donor weights, the complete low-connectivity days July 19–22 yield actual activity of 0.212 percent, a 14.687 percent synthetic counterfactual, and a gap of −14.475 percentage points (−98.55 percent), alongside the primary July 18–23 estimate of −13.256 percentage points (−86.77 percent). The narrower window is a reported sensitivity, not a replacement for the primary estimand. Its residual activity averages 21.75 accounts per day. If every such active account were outside Bangladesh, location error among those observed active would account for at most 0.212 percent of the eligible cohort per day. This is not an upper bound on all incorrectly

Table 4: Synthetic control: main estimates and placebo inference

Statistic	Value
A. Estimates, July 18–23, 2024	
Bangladesh, actual (%)	2.021
Synthetic counterfactual (%)	15.277
ATT (percentage points)	−13.256
Relative effect	−86.8%
B. Fit and in-space placebo inference	
Pre-treatment RMSPE (pp)	3.260
Post-treatment RMSPE (pp)	13.666
Post/pre RMSPE ratio	4.192
Placebo countries	14
Rank p-value, absolute ATT	0.0667
Rank p-value, RMSPE ratio	0.0667
C. Donor weights	
Pakistan	47.6%
Ghana	32.5%
Kenya	19.9%

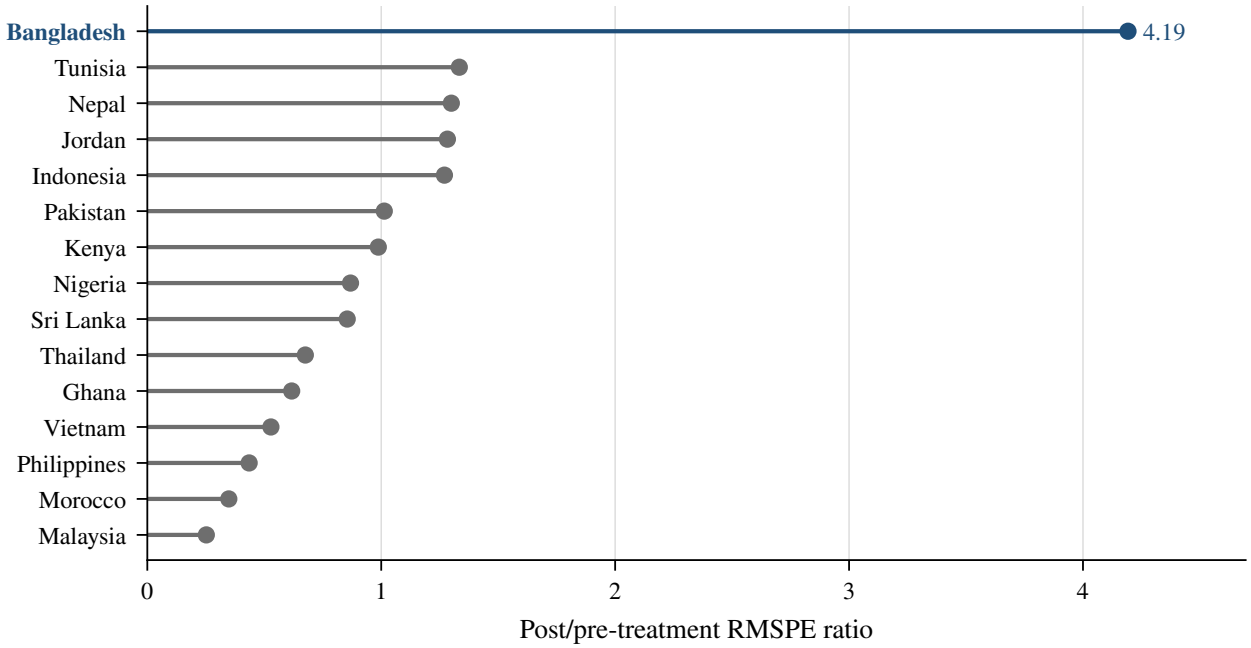
Notes: Outcome: daily percentage of eligible users with observable productive GitHub activity. Weights are chosen on May 1–July 17, 2024; the other eleven donors receive zero weight. Rank p-values compare Bangladesh with 14 in-space placebos; $1/15 = 0.0667$ is the minimum attainable value, so it indicates that Bangladesh is the most extreme unit, not $p < 0.05$. Full placebo ranking in Table A3. Sources: `results/synthetic_control/sc_inspace_placebos.csv`, `results/final/table_primary_SC_weights.csv`.

located accounts: inactive accounts' physical locations remain unobserved, and exceptional connectivity or automated publication could also explain residual activity. The case audit in Section 7.4 directly documents automation in part of this residual.

The synthetic control places 47.6 percent weight on Pakistan, 32.5 percent on Ghana, and 19.9 percent on Kenya (Table 4). No other donor receives positive weight in the baseline optimization. After restoration, shown descriptively in Figure 1, the average weekly gap between Bangladesh and its synthetic counterpart narrows to about −3 percentage points in the first two weeks (−3.11 and −2.89 percentage points) and turns positive in the third week; the positive average gaps in weeks +4 and +5 (4.24 and 4.67 percentage points) lie within the range of pre-treatment weekly gaps and are not interpreted as effects of the shutdown or of restoration. Bangladesh is more extreme than all fourteen in-space placebos in both absolute ATT and RMSPE-ratio comparisons, yielding rank statistics of 0.0667, the smallest value attainable with fourteen placebo units (Figure 2 and Table A3).

The estimate changes little when donors are removed one at a time (Table 11 in Section 8.1): the ATT ranges from −13.256 to −12.886 percentage points. In particular, excluding Pakistan

Figure 2: Synthetic-control in-space placebo inference: post/pre-treatment RMSPE ratios



Notes: Each donor is assigned the July 18–23 treatment in turn, with its synthetic control rebuilt from the remaining donors. Bangladesh has the largest post/pre RMSPE ratio and the largest absolute ATT of the 15 units, so the exact rank p-value is $1/15 = 0.0667$ for both, the minimum attainable with 14 placebo countries. Bangladesh is the most extreme unit, but the resolution of this inference is limited by the size of the donor pool. Source: Author's calculations using GH Archive.

produces −12.905 percentage points, excluding Ghana produces −13.209 percentage points, and excluding Kenya produces −12.886 percentage points. This matters because both Pakistan and Kenya experienced localized connectivity disruptions near the broader pre-treatment period: Internet Society reports regional restrictions in Pakistan around July 14–17 (Internet Society, 2024b,c), while Cloudflare documents a major Kenyan traffic disruption on June 25 (Belson, 2024b). The limited movement in the treatment estimate when either donor is removed reduces, though does not eliminate, concern that these donor-specific episodes drive the result.

Cumulative shortfall and persistence. Using the unchanged synthetic weights and 10,241-account cohort, the July 18–23 gap sums to 8,145 active user-days. The first daily return to the synthetic level is July 27, by which point the cumulative deficit is 10,186 user-days. A daily crossing is not sustained recovery: the first nonnegative trailing seven-day mean gap is August 13 (cumulative deficit 12,068); the first seven consecutive nonnegative daily gaps end August 16. Returning to Bangladesh's own pre-period mean is a different criterion: the first daily crossing is August 14 and the first seven-day mean crossing August 22.

Through September 4, negative daily gaps total 13,526 user-days and positive gaps total 9,893, leaving a net deficit of 3,633. The cumulative deficit peaks at 13,237 on August 9 and is never fully

offset within observation. Positive post-restoration gaps indicate an arithmetic offset, but cannot establish causal catch-up: early-July pre-treatment elevation, later domestic events, and donor disturbances also affect this counterfactual. Fifteen of 43 post-window days remain below it. These are descriptive sums of the original gap, not losses in hours worked or total software production; persistence after September 4 is not observed in this national panel.

6.2 Difference-in-differences and event study

The international DiD estimate is −13.088 percentage points, nearly identical to the synthetic-control ATT (Table 5, Panel A). Removing national holidays yields −13.379 percentage points. The agreement is informative because the two methods weight donors differently: synthetic control chooses donor weights to reproduce Bangladesh before the shutdown, whereas the international DiD gives equal weight to donor-country changes. Bangladesh's estimate is also more extreme than that of every in-space placebo, whose estimates range from −1.72 to 1.60 percentage points; as for the synthetic control, the resulting rank value of 0.0667 is the minimum attainable with fifteen countries.

Figure 3 reports the weekly event study from Equation (5). Its shutdown-week coefficient, −13.09 percentage points, reproduces the aggregate DiD estimate and is more than twice as large in absolute value as any pre-shutdown weekly coefficient. The pre-shutdown period, however, is volatile and does not display a clean conventional parallel-trends pattern. Weekly coefficients range from −4.89 percentage points (week −5, June 13–19) to 6.10 (week −2, July 4–10), and Bangladesh was above its pre-period position relative to the donors in the two weeks immediately before the shutdown. This mirrors the imperfect early-July fit of the synthetic control (Figure 1). Because the reference is the full pre-period mean, this elevated activity does not generate the shutdown estimate; measured against the immediately preceding week alone, the decline would be larger. Given this pre-period volatility, the weekly coefficients are not interpreted as an identified dynamic causal treatment path. They are diagnostic evidence that the largest break in the series is concentrated in the shutdown week.

The donor-placebo reference intervals in Figure 3 should be read with care. They are not conventional sampling confidence intervals: they scale the spread of the fourteen donor placebo estimates in each week. Bangladesh's pre-shutdown weekly coefficients have a standard deviation of 3.04 percentage points, compared with 0.70 to 2.18 percentage points for the donors treated as placebo units. Several pre-period coefficients for Bangladesh are therefore as extreme as any placebo, and the reference intervals are not fully calibrated measures of uncertainty for Bangladesh. The more informative comparison is with Bangladesh's own pre-shutdown range: the shutdown coefficient lies far outside it, whereas every post-restoration coefficient lies within it.

Table 5: International difference-in-differences and aggregate event study

Specification	BD pre (%)	BD shut-down (%)	Donor change	ATT (pp)	Relative effect	Placebo SD	Rank p
A. Difference-in-differences, July 18–23 vs. May 1–July 17							
All days	17.25	2.02	−2.15	−13.09	−86.6%	1.17	0.0667
Holidays excluded	17.58	2.02	−2.18	−13.38	−86.9%	1.19	0.0667
B. Event study, selected weeks (all weeks in Table A11)							
Event week (dates)				$\hat{\theta}_k$ (pp)	Placebo-based interval	Placebo SD	Rank p
−5 (Jun 13–Jun 19)				−4.89	[−9.31, −0.47]	2.26	0.0667
−2 (Jul 4–Jul 10)				6.10	[3.76, 8.43]	1.19	0.0667
−1 (Jul 11–Jul 17)				3.86	[2.15, 5.58]	0.88	0.0667
0 (Jul 18–Jul 23)				−13.09	[−15.39, −10.79]	1.17	0.0667
+1 (Jul 24–Jul 30)				−2.66	[−5.35, 0.02]	1.37	0.0667
+2 (Jul 31–Aug 6)				−2.76	[−4.87, −0.65]	1.08	0.0667
+3 (Aug 7–Aug 13)				0.52	[−1.99, 3.02]	1.28	0.9333
+4 (Aug 14–Aug 20)				4.31	[1.99, 6.64]	1.18	0.0667
+5 (Aug 21–Aug 27)				4.84	[2.40, 7.28]	1.24	0.0667
+6 (Aug 28–Sep 3)				2.53	[0.01, 5.05]	1.29	0.0667

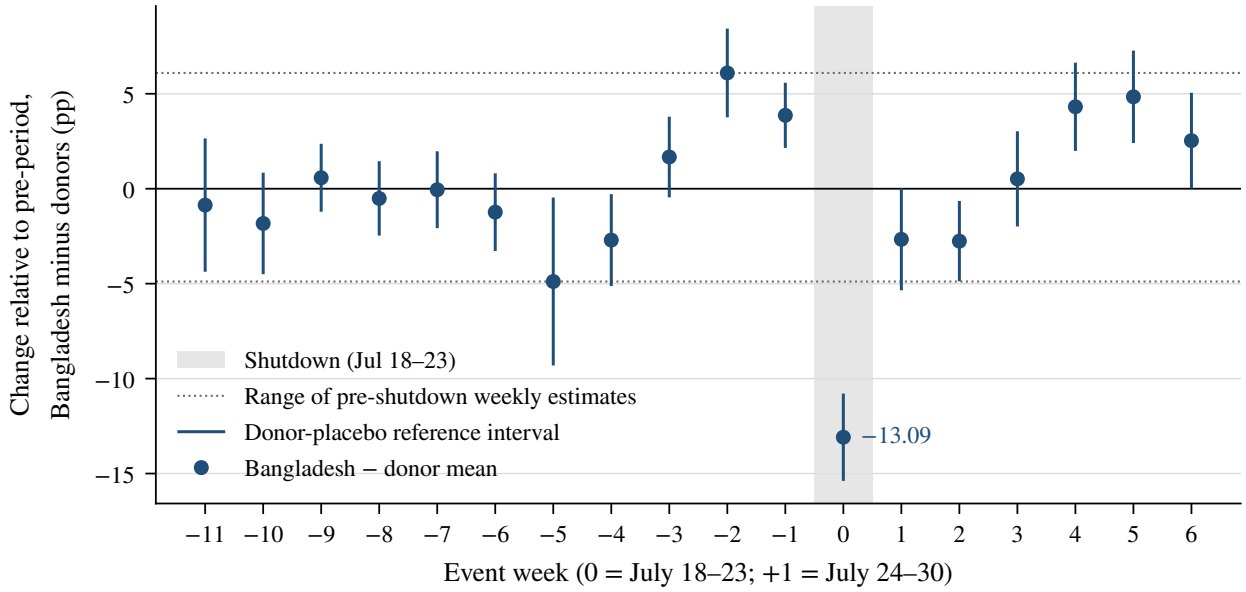
Notes: Outcome: daily percentage of eligible users with observable productive GitHub activity. Panel A compares Bangladesh's change from May 1–July 17 to July 18–23 with the equal-weighted mean change of the 14 donor countries (Equation (4)); the ATT reproduces `results/final/table_DiD_all_outcomes.csv`. Panel B reports $\hat{\theta}_k$ (Equation (5)) for weekly bins; week 0 equals the Panel A all-days ATT. Weeks −5 and −2 are the most negative and most positive pre-shutdown estimates. Inference uses in-space placebos: each donor is compared in turn with the mean of the other 13. *Placebo SD*: standard deviation of the 14 placebo estimates. *Placebo-based interval*: $\hat{\theta}_k \pm 1.96\times$SD, a donor-placebo reference interval, not a conventional sampling confidence interval. *Rank p*: share of the 15 units whose absolute estimate is at least as large as Bangladesh's (minimum 0.0667). Bangladesh's pre-period weekly estimates are more volatile than any donor's, so these placebo-based measures are not fully calibrated sampling uncertainty for Bangladesh. The pre-period does not show a clean parallel-trends pattern, and the weekly coefficients are a diagnostic, not an identified dynamic treatment path; the synthetic control remains the main design. Sources: `results/did/did_main_placebo_inference.csv`, `results/did/did_eventstudy_weekly.csv`.

After restoration, the gap narrows quickly. The coefficients are −2.66 and −2.76 percentage points in the two weeks after the shutdown and are positive from week +3 onward. The positive coefficients in weeks +4 to +6 (2.53 to 4.84 percentage points) lie within the range of pre-shutdown fluctuations and should not be interpreted as gains caused by the shutdown. Recovery dynamics within Bangladesh are examined by repository sector in Section A.7.

6.3 Productive, shipping, and collaboration outcomes

The collapse is not limited to a generic activity indicator. Holding the primary synthetic-control weights fixed, productive activity falls by 11.665 percentage points relative to a 13.469 percent counterfactual, or 86.6 percent. Shipping activity falls by 0.979 percentage points relative to a 1.049 percent counterfactual, or 93.3 percent. Collaboration activity falls by 0.611 percentage points relative to a 0.654 percent counterfactual, or 93.5 percent. Count outcomes show similarly large

Figure 3: Diagnostic event study: Bangladesh relative to the equal-weighted donor mean, by week



Notes: Points are $\hat{\theta}_k$ (Equation (5)): Bangladesh's change in the daily active-user rate relative to its May 1–July 17 mean, minus the equal-weighted mean change of the 14 donors, by seven-day bin. Week 0 is the July 18–23 shutdown (shaded) and equals the DiD estimate in Table 5. Vertical lines are donor-placebo reference intervals (±1.96 times the standard deviation of the 14 placebo estimates), not conventional confidence intervals, and are not fully calibrated because Bangladesh's pre-period is more volatile than any donor's. Dotted lines mark the range of Bangladesh's pre-shutdown coefficients. The pre-period shows no clean parallel-trends pattern; the coefficients are a diagnostic, not an identified dynamic treatment path. Source: Author's calculations using GH Archive.

relative declines in collaboration events and shipping output (Table 6). International DiD estimates for the same outcomes, which point in the same direction, are reported in Table A12.

Allowing the donor weights to re-optimize separately for each outcome produces comparable magnitudes: −86.7 percent for productive activity, −90.5 percent for shipping activity, and −91.9 percent for collaboration activity. The in-space placebo ATT rank is again 0.0667 for all three key activity-rate outcomes. The RMSPE-ratio rank is 0.0667 for productive and shipping activity but 0.1333 for collaboration activity. Among the count outcomes, shipping output has a rank of 0.0667 on both statistics, whereas collaboration events have an ATT rank of 0.2667 and an RMSPE-ratio rank of 0.1333, so Bangladesh is not the most extreme unit for that outcome. I therefore treat collaboration as consistent secondary evidence rather than giving it the same inferential weight as the primary and shipping outcomes.

6.4 International comparisons by repository domain

The international extension complements the before–after sector comparison with a separate comparison-country counterfactual for each fixed pre-treatment repository-domain cohort. The main historical-text rule assigns 2,163 of the 10,241 Bangladesh accounts (21.1 percent) to a

Table 6: Secondary outcomes

	Fixed primary weights		Outcome-specific weights		Placebo rank	
Outcome	ATT	Relative	ATT	Relative	ATT	RMSPE ratio
Productive active rate	−11.665 pp	−86.6%	−11.720 pp	−86.7%	0.0667	0.0667
Shipping active rate	−0.979 pp	−93.3%	−0.668 pp	−90.5%	0.0667	0.0667
Collaboration active rate	−0.611 pp	−93.5%	−0.480 pp	−91.9%	0.0667	0.1333
Collaboration events / 1,000	−19.956	−95.5%	−11.640	−92.5%	0.2667	0.1333
Shipping output / 1,000	−17.913	−93.0%	−13.274	−90.8%	0.0667	0.0667

Notes: Fixed-weight specifications reuse the primary active-rate SCM weights (Table A2). Outcome-specific specifications re-estimate donor weights separately for each outcome (Table A4). Rank values are exact in-space placebo ranks for the outcome-specific specifications with 14 placebo countries; 0.0667 is the minimum attainable value. All outcomes are computed over eligible users in both the numerator and the denominator.

domain. Of these, 1,822 accounts belong to 8 named domains that pass the support rule; the remaining eligible accounts are either in unsupported named groups or the separately reported unclassified group. Coverage by country and the full support screen are reported in Tables A7 and A8. The complete country-by-country appendix (Section A.10) displays all 20 groups, including unclassified accounts, across all 15 countries: 300 panels, of which 45 have no assigned users and 144 contain fewer than 30. These cells remain visible even when they cannot support estimation. A country-domain denominator of zero is not a zero activity rate, and small cells can produce large daily percentage movements.

Table 7: International synthetic controls by pre-treatment repository domain

Domain	$N_{BD,s}$	J_s	ATT (pp)	Rel. (%)	Pre RMSPE	Rank
Retail	368	11	-15.69	-91.8	2.91	0.083
Transport	44	5	-11.31	-96.8	5.85	0.167
Information	518	14	-12.74	-90.4	2.63	0.067
Finance	90	9	-12.28	-85.8	4.04	0.100
Professional services	151	11	-16.13	-89.6	3.91	0.083
Education	424	13	-15.04	-88.2	4.91	0.071
Health	155	5	-16.84	-91.8	7.21	0.167
Arts and recreation	72	7	-12.33	-94.7	5.31	0.250
Unclassified	8,078	14	-12.77	-85.8	3.41	0.067

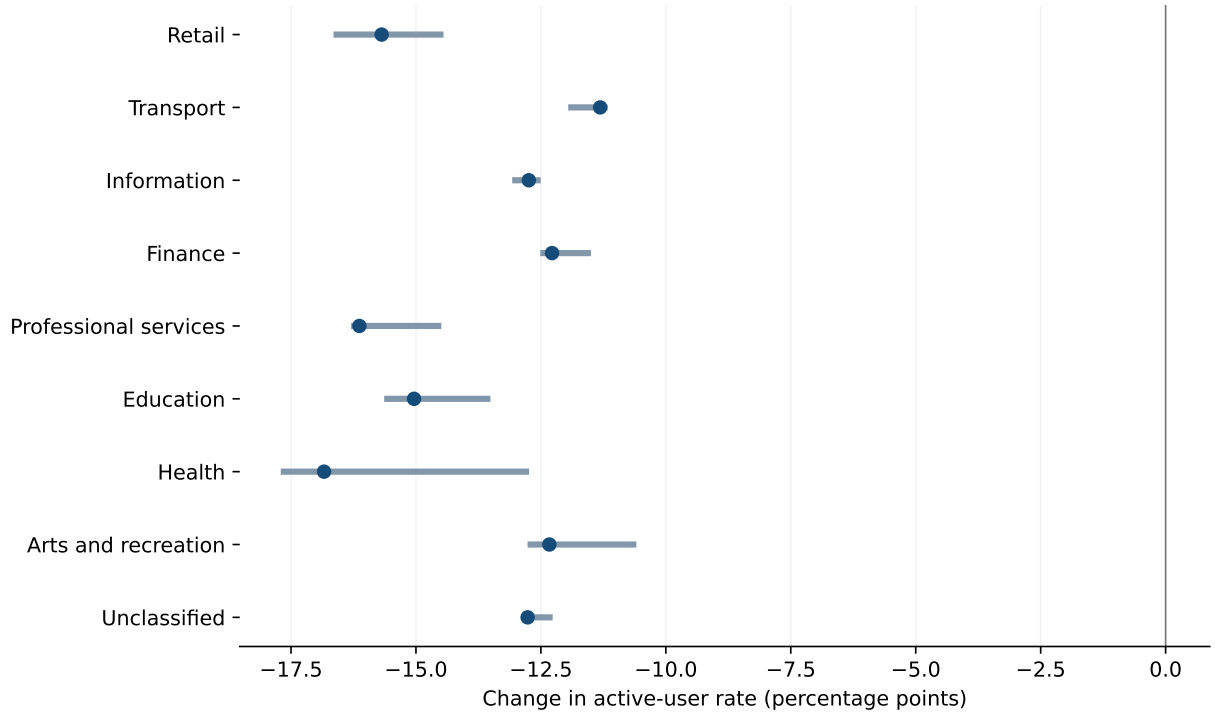
Main assignment: historical README and score at least 0.80. All supported domains and the unclassified group are shown. Outcome: primary public activity anywhere on GitHub among users assigned before treatment. ATT averages July 18–23. Rank compares absolute ATT with country placebos; its minimum depends on donor count. These are conditional comparative estimates, not effects on total industrial output or evidence that domain effects differ statistically.

Across the 8 supported named domains, estimated July 18–23 changes range from -16.84 to -11.31 percentage points. Figure 4 shows every supported estimate and the range obtained when one donor is omitted. These ranges are sensitivity summaries, not confidence intervals. Table A9 reports pre-fit and scale-adjusted placebo ranks, and Table A10 retains all classification-threshold

comparisons. Smaller domain samples and fewer eligible donors make the reference distributions coarser than in the national analysis. No domain ranking is treated as a statistically established ranking of causal exposure.

Retail- and Information-associated cohorts provide relatively closer pre-period fits: their RMSPEs are 2.91 and 2.63 percentage points, compared with estimated shutdown gaps of -15.69 and -12.74. Health has a larger point estimate (-16.84) but also a pre-period RMSPE of 7.21; Education's earlier-fit holdout error is 9.12 points. These diagnostics weaken a comparison of the largest point estimates. Every leave-one-donor-out estimate remains negative, but this robustness does not resolve imperfect fit, classifier error or common exposure to unrest.

Figure 4: Synthetic-control gaps across pre-treatment repository domains



Notes: Points show the July 18–23 ATT for all supported groups. Horizontal segments span the estimates from omitting each donor country in turn; they are not sampling confidence intervals. Assignment uses pre-shutdown activity and historical README classification at score 0.80. The outcome follows assigned users' public activity across repositories.

The supported named groups comprise 17.8 percent of eligible Bangladesh accounts. Weighting their estimates by their shares of the national eligible sample gives a contribution of -2.57 percentage points. This is a partial decomposition for the covered groups, not a national treatment effect: unsupported groups have no assigned zero effect, and the unclassified group is substantial. Moreover, domain-specific weights differ from the national weights. The national headline therefore remains the original −13.26 percentage points.

These comparisons strengthen the design relative to the older within-Bangladesh recovery indices because they introduce untreated-country counterfactuals and freeze group membership before the shock. They still depend on country-domain comparability, imperfect text classification and the absence of coincident group-specific shocks. They concern public activity among users associated with a repository domain; they do not establish effects on firms, employment sectors or industrial output. The separate recovery indices and their broader Bangladesh sample are preserved in the appendix (Section A.7).

7 Mechanism: Offline Work and Delayed Synchronization

7.1 Main backlog measure

A fall in public GitHub events does not necessarily imply an equal fall in code creation. Git is decentralized: developers may continue editing files and creating commits locally, then push them only after connectivity returns. The commit-level analysis examines this pattern among code that is subsequently observed publicly.

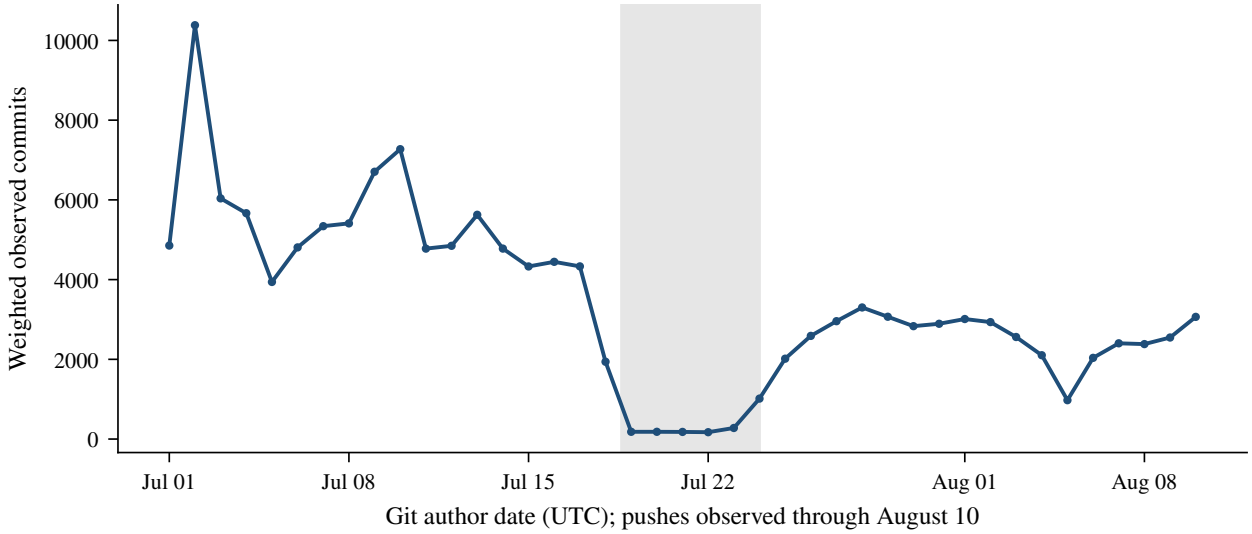
In the strict sample, 2,932 commits are authored during July 18–23. Of these, 829 first appear in a public push on or after July 24. The resulting author-time backlog share is 28.27 percent (Table 8). Replacing the author timestamp with the committer timestamp produces a nearly identical 27.16 percent backlog share (790 of 2,909 commits). The two timestamps do not validate each other: `authoredDate` and `committedDate` are Git metadata that can be set by the same local client and clock, so their agreement shows only that the result is not sensitive to the choice of field. Neither is an externally verified record of when the code was written. The mechanism evidence is therefore consistent with local work followed by delayed synchronization, but it does not directly observe the underlying production process.

Comparisons in levels sharply limit how much adaptation this backlog represents. The calendar cohort contains 2,932 commits (835 users), including 1,835 timestamped before 15:00 UTC on July 18, approximately when national connectivity collapsed (Belson, 2024a). Starting the author-time cohort at that onset leaves 1,097 commits from 250 users; 752 were first pushed on or after July 24 (68.55 percent), compared with 829 of 2,932 (28.27 percent) in the original calendar cohort. Both definitions retain pushes observed through August 10.

Inverse push-day sampling weights produce a mean of 33,787.50 commits across the six original six-day placebo windows. The onset cohort is 96.75 percent smaller, but lasts 5.375 rather than six days. Matching the placebo windows to the same 15:00 start gives a baseline of 30,038.26 and a 96.35 percent decline; scaling the six-day baseline by duration gives 96.38 percent. Creation of code that became public by August 10 therefore fell roughly 96–97 percent, and delayed publication

within that horizon offsets only a small part of the shortfall. The data cannot determine how much of the remaining shortfall reflects work that was not done, work pushed after August 10, or work never published. The original calendar cohort falls 91.32 percent relative to its six-day baseline. Backlog is small in levels: 829 commits are 2.45 percent of the six-day baseline, and the onset cohort's 752 are 2.50 percent of the matched baseline. These are descriptive level comparisons, not a second causal estimate; the early-July activity increase and unresolved metadata both affect their interpretation.

Figure 5: Observable commits by Git author date: weighted levels



Notes: Strict author-equals-pusher sample, UTC dates; counts weighted by the inverse daily metadata-request probability for pushes before July 18, and weight one thereafter. The date of authorship, not publication, defines the outcome. Zero means no observed qualifying commits. Commits never pushed publicly, pushed after August 10, omitted from incomplete payloads, or unresolved by the metadata query are unobserved; weights correct sampling, not those omissions. Later author cohorts have shorter follow-up.

The 345 onset-cohort commits pushed before July 24 involve 64 users. Of these, 246 were pushed on July 23, when fixed broadband began returning; 99 were pushed earlier. The calendar window therefore cannot be treated as complete isolation for every sampled account. Possible explanations include physical location outside Bangladesh, exceptional access, timestamp error and automated publication. The case audit below directly documents the last mechanism for some residual activity. Nor is observation complete: 35 calendar-cohort backlog commits first appear on August 5–10. Further pushes after August 10 and code never published are censored. The estimated creation decline therefore applies to code eventually observed publicly within this horizon, not all code written.

The endpoint also matters. Cloudflare's August account places the beginning of fixed-broadband recovery at 13:00 UTC on July 23, whereas its quarterly summary gives 12:00 UTC (Belson, 2024a,c). Using 12:00 and 13:00 UTC as alternative endpoints yields 899 and 904

onset-authored commits, respectively, from 211 accounts under either definition, compared with 1,097 from 250 accounts through the end of July 23. Of these narrower cohorts, 123 commits were pushed before their respective endpoint; 674 and 678 were first pushed from July 24 onward. These are alternative descriptive counts with different exposure durations, not revised level-effect estimates. The original six-day treatment window and the previously reported onset cohort remain unchanged.

Long publication delays are far more common during the shutdown than before it. Among shutdown-authored commits, 29.71 percent have an author-to-push delay above 24 hours, 25.38 percent above 48 hours, and 22.75 percent above 72 hours. In six overlapping pre-shutdown placebo windows, the corresponding averages are approximately 1.10 percent, 0.72 percent, and 0.46 percent (Table A17). Because commit metadata were queried for only a daily sample of commits pushed before July 18 but for all commits pushed from July 18 onward, the placebo rates use inverse-probability weights, while shutdown-authored commits, all pushed on or after July 18, receive weight one. The shutdown cohort therefore lies far outside the normal pre-period delay distribution.

Table 8: Commit-time mechanism

Timestamp	*N* during shutdown	Pushed Jul 24+	Backlog share	>24h	>48h	>72h	Mean delay (h)
`authoredDate`	2,932	829	28.27%	29.71%	25.38%	22.75%	40.94
`committedDate`	2,909	790	27.16%	28.60%	24.27%	21.73%	37.47

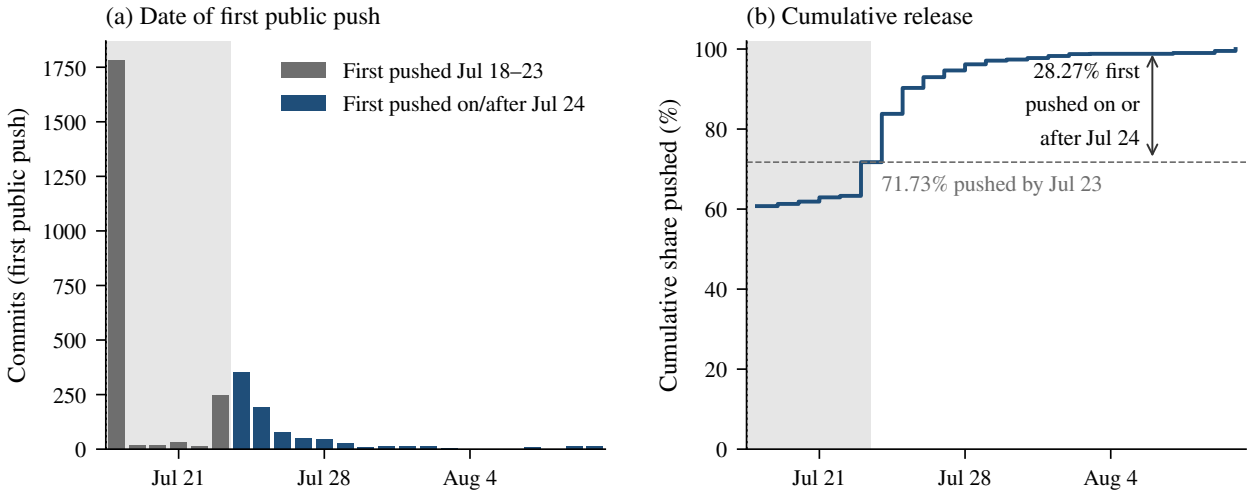
Notes: Strict sample requires author login = PushEvent actor, a non-missing timestamp, and a nonnegative delay. The backlog share is conditional on commits observed in the mechanism sample, not a share of all software production in Bangladesh. `authoredDate` and `committedDate` are Git metadata, not externally verified measurements of when code was written.

7.2 Timing and concentration of delayed pushes

The delayed commits are released quickly once connectivity begins to return (Figure 6). Of the 829 author-time backlog commits, 354 (42.7 percent) appear on July 24 and 544 (65.6 percent) by July 25. By July 27, 81.1 percent have appeared; by July 31, 92.0 percent have appeared. This temporal concentration immediately after restoration is consistent with a synchronization backlog rather than ordinary long-lived branches or sporadic publication.

The backlog is not generated by only a handful of unusually active users (Table A21). The 2,932 shutdown-authored commits come from 835 users and 977 repositories. Two hundred twenty-two users (26.6 percent) and 246 repositories (25.2 percent) contribute at least one backlog commit. The largest user accounts for 2.77 percent of backlog commits, the top five users for 12.06 percent,

Figure 6: Release timing of commits with shutdown-period Git author timestamps



Notes: Strict sample: commits whose GitHub-linked author equals the Bangladeshi pusher, with a non-missing author timestamp and nonnegative author-to-push delay; the cohort is the 2,932 commits authored July 18–23 (UTC). Panel (a): date of first public push; panel (b): cumulative share pushed. 829 commits (28.27%) first appear on or after July 24; pushes are observed through August 10. Git `authoredDate` and `committedDate` are metadata, not verified measurements of when code was written; the pattern is consistent with local work followed by delayed synchronization. The share refers only to this strict sample, not to all software production in Bangladesh. Source: Author's calculations using GH Archive and GitHub commit metadata.

and the top ten for 21.47 percent. Dropping the twenty users with the most backlog still leaves a 20.37 percent backlog share.

Backlog is, however, more common in larger pushes (Table A20). Restricting to pushes with at most 10 commits lowers the backlog share to 21.0 percent; at most 5 commits, 14.4 percent; at most 2 commits, 7.8 percent; and single-commit pushes, 3.3 percent. This pattern is economically meaningful rather than merely inconvenient: if developers accumulate local commits while offline and synchronize later, restoration should produce larger-than-normal push batches. At the same time, it means that the 28.3 percent calendar-cohort backlog share should not be interpreted as applying uniformly to every kind of push.

7.3 Who remains observable during the outage?

Publication during a calendar shutdown window is not the same as coding without connectivity. Among the 10,241 eligible Bangladesh-associated accounts, 1,128 publish a primary event at least once on July 18–23, but only 40 (0.39 percent) do so on July 19–22, the interior days that exclude both the partial first day and the beginning of broadband restoration. The denominator is users rather than user-days: this is the share active at least once across four days, not a daily activity rate. The strict onset-author cohort contains 250 accounts (2.44 percent); 191 publish at least one onset-authored commit from July 24 onward. Only 15 of the 250 also belong to the 40-account

core publication group. Public posting and timestamped local authorship thus select substantially different groups.

Table 9 characterizes them using behavior observed before the shutdown. Core publishers were active on 36.76 percent of pre-period days on average, versus 17.18 percent for other eligible users; their median pre-period number of pushes was 47.5 versus 21. Onset authors were active on 32.62 percent of pre-period days and made a median of 51.5 pushes. Delayed onset authors had a similar activity rate of 32.27 percent. Residual visibility is consequently concentrated among accounts that were already more active, rather than a representative random subset of eligible accounts. Collaboration remains uncommon in all groups; its mean daily incidence is 0.99 percent for core publishers and 0.35 percent for other accounts. These differences describe selection into observed activity, not an effect of personal characteristics on resilience. Among users with pre-period pushes, the mean share of pushes to repositories owned by the same account is 91.21 percent for core publishers and 93.19 percent for other eligible users. This ownership match characterizes their public project activity; it does not identify whether the work was personal, educational or performed for an employer.

Public account-creation dates provide a complementary historical characteristic. Among the 37 core publishers whose current account object resolves and is consistent with their observed 2024 history, median account age at shutdown was 6.28 years; it was 2.95 years among 9,723 other eligible accounts with consistent dates. Onset authors were different: their median age was 3.05 years among 249 consistent records. Thus the accounts still publishing in the interior days were older and more active, whereas the author-time group was mainly distinguished by higher prior activity. Account creation is not a measure of age, professional experience or employment. The metadata were retrieved in October 2026; 439 eligible account objects were unavailable and 42 creation dates conflicted with the historical event record, potentially reflecting changed or reused logins. Those records are excluded from account-age summaries rather than assigned a historical age. Repository-domain composition is reported in Table A16, retaining the unclassified group rather than inferring occupations from sparse labels. In particular, 32 of the 40 core publishers and 192 of the 250 onset authors remain unclassified; the small labelled cells cannot establish a distinctive occupational composition. The core group's 237 primary events are also concentrated: the largest account contributes 17.72 percent and the ten largest 60.34 percent. This concentration describes residual publication, a different quantity from the broader delayed-commit backlog.

There are 19 strict onset authors who publish before July 23, accounting for the 99 commits discussed above; another 48 publish on July 23, with some overlap. Their timestamps do not identify whether they were abroad, had exceptional access, or used an incorrectly set clock. I therefore avoid inferring occupation, employer, wealth, political affiliation or physical residence from continued posting. The mechanism cohort additionally requires eventual public publication and metadata resolution by August 10. More active users are more likely to satisfy those observation conditions,

and people who continued working but never published remain absent.

Table 9: Pre-shutdown characteristics of residual-activity cohorts

Cohort	Users	Active days (%)	Pushes (median)	Collab. days (%)
All eligible	10,241	17.25	21.0	0.35
Core publishers	40	36.76	47.5	0.99
Other eligible users	10,201	17.18	21.0	0.35
Onset authors	250	32.62	51.5	0.50
Delayed onset authors	191	32.27	52.0	0.46
Early onset publishers	19	21.12	17.0	0.67

Characteristics are measured May 1–July 17. Core publishers record at least one primary public event on July 19–22. Onset authors have strict author-equals-pusher commits timestamped July 18 at 15:00 UTC through July 23, observed in pushes by August 10. Delayed authors push at least one such commit from July 24 onward; early publishers push before July 23. Cohorts overlap. These are descriptions of selected observed users, not causal effects or verified residence. Pushes is a pre-period count; activity and collaboration are shares of 78 days.

7.4 Automation and the content of residual activity

An individual case review examines the union of the two cohorts: 275 accounts, 1,232 distinct repository–commit pairs and 321 repository snapshots. Public evidence retrieved in October 2026 resolves 1,210 commit objects and 314 repositories. The review reads author and committer identities, commit messages, available workflows at exact historical commit hashes, and changed-file lists for all 165 retrievable core push heads. Current profile labels are not treated as historical residence or proof of manual operation. Research on repository bots motivates combining identity, message and change evidence rather than relying on usernames alone (Dey et al., 2020; Golzadeh et al., 2021). This is an AI-assisted case review with a documented rubric, not a validated classifier or an independently replicated human annotation exercise. The complete case register, evidence URLs and retrieval dates are supplied with the replication files; Table A27 reports every core publisher.

Direct evidence of automated publication is present in 113 of the 180 core-window PushEvents (62.8 percent), involving 18 of the 40 accounts. Their heads comprise 55 dependency updates, 31 generated profile graphics, 15 profile-metric updates, eight deployments and four generated-data updates. Machine author or committer identities are corroborated by the messages and changed files; historical scheduled workflows provide additional evidence for the profile-metric and generated-data cases. GitHub workflows can run on schedules or remote events (GitHub); a resulting push need not require a new connection from the person associated with the repository. In particular, the account with 42 primary events, 17.72 percent of the residual total, has 41 push heads attributed to a dependency-update bot. The remaining event is retained. A normal account type therefore does not imply manual publication.

The audit also retains uncertainty. Four host-list pushes show highly regular timing and small repeated changes. An update-and-push script and example crontab exist, but deployment of that schedule is unverified; these pushes remain suggestive only. Fifteen heads cannot be recovered. The remaining 48 heads have no directly identified automation mechanism, which does not prove human authorship. Build tests, dependency files, repeated short messages and repositories about test automation are not sufficient evidence on their own. The labels apply to inspected activity, not to whether an account holder is a "bot."

Table 10: Residual publication on July 19–22: original counts and case-audit sensitivity

Definition	Events	Users	User-days	Rate (%)
All primary events	237	40	87	0.212
Excluding automated push heads only	124	29	48	0.117

The daily rate divides active user-days by $4 \times 10{,}241$. Exclusion applies only to PushEvents with directly evidenced automated head commits; all other primary events remain, including possibly related branch and pull-request events. A head audit does not classify every commit in its push. These are descriptive sensitivities, not revised national estimates or validated counts of human work. The original sample and outcome remain unchanged.

Removing only the 113 directly evidenced automated-head pushes reduces primary events from 237 to 124, accounts with any remaining event from 40 to 29, and active user-days from 87 to 48. The corresponding daily rate falls from 0.212 to 0.117 percent (Table 10). All 57 non-push primary events remain, including branch and pull-request events that may be related to automatic updates. This conservative decomposition is neither an estimate of human coding nor a corrected national outcome: inspecting a push head does not classify every commit in that push, and the audit does not cover all pre-period or donor activity. The national estimate is therefore left unchanged.

Automation also qualifies the earlier characterization of unusually active survivors. The 18 accounts with directly evidenced automated heads were active on 52.35 percent of pre-period days and made a median 79.5 pushes. The other 22 core publishers were active on 24.01 percent of days, with a median 20.5 pushes, compared with 17.18 percent and 21 pushes among other eligible accounts (Table A26). Much of the pooled intensity difference is thus concentrated in accounts that also generate automated publication. It cannot be interpreted simply as greater human resilience or professional experience.

The strict author-equals-pusher restriction removes many machine-attributed commits but does not eliminate automation. One of the 1,097 onset-cohort commits has a matching account author yet a GitHub Actions committer and an exact historical workflow that converts a bibliography into generated webpages. Another account contributes ten daily README duration updates without changes in the stated number of solved exercises; these are not new source-code changes, although an unattended publisher was not recovered. Keeping the original 1,097 visible, excluding the di-

rectly documented generated commit gives 1,096; additionally excluding those ten duration-only commits gives 1,086. Neither sensitivity estimates total human work.

Repository content offers a more defensible characterization than inferred occupations. An explicit learning/exercise marker in a repository name is present for 83 of the 250 onset authors (33.2 percent), covering 87 repositories and 384 of their 1,097 commits (35.0 percent). Examples include assignments, data-structure exercises, programming practice and bootcamp repositories. These conservative markers are documented case by case; generic fitness or commercial-looking demo sites are not automatically counted as coursework. The markers describe published projects, not whether the authors are students or professionals. Contemporary reporting also describes software firms sending developers abroad during the blackout (Moli, 2024); this establishes a possible channel of continued access, not the location of any sampled account.

8 Robustness, Identification Threats, and Limitations

8.1 Synthetic-control diagnostics

The primary synthetic control improves pre-treatment fit relative to using the Philippines alone, but the fit is not perfect. The pre-treatment RMSPE is 3.26 percentage points, and Bangladesh exhibits temporary deviations from the synthetic series in early July (Figure 1). The international event study shows the same pattern: Bangladesh's pre-shutdown weekly coefficients relative to the donor mean range from −4.89 to 6.10 percentage points and are more volatile than those of any donor (Figure 3). The paper therefore reports pre-fit explicitly rather than treating the synthetic control as a perfect replica. The design remains informative mainly because the post-treatment gap is large relative to this pre-treatment error and because the estimate changes little in leave-one-out exercises.

In-time placebos ask whether Bangladesh's own series produces large gaps in ordinary weeks (Abadie et al., 2015). Under a rule fixed in advance—six-day pseudo-treatment windows, non-overlapping and moving backward from July 17, each with at least four weeks of fitting data and with donor weights re-estimated using only earlier days—there are eight eligible pseudo-windows (Figure A1 and Table A6). The actual shutdown ATT of −13.256 percentage points is the largest absolute gap among the nine windows. The largest pseudo-gap is +7.349 percentage points (July 6–11), so the actual ATT is about 1.8 times as large in magnitude and has the opposite sign. The actual window also ranks first by the RMSPE ratio. Because the pseudo-windows differ in fit length and are not exchangeable, these ranks are descriptive rather than a formal p-value. The three early-July pseudo-windows are all positive, consistent with the elevated pre-shutdown Bangladesh activity visible in the event study (Figure 3).

The same early-July elevation explains why a synthetic difference-in-differences estimator

(Arkhangelsky et al., 2021) yields a larger estimate, −17.95 percentage points (Table A5). The difference comes from the time weights rather than the donor weights: SDID places 72 percent of its time-weight mass on July 1–17, including 27.1 percent on July 2 and 30.2 percent on July 10, so its weighted Bangladesh baseline is 22.54 percent, compared with a uniform pre-period mean of 17.25 percent; the synthetic control and the DiD instead weight all pre-treatment days equally. The comparison therefore shows how sensitive the estimated magnitude is to whether the baseline emphasizes the final pre-shutdown weeks or the full pre-period; it does not indicate that either estimator is invalid. The synthetic control remains the primary specification.

The donor pool also contains countries with their own short-lived connectivity disturbances. Pakistan experienced regional restrictions around July 14–17, immediately before Bangladesh's national shutdown, and Kenya experienced a major traffic disruption on June 25. These episodes are a reason to avoid presenting donor selection as mechanically pristine. Still, removing Pakistan or Kenya individually changes the ATT by less than 0.4 percentage points (Table 11). A more conservative extension could pre-register donor exclusion rules based on third-party connectivity data and expand the donor pool to increase the resolution of placebo inference.

Table 11: Primary leave-one-out range

Donor removed	ATT (pp)	Relative effect	Pre RMSPE	RMSPE ratio
Ghana	−13.209	−86.73%	3.297	4.131
Pakistan	−12.905	−86.46%	3.376	3.961
Kenya	−12.886	−86.44%	3.293	4.042
Any zero-weight donor (11)	−13.256	−86.77%	3.260	4.192

Notes: Removing donors that receive zero baseline weight mechanically leaves the baseline estimate unchanged.

Lagged-outcome predictors. The alternative predictor representations in Table 12 yield July 18–23 estimates between −12.00 and −15.37 percentage points. Eleven complete weekly means yield −14.14; using the final 28 or 14 daily observations yields −14.70 and −15.37; weekday means yield −12.00. These results retain the national sample and donor pool and do not add external covariates. The daily-outcome primary estimate remains −13.26. All specifications have sizeable errors when fitted through June 30 and evaluated on July 1–17 (6.40–7.52 percentage points), consistent with Bangladesh's unusual early-July activity. The persistence of a large negative shutdown gap does not erase this forecasting limitation or the threat from concurrent unrest.

Table 12: Sensitivity to pre-treatment outcome predictors

Predictors	K	ATT (pp)	Pre RMSPE	Holdout RMSPE
All 78 daily lags	78	-13.26	3.26	7.25
Complete weekly means	11	-14.14	3.46	6.80
Last 28 daily lags	28	-14.70	3.82	7.52
Last 14 daily lags	14	-15.37	4.25	7.17
Weekday means	7	-12.00	3.74	6.40

Same users, donor pool and July 18–23 estimand in every row. Weekly means use the final 77 fitting days, grouped into 11 complete weeks. Holdout errors use a separate fit through June 30 and score July 1–17, applying each predictor rule to the shorter training sample; they are not treatment-period tuning statistics. All errors are in percentage points.

8.2 Metadata and measurement

Commit metadata resolution is similar across periods: 82.34 percent of the pre-period target commits are found, compared with 84.24 percent during the shutdown and 81.81 percent after (Table A18). The shutdown mechanism is therefore not generated by a sharp deterioration in commit-object availability during the treatment window.

A related limitation concerns push-listing completeness: although 99.5–99.8 percent of pushes in every period expose a complete commit list, the share of nominal commits captured within complete pushes falls from 90.27 percent before the shutdown to 73.11 percent during the shutdown and 71.45 percent afterward (Table A19). This occurs because a small number of large pushes account for a disproportionate share of commits and are more likely to have incomplete commit listings. Since the strict mechanism sample restricts to complete pushes, this does not mechanically invalidate the 28.27 percent backlog estimate, but it does imply uneven commit-level coverage across periods. The lower coverage during and after the shutdown is therefore a measurement limitation that should be considered when interpreting the mechanism results.

Location is another measurement limitation. GitHub location fields are user-entered and may be missing, stale, or imprecise. The estimates should be read as applying to accounts associated with Bangladesh and the donor countries, not as a verified census of developers physically located in each country on every date. Public-event data additionally exclude private repositories. If private and public work respond differently to a shutdown, the paper estimates the effect on the observable public margin rather than on all software production.

8.3 Sector classification and the sectoral recovery analysis

Table A23 repeats the sectoral recovery analysis keeping only repositories whose classifier confidence is at least 0.70 or 0.90 instead of 0.80. The qualitative ordering of the main sectors is stable.

At all three thresholds, Finance and Information begin their first run of two consecutive weeks at or above 80 in week +3; Education and Health do so between weeks +5 and +8; and Accommodation/Food never does, with week +8 indices between 60.3 and 64.9. Real Estate's late recovery is broadly but not fully robust: its week +8 index is 90.0 and 85.5 at the 0.70 and 0.80 thresholds but 75.5 at 0.90. Retail's week +4 overshoot appears at every threshold (indices between 153.7 and 159.1), but the start of its first two-week run at or above 80 varies from week +4 (0.70) to week +9 (0.80) and week +7 (0.90). I therefore do not emphasize a precise recovery date for Retail.

Three sectors with substantial activity are excluded from the main results. Professional/ Scientific and Administrative Support have 577 and 196 sampled users active in the baseline period at the 0.80 threshold, but only 125 and 48 at 0.90, so their indices depend heavily on the threshold. In Arts, Entertainment, and Recreation, a single repository accounts for 82.5 percent of pre-shutdown events among the sector's classified repositories, so a sector index would largely track one project.

The classification has further limitations. First, the sector describes the application domain inferred from a repository's text, not the employer or industry of the developer: a developer employed in banking may work on a general-purpose library, and a student exercise may be classified as Retail because it implements an online shop. Second, classifier errors are possible, and this paper does not validate the classifier's accuracy on Bangladeshi repositories. The model card reports that the training labels are GPT-4.1 judgements rather than human annotations and that, on public GitHub repositories, roughly half of what the model labels confidently arguably belongs to no sector at all, such as coursework, portfolios, exercises, and generic tooling (Quispe and Xu, 2026); the score threshold mitigates but does not eliminate this. Third, coverage is incomplete. Only 55.1 percent of repositories have a recoverable historical README, and activity in unclassified repositories does not enter any sector. If newly created repositories are less likely than established ones to have a README at their first observed activity date, post-shutdown recovery through new projects would be understated. Fourth, a user may contribute to several sectors, so sector indices are not shares of a fixed workforce. Finally, the sectoral confidence intervals do not adjust for the many sector-week comparisons reported, and they reflect sampling variation across users rather than common shocks.

8.4 Concurrent unrest and causal interpretation

The most important identification threat is that the Internet shutdown was itself a response to political unrest. Protests, curfews, transportation disruptions, workplace closures, fear, or other domestic conditions could directly reduce software work even if connectivity were available. Comparisons with donor countries, whether through synthetic-control weights or the equal-weighted DiD, cannot absorb shocks unique to Bangladesh that coincide with the shutdown. For this reason, the headline

estimate should be interpreted as the effect of the shutdown episode under the identifying assumption that the abrupt Bangladesh-specific deviation is primarily attributable to lost connectivity.

Several findings make the connectivity interpretation more plausible, although none establishes it. First, the break is tightly aligned with the network outage. Second, activity begins to recover as connectivity returns (Figures 3 and A4). Third, downstream online tasks such as shipping and collaboration fall at least as strongly as the general activity rate. Fourth, commit metadata show a cluster of commits authored during the outage and first pushed immediately after restoration. An account in which developers stopped working only because of unrest does not naturally predict this last pattern, although the cluster is small relative to normal commit volume (Section 7.1). Nevertheless, the mechanism evidence is complementary, not a substitute for randomized assignment.

Selection on at least four pre-period event days can mechanically favor a temporarily active cohort and induce subsequent mean reversion. The equally weighted donor mean falls by 2.146 percentage points between the pre-period and July 18–23. This decline is compatible with selection and common time effects; it does not separately identify mean reversion. The same eligibility rule across countries mitigates but does not eliminate differential selection. The main results retain the original sample; diagnostics that fix eligibility before the estimation period are not part of the main specification.

Pakistan, which receives 47.6 percent of the original synthetic weight, experienced a national connectivity disruption on July 31 (about 08:30–10:30 UTC; Belson, 2024a). This occurs after the primary treatment window but within the recovery comparison. It can depress the synthetic post-period series and inflate Bangladesh's apparent rebound. The primary July 18–23 estimate is unaffected by that date; extrapolated recovery is less secure.

The expanded connectivity audit in Table A25 also documents Kenya's August 30–31 disruption and provider-specific pre-period events in Nepal and Malaysia. The country and domain trajectories are all shown in Section A.10, including unsupported cells. Donor outages can affect both pre-fit and apparent recovery; the source review does not justify excluding countries after observing their outcomes, and it does not establish that unmentioned countries were unaffected.

9 Discussion

The results illustrate why high-frequency digital trace data can add information beyond aggregate estimates of shutdown costs. A national GDP series cannot show whether digital workers cease production, shift to offline tasks, or accumulate work that is released later. GitHub events and metadata measure observed public activity and delayed publication; they cannot observe work that never becomes public or distinguish all unobserved adaptation from lost production.

The main estimate—an 86.8 percent decline in the daily active-user rate relative to the synthetic

counterfactual (Table 4)—is best interpreted as a collapse in observable online software-production activity. The mechanism shows a small observed backlog alongside a collapse in author-time levels: for the cohort authored after the July 18 cutoff, creation of code observed publicly by August 10 falls roughly 96–97 percent from the pre-shutdown comparison, and that cohort's backlog represents 2.50 percent of normal volume (2.45 percent for the calendar cohort). The original 28.27 percent backlog share conditions on a much smaller calendar-cohort denominator. It does not establish that missing activity was displaced across time, or quantify destruction of work that never became public.

At the same time, delayed synchronization observed through August 10 accounts for only a small part of the collapse. Most eligible users record no productive public activity during the shutdown, and the observed backlog is much smaller than the counterfactual volume implied by the active-rate and shipping estimates. Connectivity therefore appears to matter not only for publishing already-written code but also for complementary tasks: accessing remote dependencies, coordinating with collaborators, reviewing code, resolving issues, merging pull requests, and releasing software. The especially large declines in shipping and collaboration are consistent with these channels.

The older within-Bangladesh sector-recovery results add a descriptive dynamic dimension. In aggregate, Bangladesh's gap relative to the donor mean returned within its pre-shutdown range in the first week after restoration (Figure 3), but that wide and volatile range is a weak recovery criterion. The separate daily, rolling-mean and cumulative measures in Section 6 give different return dates and do not show full compensation by September 4. Reconnection did not restore observable activity at once: weeks after the shutdown, activity in Education-, Health-, and especially Accommodation/Food-related repositories remained below its pre-shutdown level, while Finance- and Information-related activity recovered within about three weeks. This pattern is consistent with the costs of a shutdown extending beyond the days without connectivity and falling unevenly across types of projects. The weaker recovery of pre-existing repositories, relative to sector-wide activity, further suggests that part of the recovery took the form of new projects or switching across repositories, which an analysis restricted to pre-existing projects would miss. Because that recovery analysis has no untreated comparison group and classifies repositories rather than workers, these interpretations remain suggestive. It is distinct from the new international cohort comparisons in Section 6.4.

The international domain analysis addresses a different question: whether the shutdown gap persists when Bangladesh-associated users with a given pre-treatment project domain are compared with corresponding users elsewhere. It introduces a counterfactual and prevents post-shutdown project switching from determining group membership, but a classifier does not turn repository descriptions into verified employment data. Incomplete historical-text coverage, generic learning projects, small country-domain samples and concurrent unrest remain substantive limits. The

residual-user analysis is similarly selective. Direct automation evidence in 113 of 180 core-window pushes materially changes the interpretation of the older, more active accounts that remain visible. Their pre-period intensity partly reflects accounts that also publish automated output; it cannot be read as greater human resilience. Timestamped authorship and delayed publication involve a different group, with explicit learning or exercise markers for at least 83 of 250 onset authors. These are characteristics of observed projects, not identified occupations or verified continued work inside Bangladesh.

The national cumulative comparison is more informative about recovery than a first return inside a wide pre-period range. Relative to the unchanged synthetic series, 8,145 active user-days are missing in the six-day shutdown window; later positive gaps partly offset, but never fully repay, the cumulative shortfall, which is 3,633 user-days at September 4. Neither this arithmetic offset nor sectoral baseline crossings identify a causal rebound. The data do not evaluate particular resilience policies, so no comparative policy recommendation is drawn.

10 Conclusion

Bangladesh's July 2024 Internet shutdown provides a high-frequency case study of what happens to digitally mediated production when national connectivity is abruptly removed. Using public GitHub events for 59,158 pre-active accounts across fifteen countries, the paper finds that Bangladesh's daily productive-activity rate fell by 13.26 percentage points relative to a synthetic counterfactual during July 18–23, an 86.8 percent decline. International difference-in-differences, secondary outcomes, placebos, and temporal discontinuity analyses point in the same direction (Table 3). The cumulative synthetic-control deficit is 8,145 active user-days in the shutdown window and 3,633 by September 4 after partial offsets. A first daily crossing on July 27 is weaker evidence of recovery than the first nonnegative seven-day mean gap on August 13. These descriptive sums inherit imperfect pre-fit and concurrent-shock limitations.

The commit-level evidence shows that the observed backlog was small relative to normal creation. After the approximately 15:00 UTC July 18 onset, 1,097 commits were authored and subsequently observed publicly by August 10, a 96.35 percent fall against duration-matched pre-shutdown windows (96.75 percent against the original six-day windows). That cohort's 752 delayed commits equal 2.50 percent of normal volume. Because later pushes and unpublished code are unobserved, these figures do not show how much of the shortfall was work not done rather than work published after August 10 or never published. The original calendar-cohort backlog share of 28.27 percent and onset-cohort share of 68.55 percent describe delayed publication among the surviving observed commits; neither establishes that total software work continued at a comparable rate.

The international extension provides separate synthetic comparisons for fixed repository-

domain cohorts rather than treating a national before–after recovery index as a sector-specific causal effect. Its interpretation is conditional on common measurement and counterfactual stability; it does not identify effects on industry value added or developers' employers. The 40 accounts that remain publicly active on July 19–22 are not interchangeable with the 250 users in the strict onset-author cohort. Direct automation evidence in 113 of their 180 pushes shows that residual posting includes machine-generated publication. Excluding only those pushes lowers the residual daily activity rate from 0.212 to 0.117 percent; the remainder still cannot be equated with verified human work inside Bangladesh.

A within-Bangladesh analysis of the restoration period adds a further qualification: reconnection did not immediately restore observable activity. Recovery was faster in Finance- and Information-related repositories than in Education- and Health-related repositories, and activity in Accommodation/Food-related repositories remained depressed throughout the observed window. These sectoral patterns are descriptive comparisons with each sector's own pre-shutdown level rather than causal estimates against an untreated control.

The paper's main limitation is the coincidence of the shutdown with severe domestic unrest. Future work could combine network telemetry at subnational or provider level with developer location at finer geographic resolution, or exploit variation in restoration timing across networks and regions. Such designs could more sharply separate the direct effect of connectivity from the broader social environment. Even with this limitation, the evidence is consistent with observable software-development activity being highly exposed to network availability and with public development platforms providing unusually granular measures of that exposure.

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A Additional Empirical Diagnostics

This appendix reports the detailed tables behind the synthetic-control, difference-in-differences, mechanism, and sectoral recovery results discussed in the main text. Tables are generated from stored result files by the original table script and the advisor-revision analysis scripts.

A.1 Synthetic control

Table A1: The full country sample and fitted national donor weights

Country	Users	Pre mean (%)	Pre SD (pp)	Weight (%)
Bangladesh	10,241	17.25	3.60	–
Ghana	1,373	17.03	2.24	32.49
Indonesia	10,965	15.71	2.09	0.00
Jordan	446	14.07	2.89	0.00
Kenya	3,553	18.58	3.70	19.93
Malaysia	1,396	14.87	1.85	0.00
Morocco	1,651	15.80	2.93	0.00
Nepal	2,373	16.56	1.62	0.00
Nigeria	6,961	16.52	2.07	0.00
Pakistan	7,276	14.48	1.96	47.58
Philippines	2,926	17.10	1.53	0.00
Sri Lanka	3,068	15.68	1.45	0.00
Thailand	2,317	15.49	1.47	0.00
Tunisia	1,012	14.21	2.54	0.00
Vietnam	3,600	16.76	1.82	0.00

The sample is unchanged. Pre-period: May 1–July 17. Every comparison country enters the fit; zero weight is an optimization result, not an exclusion. Bangladesh is the treated unit. Source: stored eligible-user and country-day panels; regenerated donor audit.

Table A2: Primary synthetic-control donor weights

Country	Weight
Pakistan	47.579%
Ghana	32.488%
Kenya	19.933%

Notes: Weights are estimated on the pre-treatment (May 1–July 17, 2024) daily active-user rate. The remaining eleven donor countries receive zero weight.

Table A3: Synthetic-control in-space placebo inference

Rank	Unit	ATT (pp)	Pre RMSPE	Post RMSPE	Ratio
1	**Bangladesh**	−13.256	3.260	13.666	4.192
2	Tunisia	−2.563	2.011	2.682	1.334
3	Nepal	0.965	0.936	1.216	1.300
4	Jordan	−0.194	2.457	3.151	1.283
5	Indonesia	−1.393	1.151	1.462	1.270
6	Pakistan	0.941	1.585	1.606	1.013
7	Kenya	2.439	2.873	2.837	0.987
8	Nigeria	0.527	0.948	0.823	0.869
9	Sri Lanka	−0.748	1.382	1.181	0.855
10	Thailand	−0.215	0.965	0.652	0.675
11	Ghana	0.268	1.144	0.706	0.617
12	Vietnam	−0.097	0.867	0.458	0.528
13	Philippines	0.091	0.955	0.416	0.435
14	Morocco	0.050	1.794	0.625	0.348
15	Malaysia	0.131	1.177	0.297	0.252

Notes: Each donor country is assigned the July 18–23 treatment in turn and its synthetic control is rebuilt from the remaining donors. Units are ordered by the post/pre RMSPE ratio. Bangladesh has both the largest ratio and the largest absolute ATT among the 15 units, so the exact rank p-value is $1/15 = 0.0667$ for both statistics. This is the minimum attainable value with 14 placebo countries; inference resolution is limited by the size of the donor pool.

Table A4: Outcome-specific synthetic-control donor weights

Outcome	Country	Weight
Productive active rate	Pakistan	30.0%
	Ghana	26.1%
	Philippines	23.8%
	Kenya	18.0%
	Morocco	2.2%
Shipping active rate	Pakistan	72.4%
	Morocco	10.8%
	Ghana	9.2%
	Malaysia	4.8%
	Jordan	2.4%
	Tunisia	0.5%
Collaboration active rate	Pakistan	73.6%
	Morocco	26.4%
Collaboration events / 1,000	Morocco	84.0%
	Indonesia	13.5%
	Pakistan	2.5%
Shipping output / 1,000	Pakistan	57.2%
	Morocco	17.5%
	Ghana	8.9%
	Nigeria	8.1%
	Nepal	4.1%
	Malaysia	1.8%
	Jordan	1.4%
	Tunisia	0.7%
	Philippines	0.2%

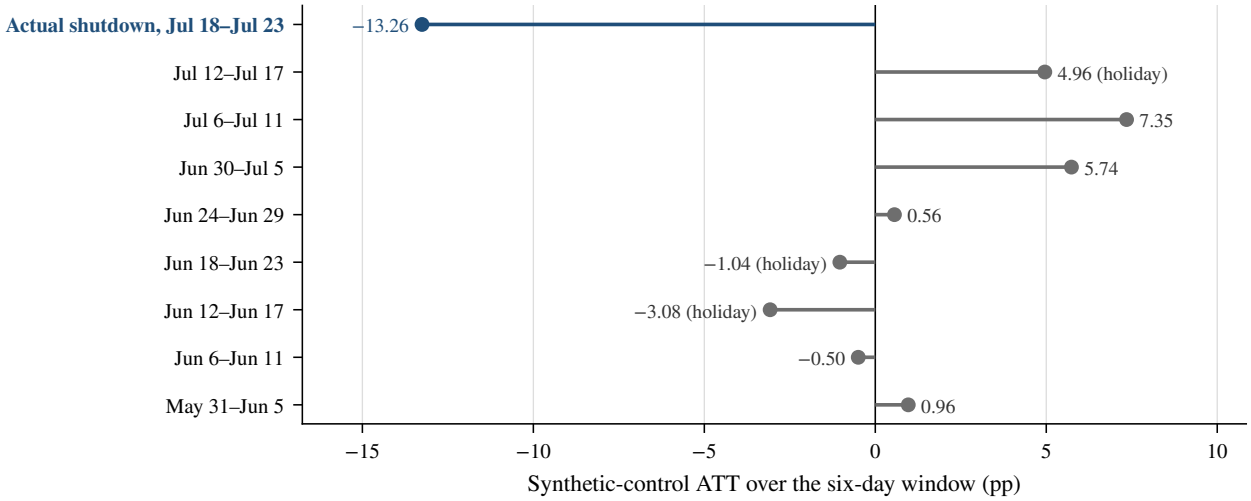
Notes: Donors with zero weight are omitted. These weights are reported for transparency; the main secondary-outcome comparison holds the primary weights in Table A2 fixed.

Table A5: Synthetic difference-in-differences: sensitivity to the pre-treatment baseline

Statistic	Value	
A. Estimates, July 18–23, 2024 (pp)		
Synthetic DiD (archived)	−17.9495	–
Canonical synthetic control (primary)	−13.256	–
B. Decomposition by weighting scheme (pp)		
Unit weights	SDID time weights	Uniform time weights
SDID unit weights	−17.95	−13.86
SC unit weights	−17.99	−14.38
Equal unit weights	−17.87	−13.09
C. SDID time weights		
Share of weight on July 1–17	72.2%	
Weight on July 2 / July 10	27.1% / 30.2%	
Days with positive weight (of 78)	14	
Bangladesh baseline, SDID-weighted (%)	22.54	
Bangladesh baseline, uniform pre-period mean (%)	17.25	

Notes: Panel A: the archived SDID estimate (Arkhangelsky et al., 2021) and the primary synthetic control. The archived SDID standard error and p-value are not reported because their computation is undocumented. Panel B recomputes the point estimate from the archived SDID unit and time weights, replacing one set at a time; equal unit weights with uniform time weights reproduce the aggregate DiD. Panel C summarizes the archived time weights over May 1–July 17. Nothing is re-estimated. Sources: `results/synthetic_control/main_synthetic_results.csv`, `sdid_time_weights.csv`, `donor_weights.csv`; computed by `analysis/synthetic_control/02_sdid_archived_check.py`.

Figure A1: Synthetic-control in-time placebos: ATT over six-day pseudo-treatment windows



Notes: Each grey row is a six-day pre-treatment window chosen by the rule in Section 8.1; donor weights are re-estimated using only the days before each window, and the ATT is the mean gap between Bangladesh and its synthetic control over the window, as in the primary specification (blue row). "Holiday" marks windows containing a Bangladesh national holiday. All eight eligible windows are shown. Details in Table A6. Source: `results/synthetic_control/reproduced/sc_intime_placebos.csv`.

Table A6: Synthetic-control in-time placebos

Window (2024)	Fit days	ATT (pp)	Pre RMSPE	Ratio	Rank |ATT|	Rank ratio	Holiday
Jul 18–Jul 23 (actual)	78	−13.256	3.260	4.192	1	1	
Jul 12–Jul 17	72	4.958	3.032	1.787	4	5	Ashura
Jul 6–Jul 11	66	7.349	2.279	3.376	2	3	
Jun 30–Jul 5	60	5.736	1.634	3.793	3	2	
Jun 24–Jun 29	54	0.560	1.685	0.654	8	9	
Jun 18–Jun 23	48	−1.036	1.678	1.227	6	6	Eid al-Adha
Jun 12–Jun 17	42	−3.076	1.546	2.487	5	4	Eid al-Adha
Jun 6–Jun 11	36	−0.498	1.588	0.844	9	7	
May 31–Jun 5	30	0.964	1.683	0.720	7	8	

Notes: Pre-specified rule: six-day pseudo-treatment windows, non-overlapping, moving backward from July 17, each with at least 28 days of fitting data from May 1. Donor weights are re-estimated for each window using only days before it; ATT and RMSPE definitions are those of the primary specification. Ranks are among the nine windows. The windows differ in fit length and are not exchangeable, so the ranks are descriptive and not a formal p-value. Source: `results/synthetic_control/reproduced/sc_intime_placebos.csv`, computed by `analysis/synthetic_control/03_intime_placebos.py`.

A.2 International repository-domain diagnostics

The following tables retain every country's coverage, all domain support decisions, and the fit and threshold diagnostics for the international extension. Missing estimates indicate insufficient support under the stated rule.

Table A7: International historical-text classification coverage

Country	Eligible	README	Assigned	Assigned (%)
Bangladesh	10,241	5,749	2,163	21.1
Ghana	1,373	892	318	23.2
Indonesia	10,965	6,810	2,678	24.4
Jordan	446	281	102	22.9
Kenya	3,553	2,264	870	24.5
Malaysia	1,396	916	343	24.6
Morocco	1,651	984	325	19.7
Nepal	2,373	1,272	444	18.7
Nigeria	6,961	4,442	1,552	22.3
Pakistan	7,276	4,209	1,512	20.8
Philippines	2,926	1,779	678	23.2
Sri Lanka	3,068	1,954	711	23.2
Thailand	2,317	1,526	581	25.1
Tunisia	1,012	577	230	22.7
Vietnam	3,600	2,182	898	24.9

Counts refer to users and their dominant pre-treatment repository. Assigned requires an available July 17 README and classifier score at least 0.80. Other eligible users remain unclassified; they are not removed from national estimates. Retrieval takes place in October 2026 and cannot recover every repository available in 2024.

Table A8: Pre-specified support screen for every domain

Domain	Bangladesh users	Eligible donors	Estimated
Agriculture	25	3	No
Mining	0	0	No
Utilities	8	0	No
Construction	1	0	No
Manufacturing	8	0	No
Wholesale	0	0	No
Retail	368	11	Yes
Transport	44	5	Yes
Information	518	14	Yes
Finance	90	9	Yes
Real estate	49	1	No
Professional services	151	11	Yes
Administrative support	36	1	No
Education	424	13	Yes
Health	155	5	Yes
Arts and recreation	72	7	Yes
Accommodation and food	192	4	No
Other services	18	1	No
Public administration	4	0	No
Unclassified	8078	14	Yes

At least 30 Bangladesh users and five donor countries with at least 30 users each are required. The rule uses sample support, not treatment-period effect size. Unreported estimates in the main table correspond to this screen, not a zero estimated effect.

Table A9: Sector-cohort fit and scale-adjusted placebo diagnostics

Domain	Pre RMSPE / mean	Holdout (pp)	Post/pre RMSPE	Ratio rank
Retail	0.15	3.57	5.43	0.083
Transport	0.51	5.16	1.96	0.167
Information	0.16	3.61	4.98	0.067
Finance	0.27	6.27	3.10	0.100
Professional services	0.24	5.24	4.18	0.083
Education	0.24	9.12	3.13	0.071
Health	0.32	6.20	2.40	0.167
Arts and recreation	0.31	6.65	2.32	0.125
Unclassified	0.20	7.50	3.89	0.067

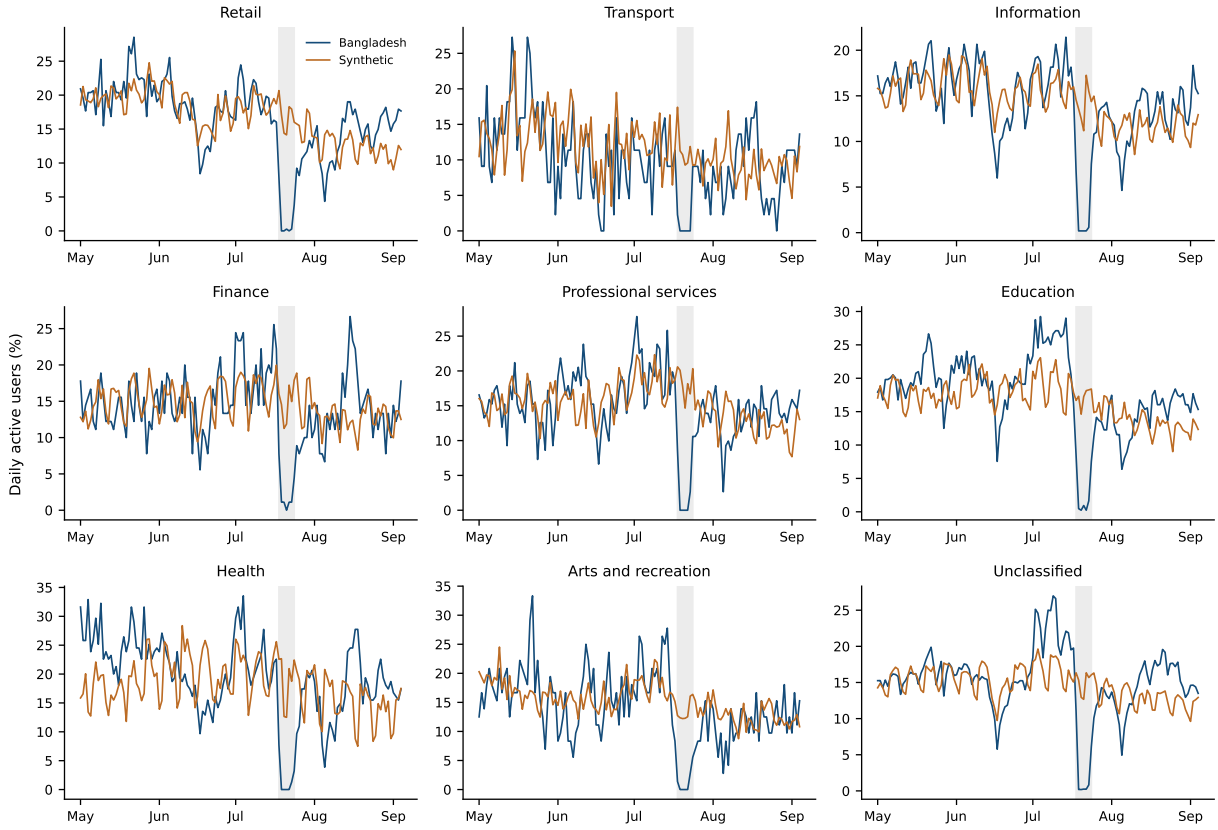
The normalized fit divides pre-period RMSPE by mean pre-period activity. Holdout weights are fitted through June 30 and evaluated July 1–17, conditional on July 17 cohort definitions. Large errors limit counterfactual credibility. Ratio rank compares the shutdown/pre-period RMSPE ratio with sector-specific donor placebos.

Table A10: Sector-cohort ATT sensitivity to classification rules

Domain	0.70	0.80	0.90	Name fallback
Accommodation and food	-14.56	–	–	-15.71
Arts and recreation	-11.61	-12.33	–	-10.90
Education	-16.30	-15.04	-15.83	-14.15
Finance	-10.97	-12.28	-12.10	-12.59
Health	-16.10	-16.84	-16.65	-15.89
Information	-13.07	-12.74	-12.50	-12.70
Professional services	-14.34	-16.13	–	-16.13
Retail	-16.47	-15.69	-15.16	-14.31
Transport	-10.95	-11.31	–	-11.90
Unclassified	-12.76	-12.77	-12.92	-12.58

Effects in percentage points. The first three columns require a historical README. The final column admits classifications from names alone at score 0.80. Each rule changes the fixed cohort and may change donor support. A dash means insufficient support, not a zero effect; columns are therefore not estimates for an identical population.

Figure A2: Observed and synthetic paths for all supported repository-domain cohorts



Notes: Fixed pre-treatment user cohorts; historical README and score at least 0.80. Grey marks July 18–23, the primary estimand. Paths after July 23 are descriptive and use the same pre-fitted weights. Panel scales vary. Full domain-country weights, donor placebos and leave-one-out estimates are retained in the replication outputs.

A.3 Difference-in-differences and event study

Table A11: Aggregate event study: all weekly estimates

Week	Dates (2024)	Days	$\hat{\theta}_k$ (pp)	Placebo-based interval	Placebo range	Rank p
−11	May 1–May 8	8	−0.86	[−4.37, 2.65]	[−1.68, 4.62]	0.6000
−10	May 9–May 15	7	−1.83	[−4.50, 0.84]	[−2.23, 2.49]	0.2667
−9	May 16–May 22	7	0.58	[−1.21, 2.36]	[−1.06, 2.46]	0.4667
−8	May 23–May 29	7	−0.51	[−2.47, 1.45]	[−2.36, 1.92]	0.4667
−7	May 30–Jun 5	7	−0.05	[−2.08, 1.97]	[−1.72, 2.02]	1.0000
−6	Jun 6–Jun 12	7	−1.23	[−3.28, 0.81]	[−0.99, 2.54]	0.2000
−5	Jun 13–Jun 19	7	−4.89	[−9.31, −0.47]	[−3.64, 2.34]	0.0667
−4	Jun 20–Jun 26	7	−2.71	[−5.13, −0.29]	[−3.06, 1.62]	0.1333
−3	Jun 27–Jul 3	7	1.67	[−0.46, 3.80]	[−1.31, 1.45]	0.0667
−2	Jul 4–Jul 10	7	6.10	[3.76, 8.43]	[−1.74, 1.62]	0.0667
−1	Jul 11–Jul 17	7	3.86	[2.15, 5.58]	[−1.63, 0.98]	0.0667
0	Jul 18–Jul 23	6	−13.09	[−15.39, −10.79]	[−1.72, 1.60]	0.0667
+1	Jul 24–Jul 30	7	−2.66	[−5.35, 0.02]	[−1.95, 2.23]	0.0667
+2	Jul 31–Aug 6	7	−2.76	[−4.87, −0.65]	[−1.22, 2.56]	0.0667
+3	Aug 7–Aug 13	7	0.52	[−1.99, 3.02]	[−1.87, 2.73]	0.9333
+4	Aug 14–Aug 20	7	4.31	[1.99, 6.64]	[−1.82, 2.77]	0.0667
+5	Aug 21–Aug 27	7	4.84	[2.40, 7.28]	[−1.48, 2.71]	0.0667
+6	Aug 28–Sep 3	7	2.53	[0.01, 5.05]	[−2.32, 2.31]	0.0667

Notes: $\hat{\theta}_k$ from Equation (5), relative to the May 1–July 17 mean. Week −11 includes May 1; September 4 (a partial week) is omitted. *Placebo range*: minimum and maximum of the 14 in-space placebo estimates. *Placebo-based interval*: $\hat{\theta}_k \pm 1.96$ times the placebo standard deviation; a donor-placebo reference interval, not a conventional confidence interval. Other definitions as in Table 5. Source: `results/did/did_eventstudy_weekly.csv`.

Table A12: International difference-in-differences by outcome

	All days		Holidays excluded	
Outcome	ATT	Relative	ATT	Relative
Active-user rate (primary)	−13.088	−86.6%	−13.379	−86.9%
Productive active rate	−11.748	−86.7%	−12.007	−86.9%
Shipping active rate	−0.565	−89.0%	−0.590	−89.4%
Collaboration active rate	−0.266	−86.3%	−0.269	−86.4%
Collaboration events / 1,000	−10.575	−91.8%	−10.685	−91.9%
Shipping output / 1,000	−10.380	−88.5%	−10.827	−88.9%

Notes: Aggregate DiD as in Table 5, Panel A, for each outcome; rates in percentage points, counts per 1,000 eligible users. No placebo inference is stored for these specifications. Source: `results/final/table_DiD_all_outcomes_eligible.csv`.

A.4 Regression discontinuity in time

A regression discontinuity in time provides a third temporal comparison. The archived preferred specification uses a cutoff at July 17, 2024, a bandwidth of $h = 7$ days (July 10–24, fifteen daily observations), and a local linear fit on each side. With $\tilde{t} = t - \text{July 17}$ and $D_t = \mathbf{1}\{\tilde{t} \geq 0\}$, so that July 17 itself lies on the post-cutoff side, I estimate

$$Y_t = \alpha + \tau D_t + \gamma_1 \tilde{t} + \gamma_2 D_t \tilde{t} + u_t, \qquad |\tilde{t}| \leq h, \tag{7}$$

where τ is the discontinuity and Y_t is the daily share of users with at least one productive GitHub event in a balanced user-day panel of 10,329 Bangladesh-associated and 3,025 Philippines-associated accounts. This eligible sample differs slightly from the 10,241 Bangladesh accounts of the country-day panel used for the synthetic control. I estimate Equation (7) for Bangladesh, for the Bangladesh-minus-Philippines gap as a controlled specification, and for the Philippines alone as a geographic placebo. Inference uses heteroskedasticity-and-autocorrelation-consistent (HAC) Newey–West standard errors with two lags, following the cautions in Hausman and Rapson (2018) regarding serial correlation in time-series discontinuity designs. Because time itself is the running variable and other contemporaneous events can occur at the same date, the RDiT is interpreted as a robustness check rather than a standalone source of causal identification.

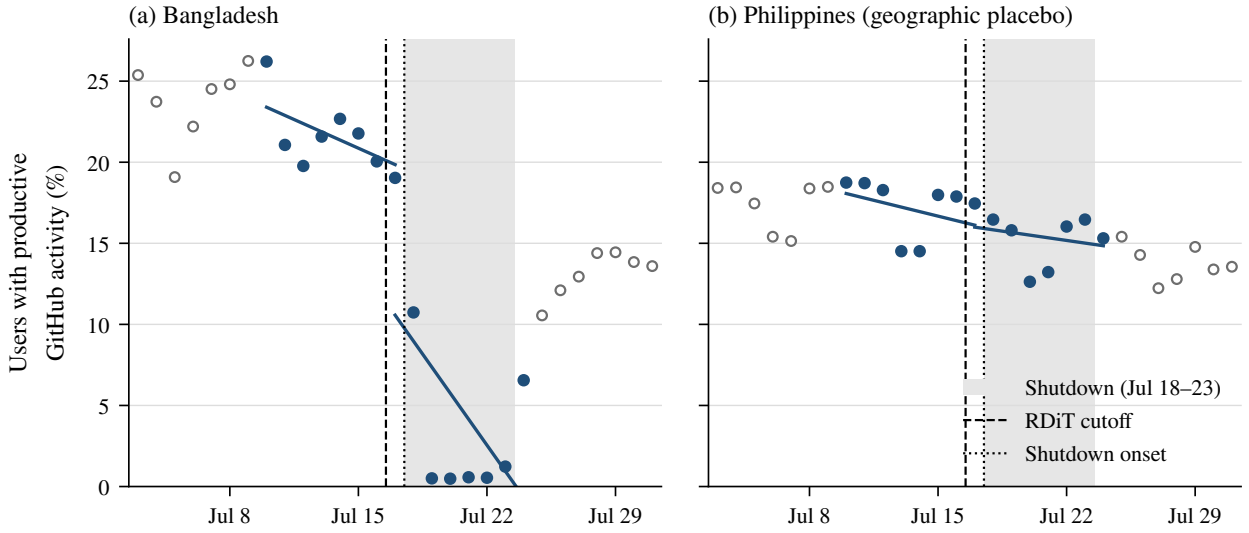
The July 17 cutoff reproduces the original preferred specification; it is not chosen because it yields a preferred estimate. No daily cutoff matches the event exactly: provider-level mobile outages began late on July 17 UTC, whereas the nationwide near-total cutoff occurred around 15:00 UTC on July 18 (Belson, 2024a). I therefore report a complete grid of July 17, 18, and 19 cutoffs crossed with 4-, 5-, 7-, 10-, and 14-day bandwidths in Table A14, together with the separate archived 16-

specification grid in Table A15. An hourly design would require a separate specification and is not pursued here.

A.4.1 Results

Figure A3 shows the daily series used in the RDiT. In Bangladesh, the share of users with productive activity is 19.03 percent on July 17, falls to 10.74 percent on July 18, and lies between 0.48 and 1.23 percent from July 19 through July 23 before recovering gradually from July 24. The Philippines, used as a geographic placebo, shows no comparable break.

Figure A3: Regression discontinuity in time: daily share of users with productive activity around the July 17 cutoff (visual companion to Table A13)



Notes: Daily percentage of users with productive GitHub activity in the balanced user-day panels used for the RDiT (10,329 Bangladesh-associated and 3,025 Philippines-associated accounts). Filled markers are the fifteen days in the 7-day bandwidth of Equation (7). Lines are the fits of the archived preferred specification and correspond exactly to Table A13. Dashed line: cutoff, with July 17 on the post-cutoff side; dotted line: shutdown boundary between July 17 and 18; shaded: July 18–23 shutdown. The RDiT is a temporal robustness check, not the primary design. Source: Author's calculations using GH Archive.

The RDiT produces a smaller point estimate but the same qualitative conclusion (Table A13). The archived preferred local linear specification estimates a discontinuity of −9.28 percentage points in Bangladesh's active-user rate; a controlled Bangladesh-minus-Philippines version yields −9.16 percentage points. HAC p-values are approximately 0.065 in both cases, while the Philippines placebo is near zero. The smaller magnitude than in the synthetic control and DiD is consistent with the design visible in Figure A3: the cutoff day, July 17, precedes the shutdown but lies on the post-cutoff side, and the linear post-cutoff fit spans both the collapse and the beginning of the recovery on July 24. Because Hausman and Rapson (2018) urge caution in interpreting RDiT estimates, I emphasize the direction and timing of the break rather than treating the RDiT p-value

as the decisive inferential statistic.

Table A13: Regression discontinuity in time: preferred specification

Specification	Outcome	Effect (pp)	HAC SE	95% CI	p	Days
Main local linear	Bangladesh active rate	−9.284	5.034	[−19.151, 0.584]	0.065	15
Controlled local linear	Bangladesh – Philippines active rate	−9.160	4.980	[−18.921, 0.601]	0.066	15
Geographic placebo	Philippines active rate	−0.124	1.492	[−3.048, 2.801]	0.934	15

Notes: Local linear RDiT at the July 17, 2024 cutoff, bandwidth 7 days, polynomial degree 1, no donut, HAC lags 2. Effects are reported in percentage points. RDiT is a temporal robustness check, not the primary causal design. Source: `results/rdit/main_rdit_summary.csv`.

The RDiT grid spans −4.45 to −19.52 percentage points for Bangladesh, illustrating substantial specification sensitivity (Table A14). The archived 16-row grid is also retained, but its input and cutoff provenance is unresolved. The RDiT sample includes 174 accounts absent from the main synthetic-control cohort, while 86 accounts in that cohort are absent from the RDiT sample; the difference is not simply 88 additional accounts. These facts limit a direct comparison of the two levels and estimates.

Table A14: RDiT: complete cutoff and bandwidth grid

Cutoff	h	Bangladesh	BD minus PH	Philippines
07-17	4	-4.45 (3.45)	-2.40 (3.43)	-2.05 (0.53)
07-17	5	-7.30 (4.36)	-6.56 (4.87)	-0.74 (1.54)
07-17	7	-9.28 (5.03)	-9.16 (4.98)	-0.12 (1.49)
07-17	10	-13.70 (6.05)	-13.07 (5.80)	-0.62 (1.35)
07-17	14	-16.80 (5.64)	-16.04 (5.35)	-0.76 (1.16)
07-18	4	-11.09 (2.72)	-7.47 (2.14)	-3.62 (2.01)
07-18	5	-12.98 (3.12)	-8.75 (1.94)	-4.24 (1.86)
07-18	7	-17.49 (4.48)	-15.58 (4.29)	-1.91 (1.34)
07-18	10	-18.06 (3.94)	-17.35 (4.18)	-0.71 (1.31)
07-18	14	-19.52 (3.47)	-17.82 (3.50)	-1.70 (1.07)
07-19	4	-9.00 (2.49)	-6.69 (2.12)	-2.31 (1.44)
07-19	5	-11.54 (2.91)	-7.73 (1.99)	-3.81 (1.84)
07-19	7	-16.59 (4.24)	-13.80 (4.06)	-2.79 (1.19)
07-19	10	-16.76 (3.15)	-15.52 (3.59)	-1.23 (1.17)
07-19	14	-17.60 (2.99)	-15.50 (3.16)	-2.10 (1.00)

Notes: All effects and HAC(2) standard errors in percentage points. Local linear fits with separate slopes, cutoff day on the right, inclusive symmetric bandwidth, $n/(n-4)$ variance correction. All 45 fits are reported; none replaces the archived July 17, $h = 7$ result. Source: new round-2 diagnostic grid.

Table A15: RDiT: preserved archived 16-specification grid

Pre bandwidth	Polynomial	Donut	Effect (pp)	SE (pp)	*n*
7	lin	False	-14.0	2.2	14
7	lin	True	-20.4	1.5	12
7	quad pre	False	-16.1	2.8	14
7	quad pre	True	-21.2	2.5	12
10	lin	False	-13.1	1.9	17
10	lin	True	-18.5	1.0	15
10	quad pre	False	-15.2	2.7	17
10	quad pre	True	-20.5	1.7	15
14	lin	False	-14.8	2.4	21
14	lin	True	-20.6	1.5	19
14	quad pre	False	-13.2	2.3	21
14	quad pre	True	-18.1	1.4	19
21	lin	False	-17.7	3.2	28
21	lin	True	-23.7	2.5	26
21	quad pre	False	-12.4	2.1	28
21	quad pre	True	-17.5	1.2	26

Notes: Transcribed without re-estimation from data/processed/table3_rdit_sensitivity.csv. The historical cutoff, outcome, sample and variance procedure are not documented. The row labelled linear, 7 days, no donut is -14.0 pp, compared with -9.28 pp in the reproduced preferred specification; this is an unresolved provenance conflict, not evidence that one specification should replace the other.

A.5 Mechanism

Table A16: Repository-domain composition of residual-activity cohorts

Pre-treatment domain	All eligible	Core publishers	Onset authors	Delayed authors
Accommodation and food	192	0	4	3
Administrative support	36	0	0	0
Agriculture	25	0	1	1
Arts and recreation	72	0	4	2
Construction	1	0	0	0
Education	424	4	18	13
Finance	90	1	1	1
Health	155	0	5	3
Information	518	1	11	10
Manufacturing	8	0	0	0
Other services	18	0	0	0
Professional services	151	0	5	2
Public administration	4	0	0	0
Real estate	49	0	1	1
Retail	368	1	6	5
Transport	44	0	0	0
Unclassified	8,078	32	192	149
Utilities	8	1	2	1

Counts, not effects. Historical README classification at score 0.80; all unclassified accounts remain visible. Cohorts overlap and have the definitions in the mechanism section. Small cells and substantial unclassified coverage do not establish occupational specialization.

Table A17: Mechanism placebo comparison

Metric	Real shutdown	Mean placebo	Maximum placebo
Released in next 6 days	25.38%	0.56%	0.77%
Delay >24h	29.71%	1.10%	1.31%
Delay >48h	25.38%	0.72%	0.90%
Delay >72h	22.75%	0.46%	0.56%
Delay >120h	13.88%	0.32%	0.46%

Notes: Placebo windows are six six-day authoring windows between July 1 and July 11, with six-day subsequent release windows that end before treatment. Windows overlap, so the comparison is descriptive rather than an exact independent randomization test. Commit metadata were queried for all first-push commits pushed on or after July 18 but for a daily sample of those pushed July 1–17 (7–17 percent of commits per push date). Placebo rates therefore weight each commit by the inverse of its push date's query rate (number of first-push commits divided by number queried); commits authored July 18–23 can only be pushed on or after July 18 and receive weight one, so all rows estimate the same population share. Without weights, placebo rates would over-represent late pushes. Mean placebo N is 4,144 observed commits per window. Sources: `results/final/table_mechanism_placebos.csv`, reproduced by `analysis/mechanism/01_placebo_weighting_audit.py`.

Table A18: Metadata coverage by push period

Period	Target N	Commit found	Author timestamp	Author login	Author = pusher
Pre	17,000	82.34%	82.34%	77.95%	71.49%
Shutdown	3,197	84.24%	84.24%	79.17%	69.69%
Post	65,868	81.81%	81.81%	77.81%	71.05%

Notes: Similar resolution rates across periods reduce concern that the mechanism result is generated by treatment-specific metadata missingness.

Table A19: Push-listing completeness by period

Period	Pushes	Mean push size	Complete pushes	Nominal commits	Commits in complete pushes
Pre	121,960	1.29	99.78%	157,834	90.27%
Shutdown	2,738	1.70	99.53%	4,667	73.11%
Post	58,110	1.69	99.66%	98,022	71.45%

Notes: A push is complete when its historical payload lists every commit. Large pushes account for a disproportionate share of commits and are more likely to be incomplete. Source: `results/mechanism/robust_push_size_period.csv`.

Table A20: Mechanism estimates by push size

Sample	N	Backlog N	Backlog share	>24h	>48h	>72h
All complete pushes	2,932	829	28.27%	29.71%	25.38%	22.75%
Push size ≤ 10	2,503	526	21.01%	21.81%	18.98%	17.14%
Push size ≤ 5	2,217	320	14.43%	15.07%	13.62%	12.45%
Push size ≤ 2	1,947	151	7.76%	7.96%	7.29%	6.93%
Single-commit pushes	1,707	57	3.34%	3.28%	2.93%	2.75%

Notes: Strict `authoredDate` sample of commits authored July 18–23. The backlog share is not invariant to push size.

Table A21: Concentration of the backlog across users and repositories

Statistic	Value
Shutdown-authored commits	2,932
Backlog commits	829 (28.27%)
Users / with backlog	835 / 222 (26.59%)
Repositories / with backlog	977 / 246 (25.18%)
Largest user's share of backlog	2.77%
Top 5 users' share of backlog	12.06%
Top 10 users' share of backlog	21.47%
Backlog share after dropping top-k backlog users	
$k = 1$ ($N = 2,909$)	27.71%
$k = 3$ ($N = 2,868$)	26.67%
$k = 5$ ($N = 2,831$)	25.75%
$k = 10$ ($N = 2,739$)	23.77%
$k = 20$ ($N = 2,577$)	20.37%

Notes: Strict `authoredDate` sample. Users are ranked by their number of backlog commits, breaking ties by lowercase login in ascending order. The deterministic audit reproduces the archived top-10 exclusion ($N = 2,739$, 23.77 percent); the referee reports $N = 2,740$ and 23.76 percent with an unspecified tie order. Both figures are recorded as a tie-handling discrepancy; the archived result is preserved.

A.6 Within-Bangladesh sectoral recovery

The designs above estimate the effect of the shutdown during July 18–23. To describe how activity evolved after restoration, I use the fixed sample of 20,000 accounts described in Section 4.6. For

user i, sector s, and day t, let $A_{ist} = 1$ if the user records at least one PushEvent on day t in a repository classified to sector s; activity in several repositories of the same sector on the same day counts once. Days are grouped into event periods k: the final eight complete pre-shutdown weeks, $k = -8, \ldots, -1$ (May 23–July 17, 2024); the shutdown, $k = 0$ (July 18–23); and consecutive seven-day bins after it, starting with $k = +1$ (July 24–30). Let $\bar{A}_{isk}$ be user i's share of active days in period k, and $\bar{A}_{is}^{\text{pre}}$ the corresponding share over the 56 baseline days. I report

$$\beta_{sk} = \frac{100}{N} \sum_{i=1}^{N} \left(\bar{A}_{isk} - \bar{A}_{is}^{\text{pre}} \right), \qquad R_{sk} = 100 \times \frac{\sum_i \bar{A}_{isk}}{\sum_i \bar{A}_{is}^{\text{pre}}}, \tag{8}$$

where $N = 20{,}000$ and users without activity in sector s contribute zeros. The coefficient β_{sk} is the change in the daily probability of sector-s push activity relative to the baseline, in percentage points; the recovery index R_{sk} normalizes the baseline to 100. Standard errors for β_{sk} are 100 times the standard deviation of the user-level differences divided by $\sqrt{N}$, and 95 percent intervals use the normal approximation. As a summary of recovery timing, I report the first post-shutdown week that begins two consecutive weeks with $R_{sk} \geq 80$.

Because the activity measure counts pushes to any classified repository, including repositories first observed after the shutdown, it captures recovery through new projects and through switching across repositories. As a complementary specification, I also follow only the repositories pushed to during the baseline weeks, whose activity cannot be replaced by new projects.

This analysis compares each sector with its own pre-shutdown baseline and has no untreated comparison group. It therefore describes the heterogeneity and timing of recovery within Bangladesh; it does not estimate sector-specific causal effects. Post-shutdown deviations from the baseline may also reflect trends, seasonality, or other Bangladesh-specific developments. The confidence intervals capture sampling variation across users but not common shocks that affect all users in a given week. The pre-shutdown coefficients ($k < 0$) indicate how much weekly activity fluctuates around the baseline in the absence of a shutdown and serve as a benchmark for interpreting post-shutdown deviations.

A.7 Sectoral recovery after restoration

The synthetic-control estimates describe the collapse during the shutdown window. They do not show how activity evolved once connectivity returned, or whether recovery was common across the kinds of projects on which Bangladeshi developers work. This subsection addresses both questions using the fixed sample of 20,000 Bangladesh-associated accounts and the repository-sector classification of Section 4.6, estimated as described in Section A.6. The analysis compares each sector with its own pre-shutdown baseline. It documents heterogeneity in recovery dynam-

ics within Bangladesh; it does not estimate sector-specific causal effects relative to an untreated international control, and the synthetic-control estimate in Section 6.1 remains the paper's headline result. Throughout, a sector refers to the economic sector associated with the repositories in which observable GitHub activity occurs, not to developers' employers or industries of employment.

Table A22: Recovery of observable GitHub push activity by repository sector

		Pre	Pre rate	Recovery index (pre-period mean = 100)					First two-
Sector	NAICS	users	(%)	Shutdown	+1	+2	+4	+8	week run ≥80
Finance	52	382	0.078	30.9	92.2	67.6	89.5	91.3	3
Information	51	1,528	0.442	11.7	75.9	72.2	106.0	94.5	3
Education	61	1,234	0.385	11.7	59.1	53.4	75.4	81.9	6
Health	62	514	0.132	6.3	40.7	51.5	77.5	89.4	6
Retail	44–45	1,012	0.295	6.2	52.0	52.3	153.7	79.6	9
Real Estate	53	213	0.063	4.0	33.0	25.1	44.4	85.5	5
Accommodation/Food	72	480	0.144	3.5	44.3	42.8	54.2	62.6	–

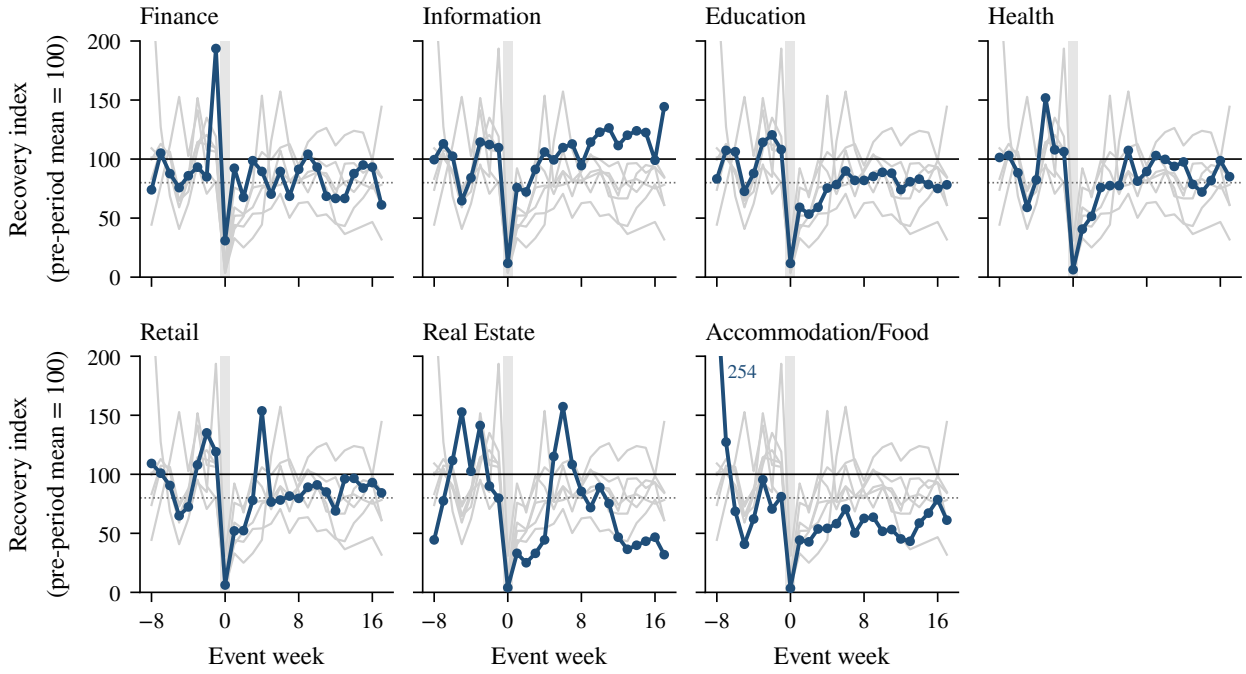
Notes: Fixed sample of 20,000 Bangladesh-associated accounts. Repository sectors from the historical-README classifier with confidence ≥ 0.80. The outcome is the share of sampled users with at least one PushEvent to a repository of the sector on a given day. *Pre users*: sampled users with such activity in the baseline period. *Pre rate*: mean daily outcome over the baseline period (May 23–July 17, 2024, the final eight complete pre-shutdown weeks), in percent of all 20,000 users. Shutdown: July 18–23; week +1: July 24–30; subsequent 7-day bins. *First two-week run ≥80*: first post-shutdown week that begins two consecutive weeks with an index of at least 80 (observed through week +17, November 13–19); it does not imply that the index stays above 80 thereafter. – indicates no such week. Within-Bangladesh comparison with the sector's own pre-period; not a causal effect relative to an untreated control. Sources: `results/sector_recovery/sectorwide/sectorwide_recovery_summary.csv`, verified against `results/sector_recovery/final/table_sector_recovery_main.csv`.

All seven sectors show a sharp fall in observable push activity during July 18–23 (Table A22 and Figure A4). The shutdown-period recovery index ranges from 3.5 in Accommodation/Food to 11.7 in Information and Education and 30.9 in Finance. In every sector, the 95 percent confidence interval for the change relative to the baseline excludes zero (Appendix Figure A5).

Restoration of connectivity did not imply an immediate or uniform recovery. In the first week after the shutdown (July 24–30), the recovery index ranges from 33.0 in Real Estate to 92.2 in Finance. Mobile connectivity returned only during this week, later than fixed broadband (Belson, 2024a), so part of the week +1 shortfall may reflect incomplete restoration. Recovery was also not monotonic: in Finance, Information, Education, Real Estate, and Accommodation/Food, the index is lower in week +2 (July 31–August 6) than in week +1. This week includes the renewed mobile shutdown on August 4 and the nationwide disruption on August 5 (Belson, 2024c), alongside unrest and initially declared holidays. These overlapping events prevent a separate attribution of the second-week decline. The stored holiday labels extend through August 7, although offices were ordered to reopen on August 6 (Bangladesh Sangbad Sangstha, 2024); I keep the stored labels and document the discrepancy in the replication materials.

Finance and Information recovered fastest. Finance returned to an index of 92.2 in week +1,

Figure A4: Recovery of observable GitHub push activity by repository sector



Notes: Each panel highlights one sector's recovery index R_{sk} (Equation (8)), 100 times its mean daily probability of push activity in event week k relative to May 23–July 17, 2024; other sectors in grey. Shaded: July 18–23 shutdown; week +1 is July 24–30, and the window ends at week +17 (November 13–19). Lines mark 100 and 80; the one value above 200 is labeled. Fixed sample of 20,000 Bangladesh-associated accounts; sectors from a README classifier with confidence ≥ 0.80 (Quispe and Xu, 2026) describe the repositories in which activity occurs, not developers' employers. Within-Bangladesh descriptive comparison, not a causal effect. Source: Author's calculations using GH Archive.

and both sectors begin a run of two consecutive weeks at or above 80 in week +3. Information exceeds its baseline by week +4 (106.0). In week +8, Finance is 91.3 and Information is 94.5. The Finance series is nonetheless noisy: its baseline includes a spike in week −1 (index 193.6), and its post-shutdown index falls back below 80 in several later weeks.

Education and Health recovered more gradually. Their week +1 indices are 59.1 and 40.7, they reach 75.4 and 77.5 by week +4, and both begin a run of two consecutive weeks at or above 80 only in week +6. By week +8, they stand at 81.9 and 89.4.

Accommodation/Food remained persistently depressed. Its index is 44.3 in week +1, 54.2 in week +4, and 62.6 in week +8, and it never records two consecutive weeks at or above 80 through the end of the observation window (week +17, November 13–19). This result should be read alongside the sector's unstable baseline: the index is 254.1 and 127.3 in the first two baseline weeks and between 40.8 and 95.5 in the remaining six. Part of the post-shutdown gap may therefore reflect a decline that began before the shutdown rather than a failure to recover.

Real Estate recovered slowly at first and later approached its baseline. Its index remains between 25.1 and 44.4 through week +4, rises above 100 in weeks +5, +6 and +7 (115.1, 157.3 and 108.3),

and is 85.5 in week +8. This later recovery is not durable within the observed window: from week +12 onward, the index lies between 31.9 and 46.7 (Figure A4). Real Estate is also the smallest of the seven sectors, with 213 sampled users active in the baseline period, so its weekly estimates are imprecise.

Retail shows a pronounced temporary rebound. Its index is about 52 in weeks +1 and +2, rises to 153.7 in week +4 (August 14–20), and falls back to 76.5 in week +5. Its first run of two consecutive weeks at or above 80 begins only in week +9. The week +4 value is therefore a transient overshoot, possibly reflecting a burst of catch-up activity, rather than evidence of completed recovery; the data do not identify what generated it. The timing of Retail's first two-week run is also sensitive to the classifier threshold (Section 8.3).

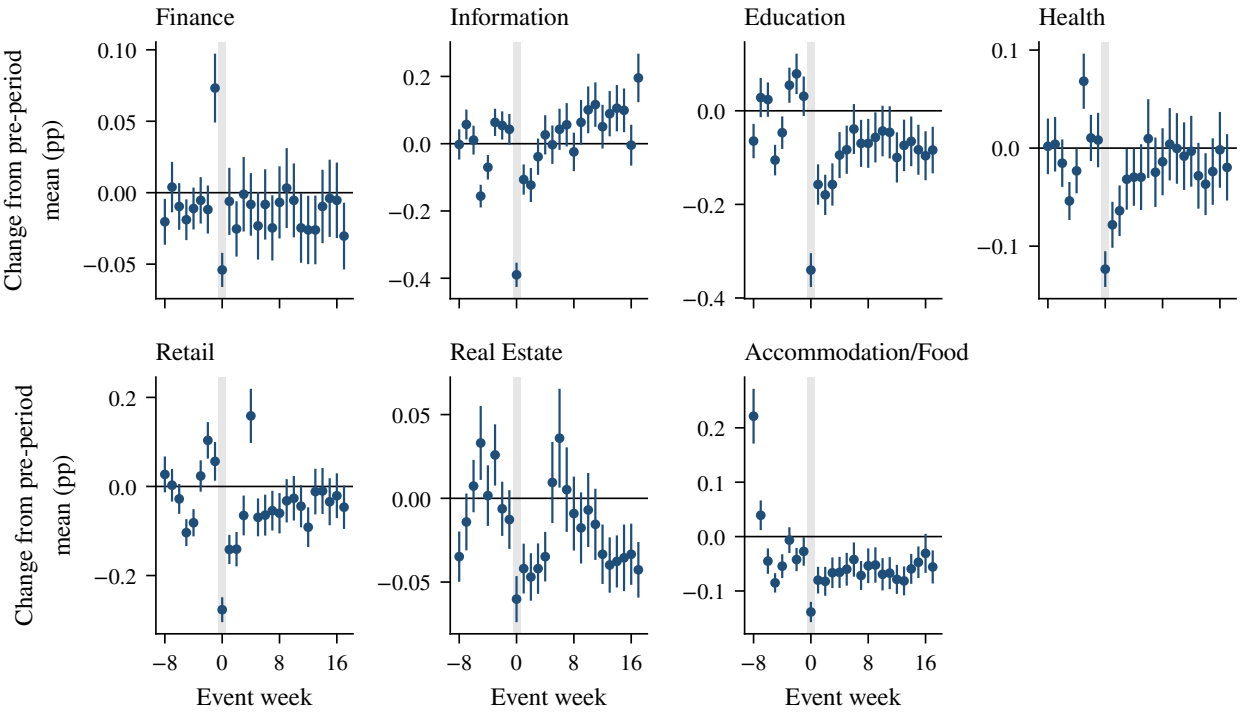
Index values above 100 later in the window should not be read as gains caused by the shutdown. Information, for example, has an index between 111.5 and 126.3 in weeks +10 to +15 and 144.3 in week +17. Without an untreated comparison group, such movements cannot be separated from trend growth, seasonality, or changes in the sampled users' activity.

The main specification counts pushes to any classified repository, including repositories first observed after the shutdown. A complementary specification follows only repositories pushed to during the eight baseline weeks (Table A24). Activity in these pre-existing repositories recovers far less: in week +8, their indices range from 10.1 (Accommodation/Food) to 31.4 (Education), compared with 62.6 to 94.5 in the sector-wide specification. Because repositories selected on recent activity would become less active over time even without a shutdown, this gap does not measure a shutdown effect. It is nevertheless consistent with recovery occurring partly through new repositories and switching across projects rather than solely through a return to pre-existing ones.

Taken together, the sectoral evidence indicates that the collapse in observable activity documented in Section 6.1 was followed by a recovery that was gradual and uneven across the economic sectors associated with developers' repositories. These are before–after comparisons within Bangladesh. They share the identification limitations discussed in Section 8.4 and, in addition, cannot separate post-shutdown recovery from other Bangladesh-specific developments between August and November 2024.

A.7.1 Additional sector-recovery diagnostics

Figure A5: Sectoral event study: change in daily push-activity probability relative to the pre-shutdown baseline



Notes: Points are β_{sk} from Equation (8): the mean change, across the fixed sample of 20,000 Bangladesh-associated accounts, in the daily probability of a PushEvent in a sector-s repository during event week k, relative to each user's mean over May 23–July 17, 2024, in percentage points. Vertical lines are 95 percent confidence intervals from the cross-user standard deviation of these differences; they capture sampling variation across users but not common weekly shocks. Shaded: July 18–23 shutdown; the window ends at week +17 (November 13–19). Vertical scales differ across panels. Within-Bangladesh descriptive comparison, not a causal effect relative to an untreated control. Source: Author's calculations using GH Archive.

Table A23: Sectoral recovery indices by classifier-confidence threshold

Sector	Threshold	Pre users	Week +1	Week +4	Week +8	First two-week run ≥80
Finance	0.70	437	91.6	92.4	89.2	3
	0.80	382	92.2	89.5	91.3	3
	0.90	275	101.7	82.5	101.7	3
Information	0.70	2,123	73.0	116.6	101.3	3
	0.80	1,528	75.9	106.0	94.5	3
	0.90	968	79.5	97.6	91.5	3
Education	0.70	1,345	61.6	75.4	79.4	5
	0.80	1,234	59.1	75.4	81.9	6
	0.90	1,029	57.8	73.3	77.0	6
Health	0.70	542	40.8	76.4	88.3	6
	0.80	514	40.7	77.5	89.4	6
	0.90	455	41.1	75.5	87.6	8
Retail	0.70	1,090	52.2	159.1	82.5	4
	0.80	1,012	52.0	153.7	79.6	9
	0.90	829	50.1	157.4	80.9	7
Real Estate	0.70	228	33.1	45.5	90.0	5
	0.80	213	33.0	44.4	85.5	5
	0.90	153	44.9	44.9	75.5	5
Accommodation/Food	0.70	514	46.0	59.3	64.9	–
	0.80	480	44.3	54.2	62.6	–
	0.90	394	39.2	45.1	60.3	–
Excluded from the main figures and table						
Professional/Scientific	0.70	1,241	57.9	66.9	91.2	7
	0.80	577	64.9	64.9	91.0	7
	0.90	125	94.4	65.6	78.7	1
Admin. Support	0.70	268	30.8	83.0	77.0	9
	0.80	196	32.2	72.9	84.7	10
	0.90	48	5.2	61.9	56.8	–

Notes: Each row repeats the main specification (Table A22) keeping only repositories whose classifier confidence is at least the stated threshold; 0.80 is the main specification. Professional/Scientific and Administrative Support are excluded from the main results because their support falls sharply at the 0.90 threshold. Source: `results/sector_recovery/final/sector_threshold_robustness_full.csv`.

Table A24: Recovery of pre-existing repositories versus sector-wide user activity

		Fixed pre-period repositories			Sector-wide (main)		
Sector	Repos	+1	+4	+8	+1	+4	+8
Finance	417	60.0	28.3	23.0	92.2	89.5	91.3
Information	2,040	46.8	37.4	27.3	75.9	106.0	94.5
Education	1,523	40.3	38.2	31.4	59.1	75.4	81.9
Health	651	25.1	28.6	23.1	40.7	77.5	89.4
Retail	1,322	28.7	30.9	19.8	52.0	153.7	79.6
Real Estate	254	16.1	25.7	30.0	33.0	44.4	85.5
Accommodation/Food	581	29.4	16.1	10.1	44.3	54.2	62.6

Notes: Recovery indices (pre-period mean = 100). *Fixed pre-period repositories*: classified repositories (confidence ≥ 0.80) with at least one PushEvent by a sampled user during the final eight pre-shutdown weeks; the outcome is each repository's daily PushEvent active-day rate, and new repositories never enter. *Sector-wide*: main specification of Table A22, which includes activity on repositories first observed after the shutdown. Because a cohort selected on recent activity would decline over time even without a shutdown, the fixed-repository indices are not a counterfactual for the sector-wide indices. Sources: `results/sector_recovery/pre8/sector_recovery_summary_pre8.csv` and `results/sector_recovery/sectorwide/sectorwide_recovery_summary.csv`.

A.8 Connectivity chronology and threats to the comparison

The connectivity record is useful both for defining exposure and for assessing the comparison series. Network traffic is not itself a measure of GitHub production, and a provider-level disruption need not affect every account in a country. The following review records documented events within the May 1–September 4 observation period. Absence from these non-exhaustive reports is not evidence of uninterrupted connectivity.

Table A25: Documented connectivity events relevant to the analysis

Country	Date (2024)	Documented event	Implication
Bangladesh	July 18	Near-zero national traffic around 15:00 UTC (Belson, 2024a)	Daily July 18 includes pre-cut hours.
Bangladesh	July 23	Broadband recovery at 13:00 UTC in the August report, 12:00 UTC in Q3 (Belson, 2024a,c)	Both endpoints retained in author-cohort counts.
Bangladesh	July 28	Major mobile networks recover around 09:00 UTC (Belson, 2024a)	Restoration differs across access networks.
Bangladesh	August 4–5	Renewed mobile outage, followed by national interruption on August 5 (Belson, 2024c)	Later dip has additional network exposure.
Kenya	May 2; May 12	Power-related disruption; submarine-cable failures (Belson, 2024b)	Pre-period donor contamination.
Kenya	June 25	Large traffic disruption during protests; cause disputed (Belson, 2024b)	Connectivity and unrest are not separated.
Kenya	August 30–31	Power-outage disruption (Belson, 2024c)	Later synthetic path may be depressed.
Pakistan	July 31	About two hours of disruption, with roughly 45% national traffic decline (Belson, 2024a)	Outside primary window; affects recovery comparison.
Nepal	May 2; June 3	Provider outages associated with upstream payments and routing (Belson, 2024b)	Exposure differs by provider.
Malaysia	May 21	Celcom disruption; a separate apparent dip was a Cloudflare processing issue (Belson, 2024b)	Telemetry artefacts must not be coded as outages.

Times are UTC. This source audit does not change the donor pool or refit weights. Pakistan's regional July restrictions are discussed in the main donor diagnostics. The August 6 reopening reported by BSS conflicts with August 6–7 holiday labels in the input panel; the original labels and separate sensitivity remain documented. Kenya and Pakistan have positive national weights, so their post-window disruptions are especially relevant to apparent catch-up. No causal GitHub effect is assigned to a listed event solely because it appears in the chronology.

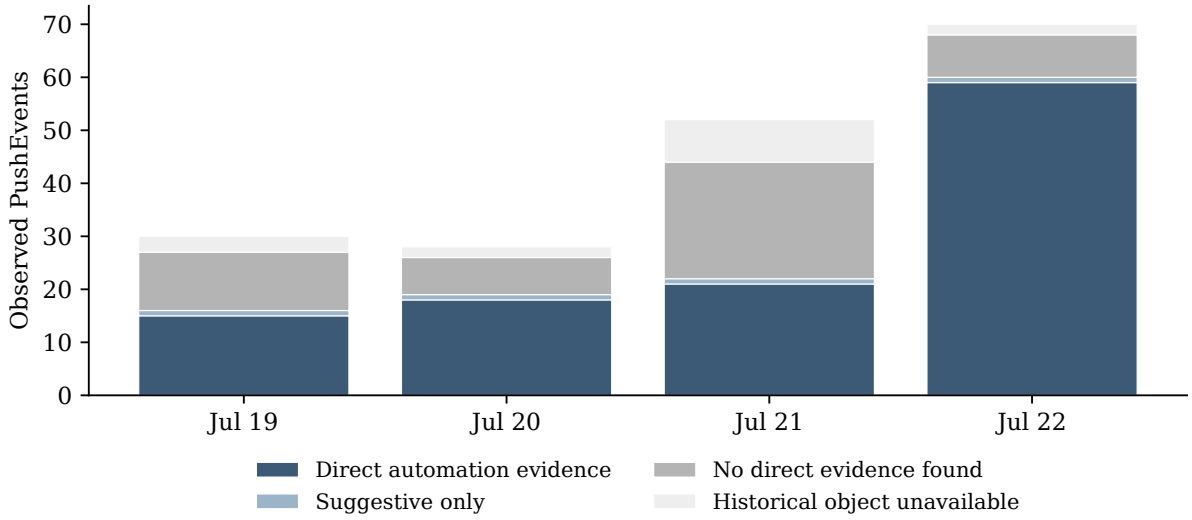
A.9 Detailed residual-publication audit

Table A26: Pre-period activity and the automation component of residual publication

Group	Users	Active days (%)	Pushes	Fri–Sat (%)
All eligible	10,241	17.25	21.0	25.96
Core publishers	40	36.76	47.5	24.08
Core with automated heads	18	52.35	79.5	23.11
Other core publishers	22	24.01	20.5	24.88
Other eligible users	10,201	17.18	21.0	25.97
Onset authors	250	32.62	51.5	24.59

May 1–July 17. Active days is the mean percentage of the 78 days; pushes is the median count. Fri–Sat is the mean account-level share of active days falling on Bangladesh's Friday–Saturday weekend (using UTC day labels), conditional on at least one active day. Grouping by automated heads concerns observed July 19–22 activity, not account type. Onset authors overlap with other groups. No causal or occupational interpretation is implied.

Figure A6: Core-window pushes by strength of automation evidence



Notes: All 180 PushEvents on July 19–22 from the 40 core publishers. Classification concerns the head commit and available historical evidence. Direct evidence requires a machine identity corroborated by messages/files or a matching generating workflow; suggestive patterns and unavailable objects remain separate. No-direct-evidence cases are not certified human activity. Non-push primary events are omitted from this figure but retained in Table 10.

Table A27: Individual review of every core-window publisher

Case	Events	Auto	Evidence	Basis and qualification
C002	4	4	Direct	GitHub Actions author; generated snake SVG files.
C003	4	4	Direct	readme-bot; scheduled WakaTime README workflow.
C006	1	0	Not identified	Kubernetes/project files; no identified automated publishing mechanism.
C008	3	3	Direct	readme-bot; historical scheduled WakaTime workflow.
C014	5	0	Not identified	Build-fix commits; CI and stale-issue workflows do not establish automation of these pushes.
C021	5	0	Not identified	IDE-file removals and a project README; no direct automation evidence.
C024	8	0	Insufficient	Eight regularly timed profile-repository pushes, but historical objects unavailable; timing alone is insufficient.
C039	21	7	Direct	Snyk bot heads; dependency-upgrade messages and manifests.
C043	16	16	Direct	GitHub Actions author; generated snake SVG files.
C047	3	0	Not identified	Favicon and application changes; no identified automatic publisher.
C049	10	0	Not identified	Stack/queue exercises and React signup edits; repetitive messages alone do not establish automation.
C051	3	3	Direct	readme-bot; historical scheduled WakaTime workflow.
C062	3	1	Direct	Snyk bot head; lucide-react dependency upgrade.
C070	4	4	Direct	github-actions identity; scheduled video-index pipeline and matching commit message.
C083	4	0	Suggestive	Four 22:00 UTC host-list updates; an update-and-push script and example crontab exist, but deployment of that schedule is unverified.
C084	8	8	Direct	github-actions[bot] author; generated deployment output, not inferred from the repository topic.
C089	4	4	Direct	readme-bot; historical scheduled WakaTime workflow.
C093	6	2	Direct	Snyk bot heads; mysql2 dependency upgrades.
C098	1	0	Not identified	Engineering scripts/documents; no identified automatic publisher; commit title uses a local calendar date.
C099	3	0	Not identified	HTML/CSS fitness-layout edits; no identified automatic publisher.
C110	2	1	Direct	pyup-bot identity; Sphinx dependency update.
C115	3	1	Direct	Snyk bot head; dependency-upgrade manifest changes.
C116	2	0	Not identified	Application project-page edits; no identified automatic publisher.
C119	2	0	Insufficient	Two CreateEvents only; no commit to inspect. Repository creation is not evidence of coding.
C123	1	0	Not identified	Java generic-type examples; no identified automatic publisher.
C133	1	0	Not identified	Three-line HTML change through GitHub UI; GitHub/web-flow is not itself bot evidence.
C139	7	0	Not identified	Authentication and interface changes; no identified automatic publisher.
C140	4	0	Mixed	Mixed activity: BibTeX edits and a merge; one onset commit is generated by github-actions[bot] and an exact historical import workflow.

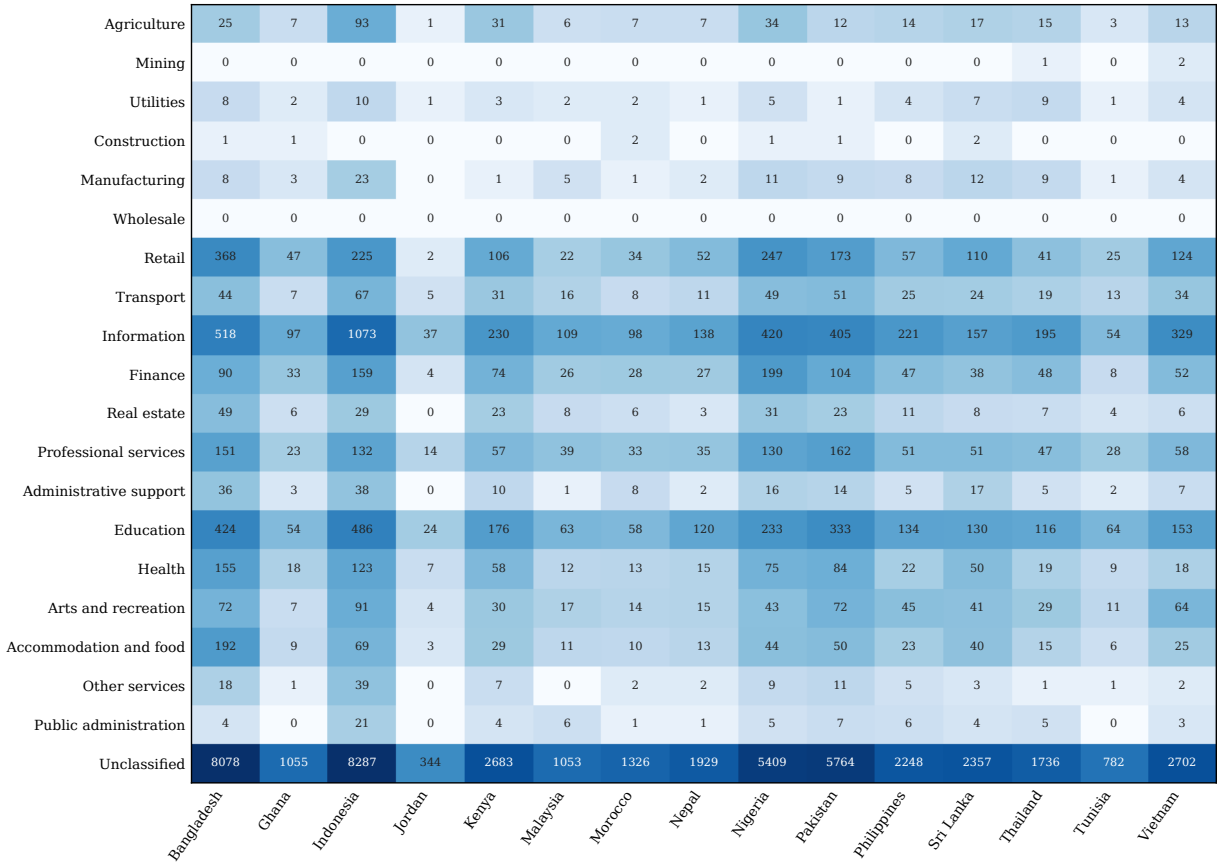
Case	Events	Auto	Evidence	Basis and qualification
C148	1	0	Not identified	Power-system code repository; inspected core head resolves a gitignore conflict, not an automatic release.
C159	1	1	Direct	readme-bot; historical weekly WakaTime workflow.
C161	7	7	Direct	GitHub Actions author; generated snake SVG files.
C177	5	0	Insufficient	Historical repository and four core heads unavailable.
C183	6	2	Direct	Snyk bot heads; Dockerfile vulnerability-fix updates.
C222	10	0	Not identified	C-sharp interview/data-structure exercises; the term Auto Stack is not evidence of a bot.
C226	1	0	Not identified	README edits via GitHub UI; no identified automatic publisher.
C229	1	0	Not identified	React router and page components; no identified automatic publisher.
C242	4	4	Direct	GitHub Actions author; generated snake SVG files.
C260	12	0	Not identified	Kotlin multiplatform examples; repeated init messages cover different files and do not establish automation.
C272	9	0	Insufficient	Three core heads in two repositories unavailable; event combinations alone are insufficient.
C273	42	41	Direct	All 41 core heads identify pyup-bot and change dependency requirements.

Events counts all primary events on July 19–22; Auto counts only directly evidenced automated push heads. "Not identified" is not proof of manual authorship. Mixed denotes the account with a generated onset-authored commit alongside other activity; that generated commit is not itself a core push head. Case IDs map to account/repository/commit URLs and dated API evidence in `results/evidence_20261020/individual_case_review.csv`. The review is AI-assisted and has not been independently validated by a second reviewer.

A.10 Country-by-country activity and complete domain coverage

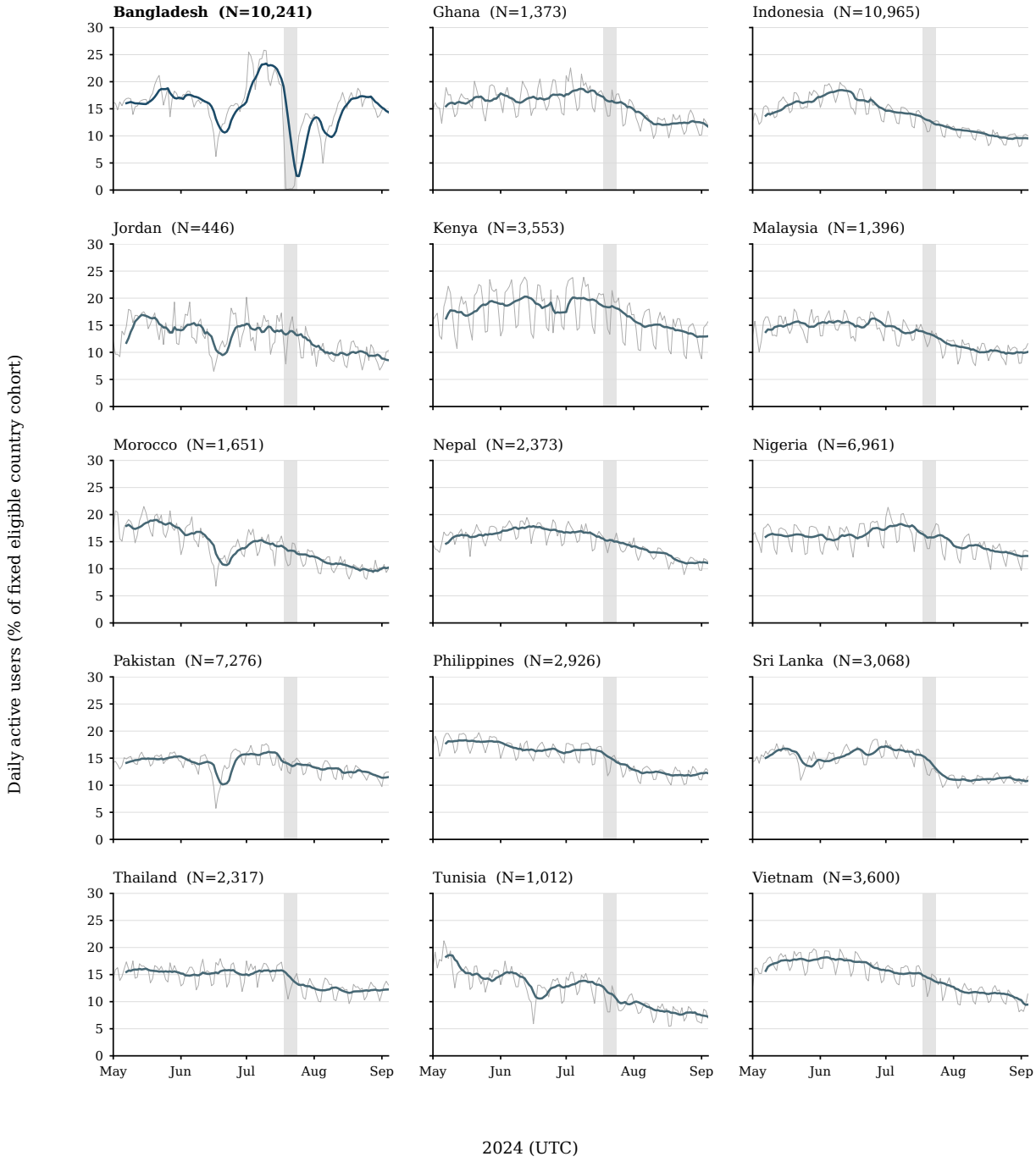
The following figures show every country and every defined repository domain, including cells that do not qualify for a synthetic-control estimate. Cohorts and denominators are fixed before July 18 using the historical-text assignment described in Section 4.5. A small or empty cell is shown explicitly; it is not replaced by a zero causal effect. These plots describe measured public activity and do not identify employment industries.

Figure A7: Eligible accounts by country and pre-treatment repository domain



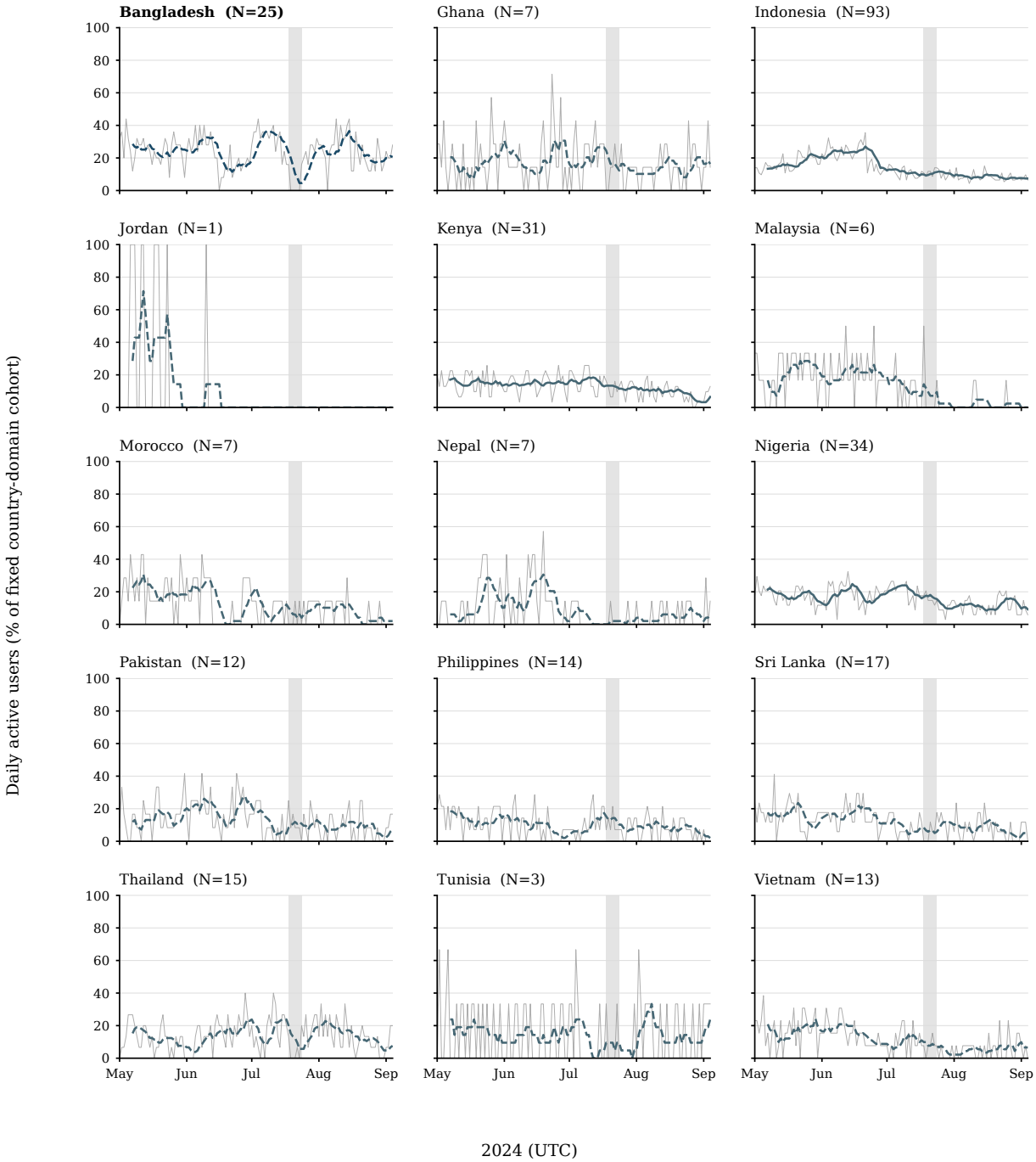
Notes: Numbers are fixed cohort sizes. Colour represents $\log_{10}(1 + N)$ for readability. All 59,158 eligible users are assigned exactly once, including the unclassified group. The existing estimation screen requires at least 30 Bangladesh users and five donor countries with at least 30 users each; these descriptive figures apply no such deletion.

Figure A8: Daily public activity in Bangladesh and every donor country



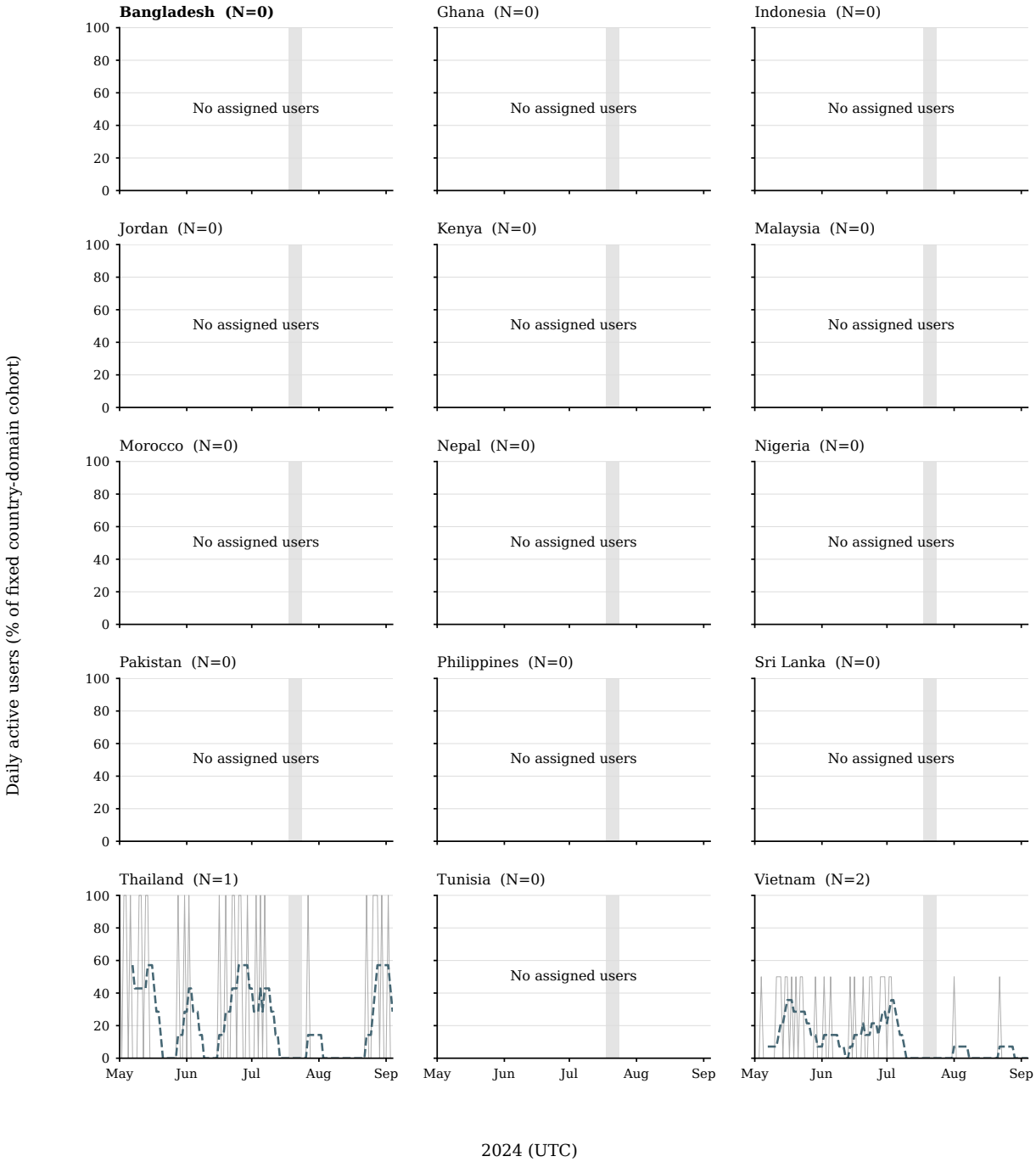
Notes: Thin grey lines: daily rates; coloured lines: trailing seven-day means, shown only after seven observations. Grey shading: July 18–23. All days and rates use UTC and the original country denominators. The smoothing is descriptive and is not used to re-estimate the primary model.

Figure A9: Country-by-country public activity: Agriculture



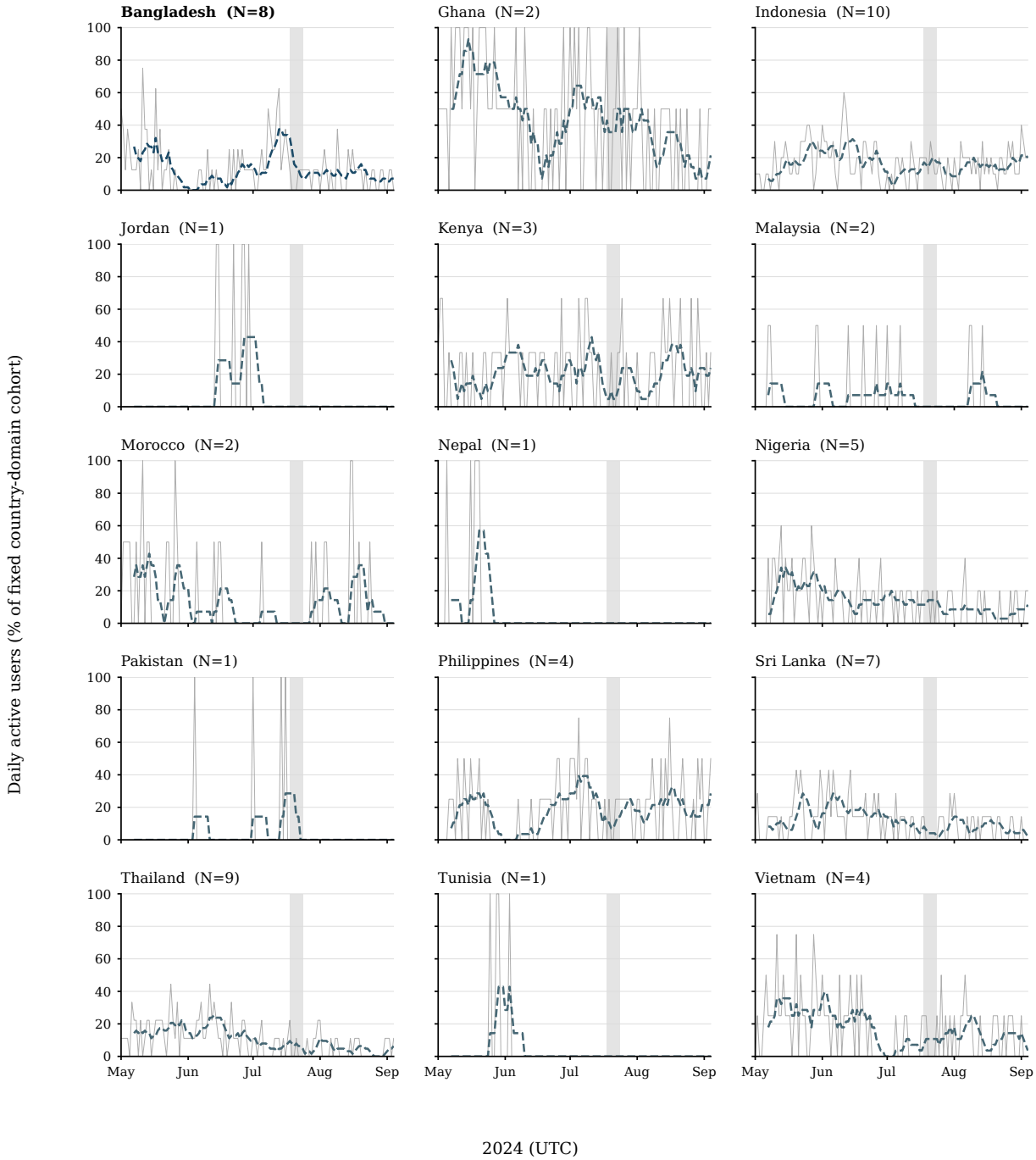
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A10: Country-by-country public activity: Mining



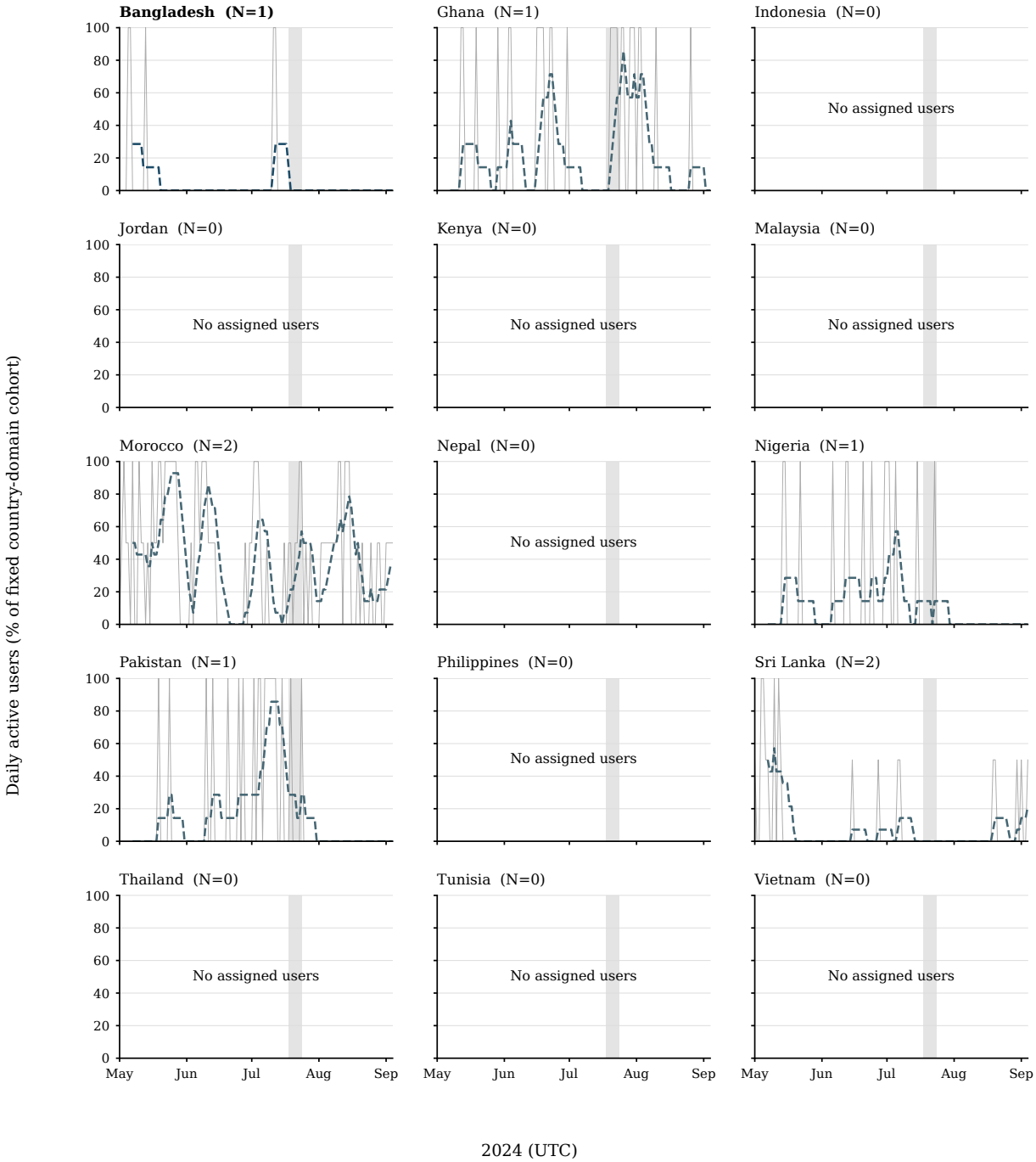
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A11: Country-by-country public activity: Utilities



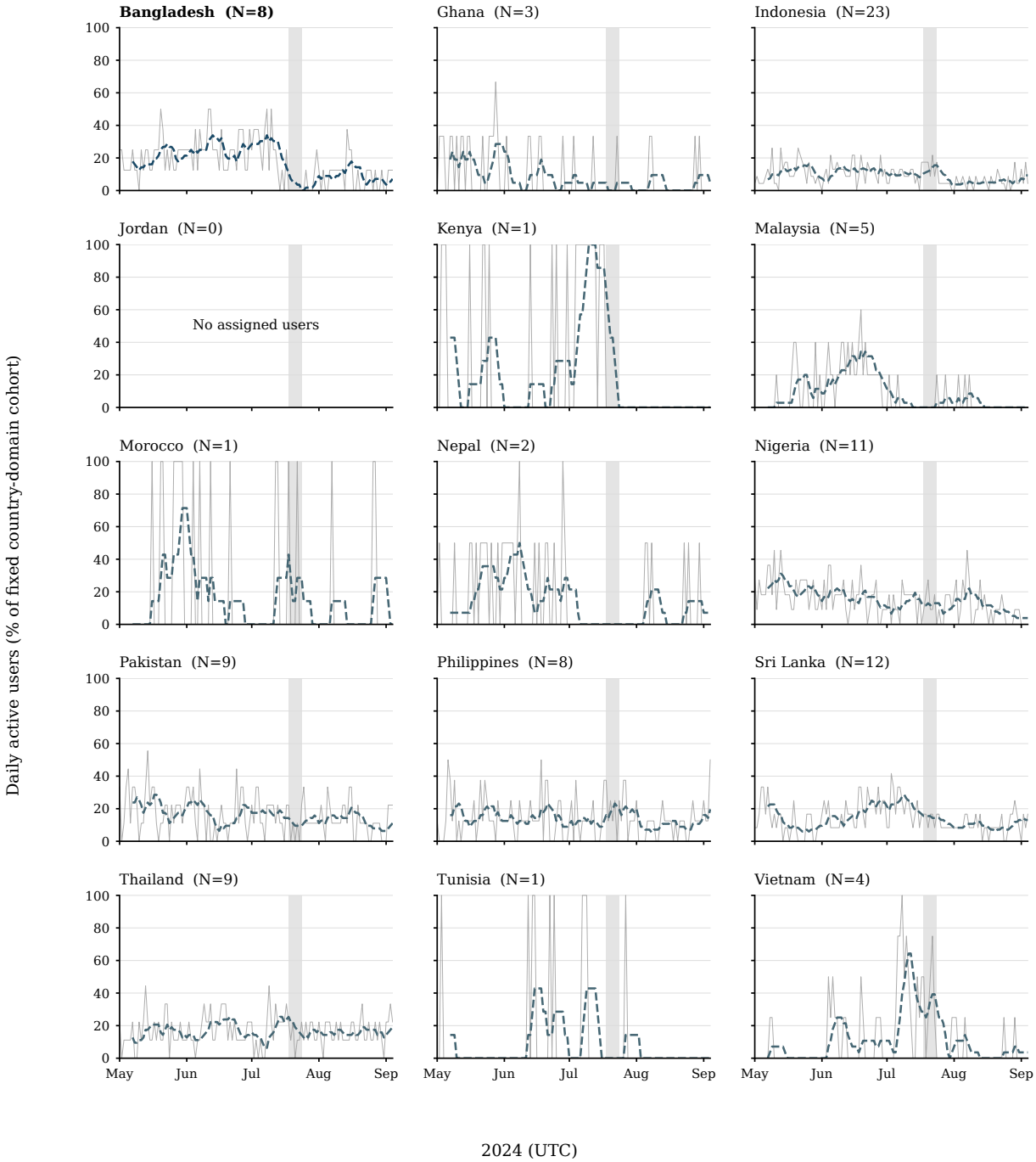
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A12: Country-by-country public activity: Construction



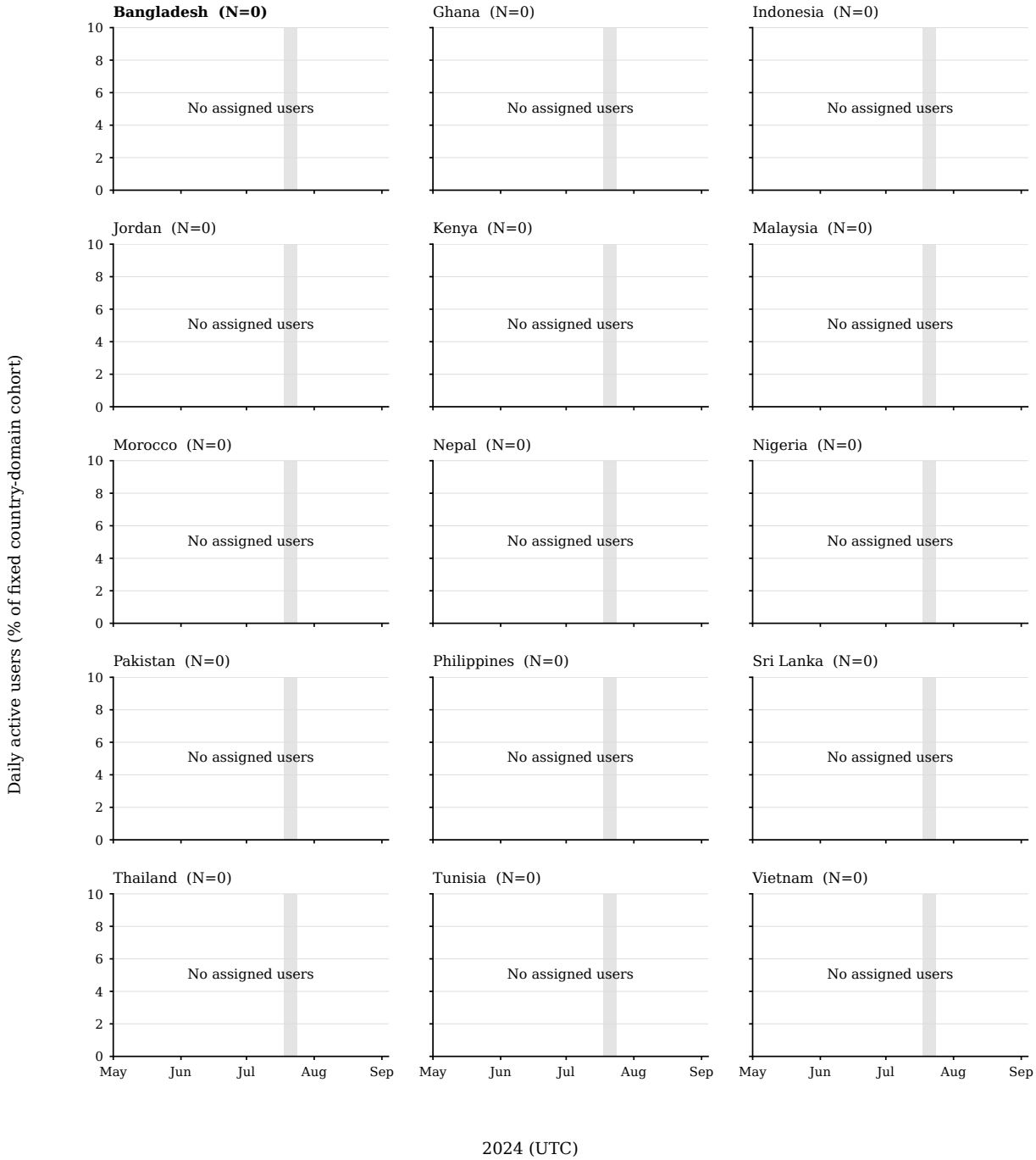
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A13: Country-by-country public activity: Manufacturing



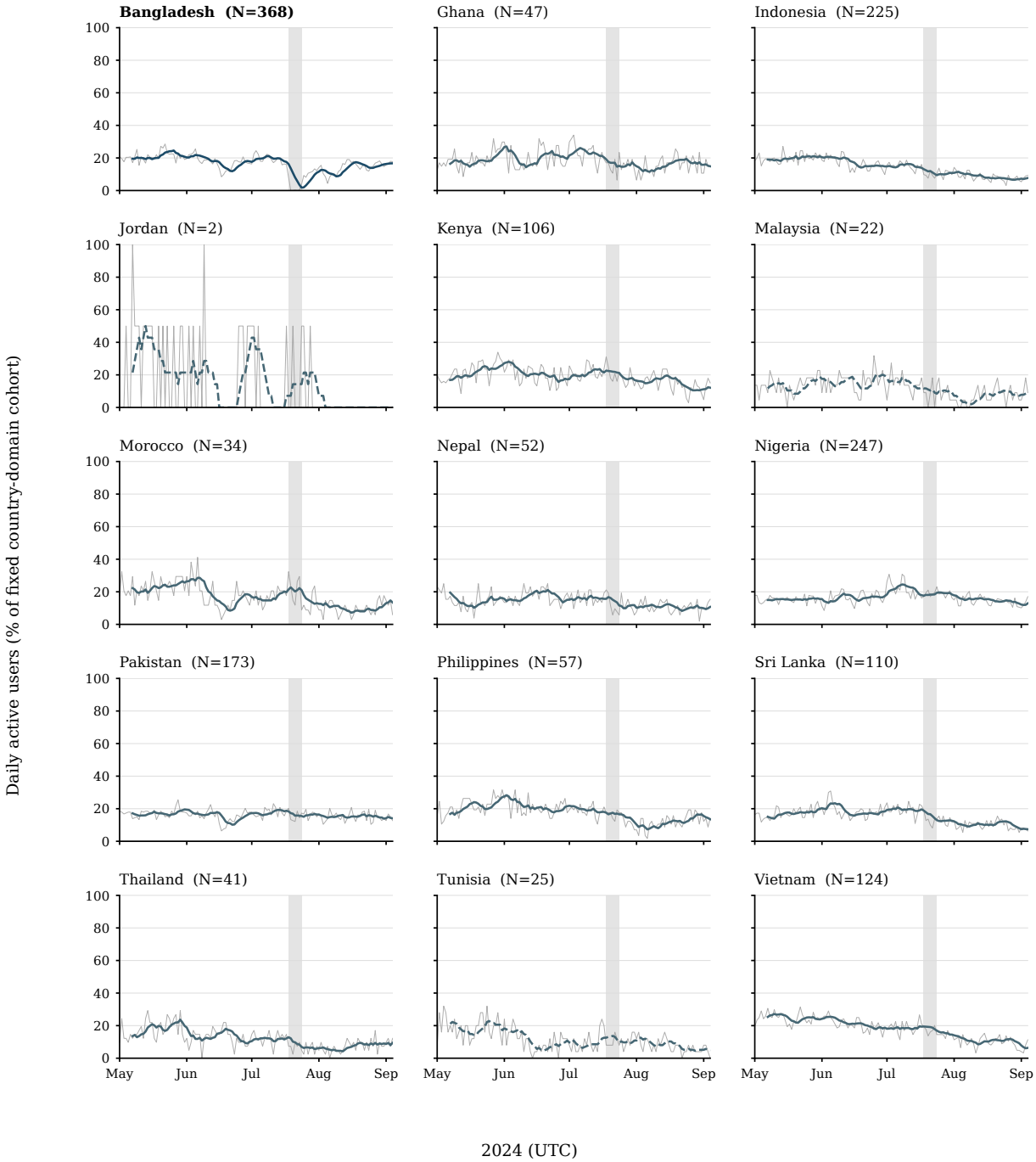
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A14: Country-by-country public activity: Wholesale



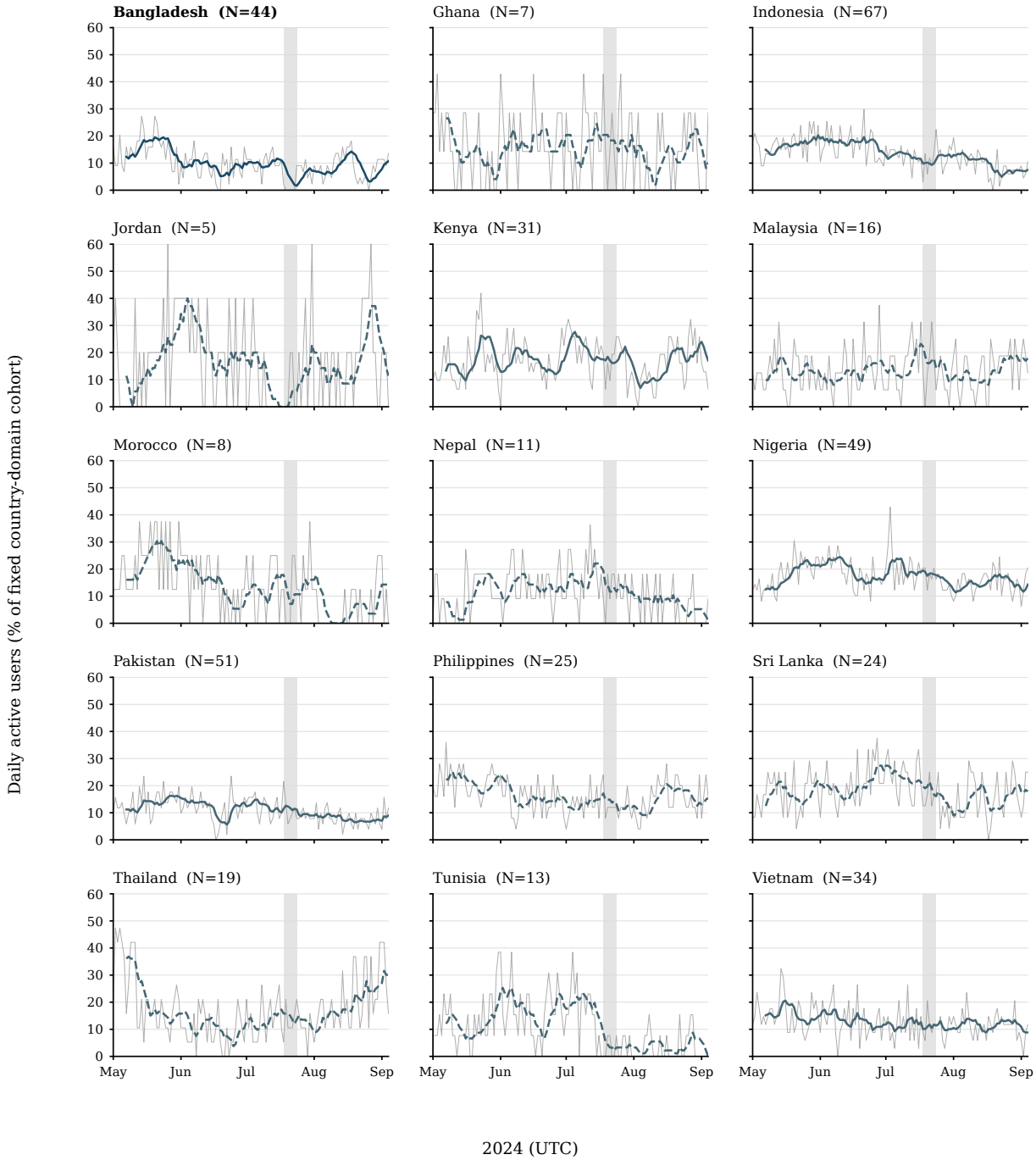
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A15: Country-by-country public activity: Retail



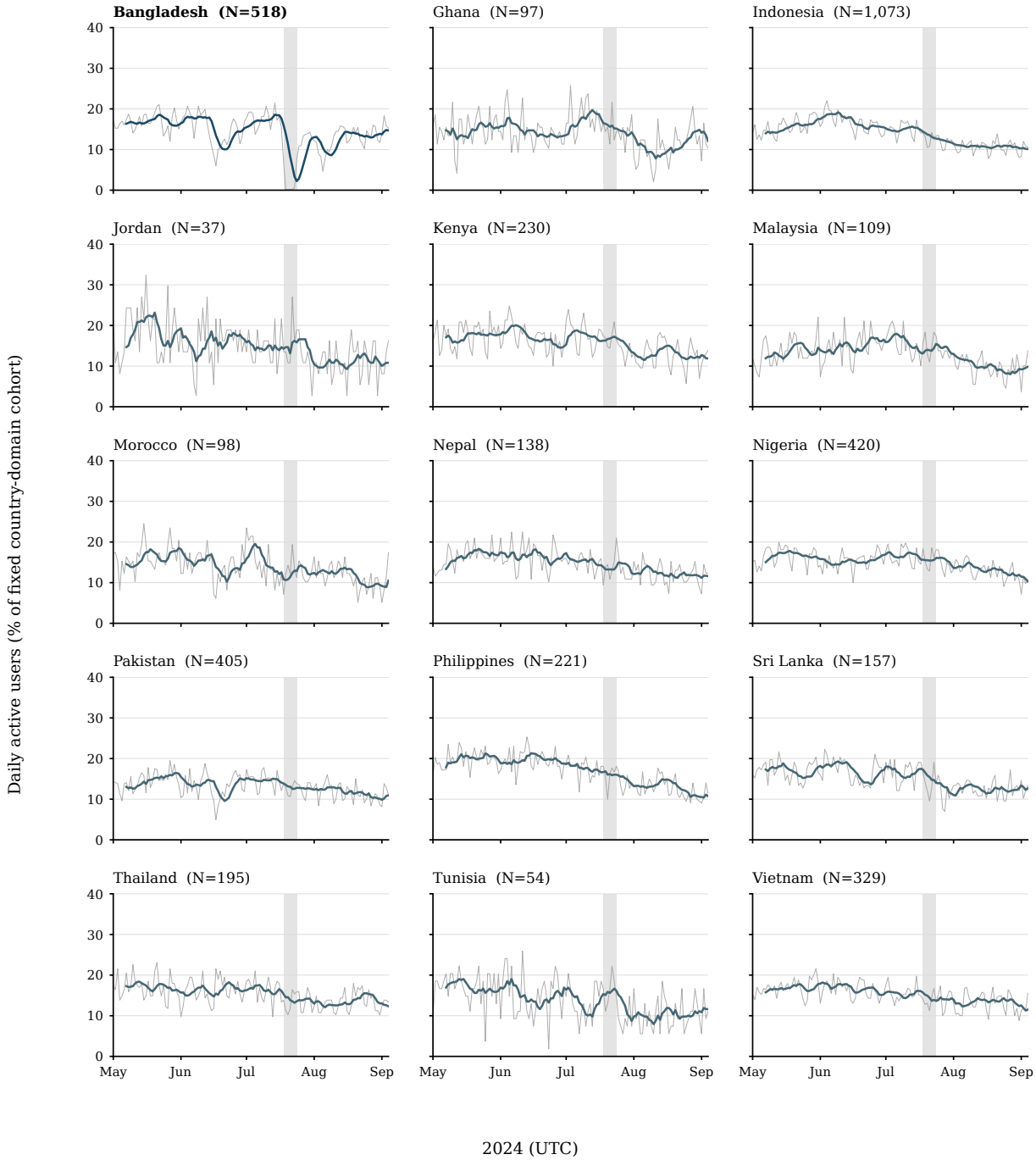
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A16: Country-by-country public activity: Transport



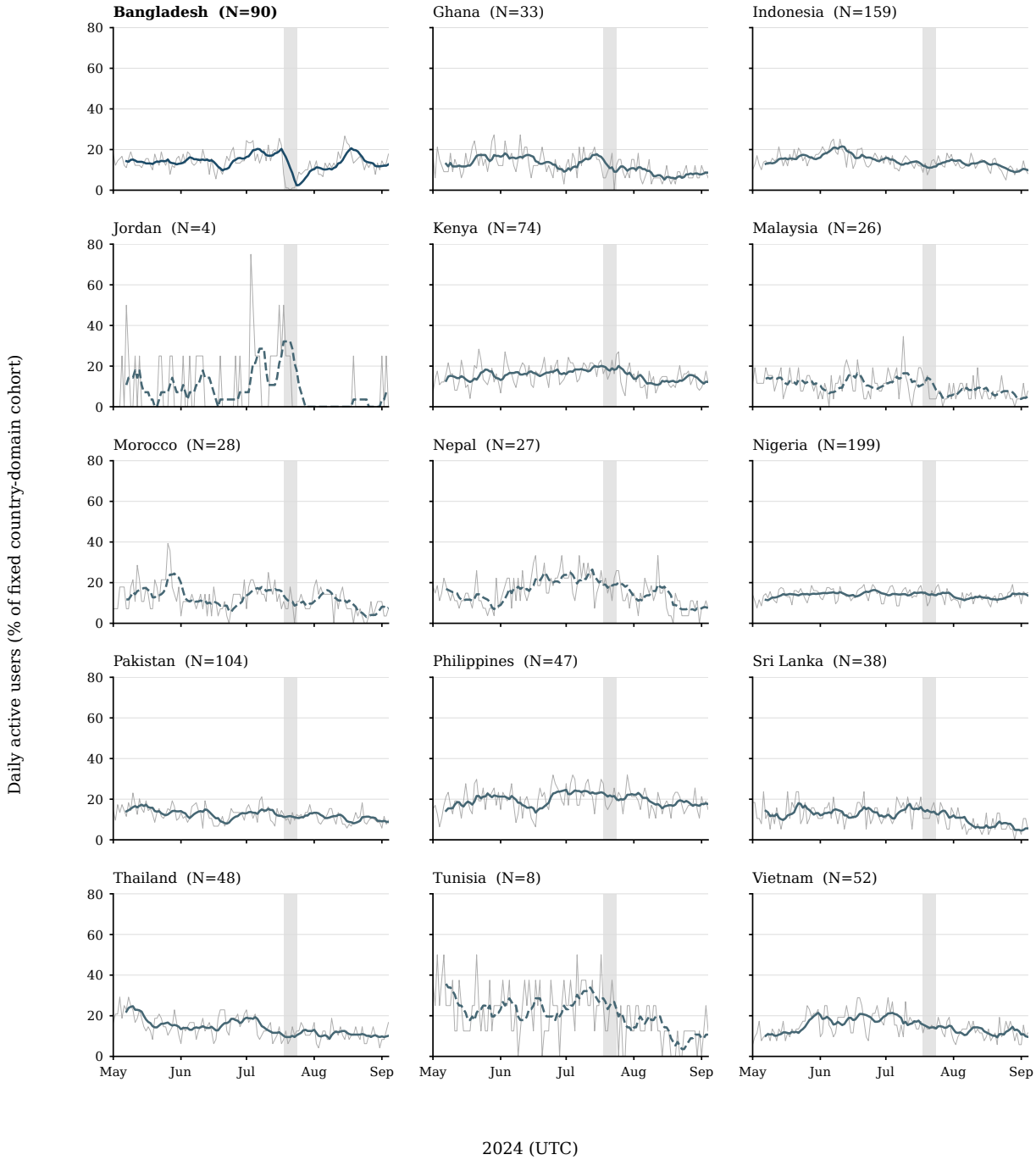
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A17: Country-by-country public activity: Information



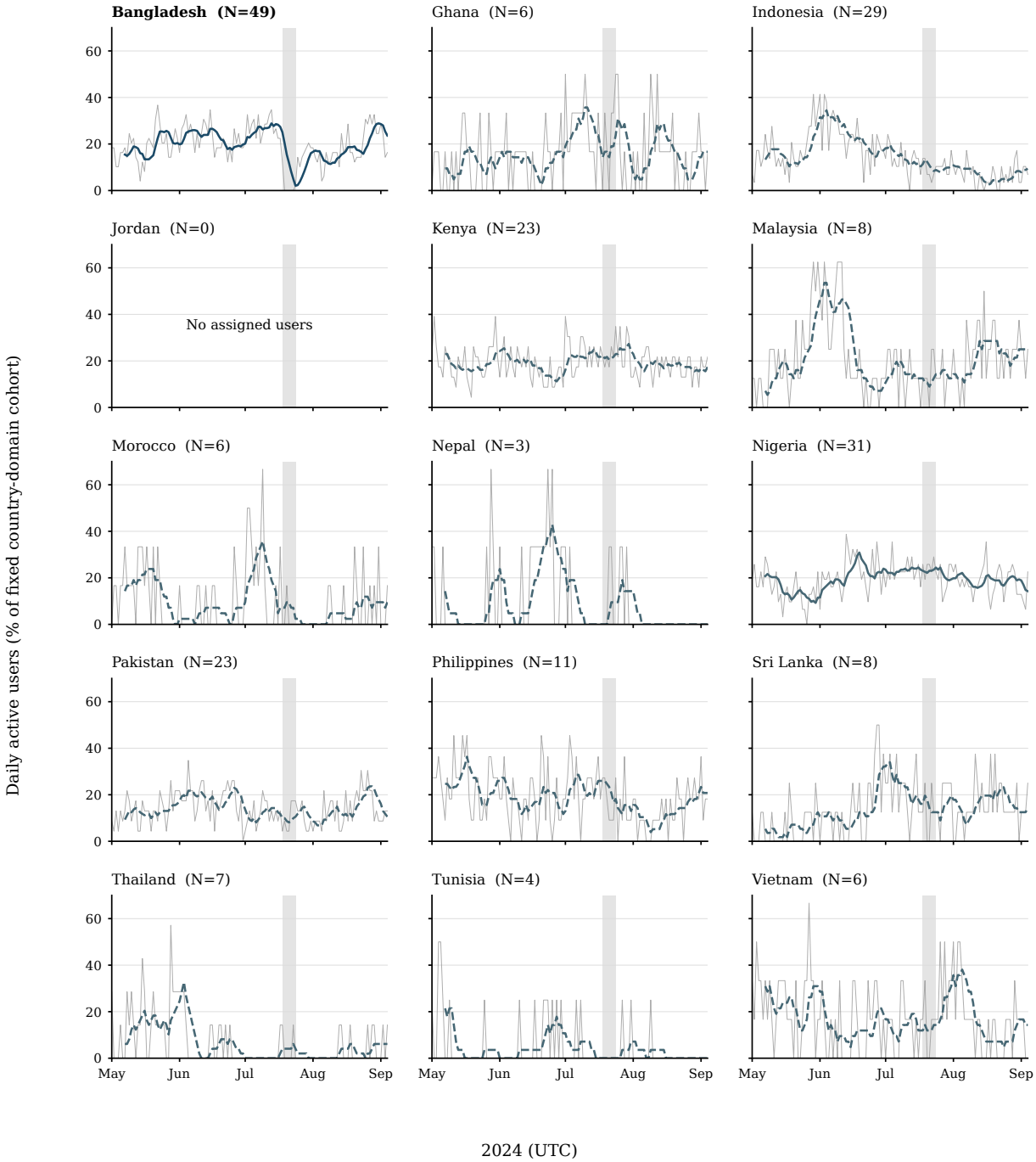
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A18: Country-by-country public activity: Finance



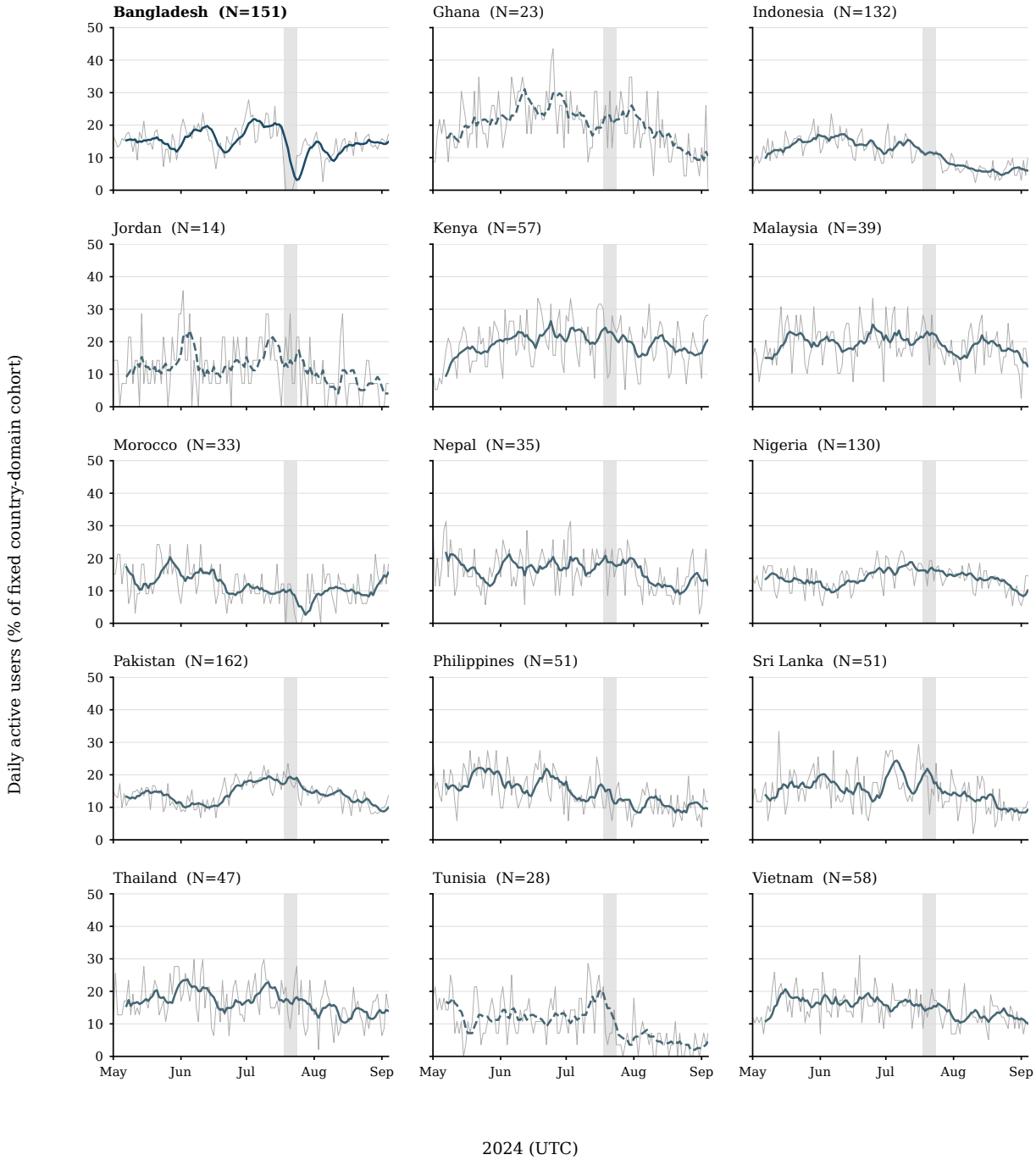
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A19: Country-by-country public activity: Real estate



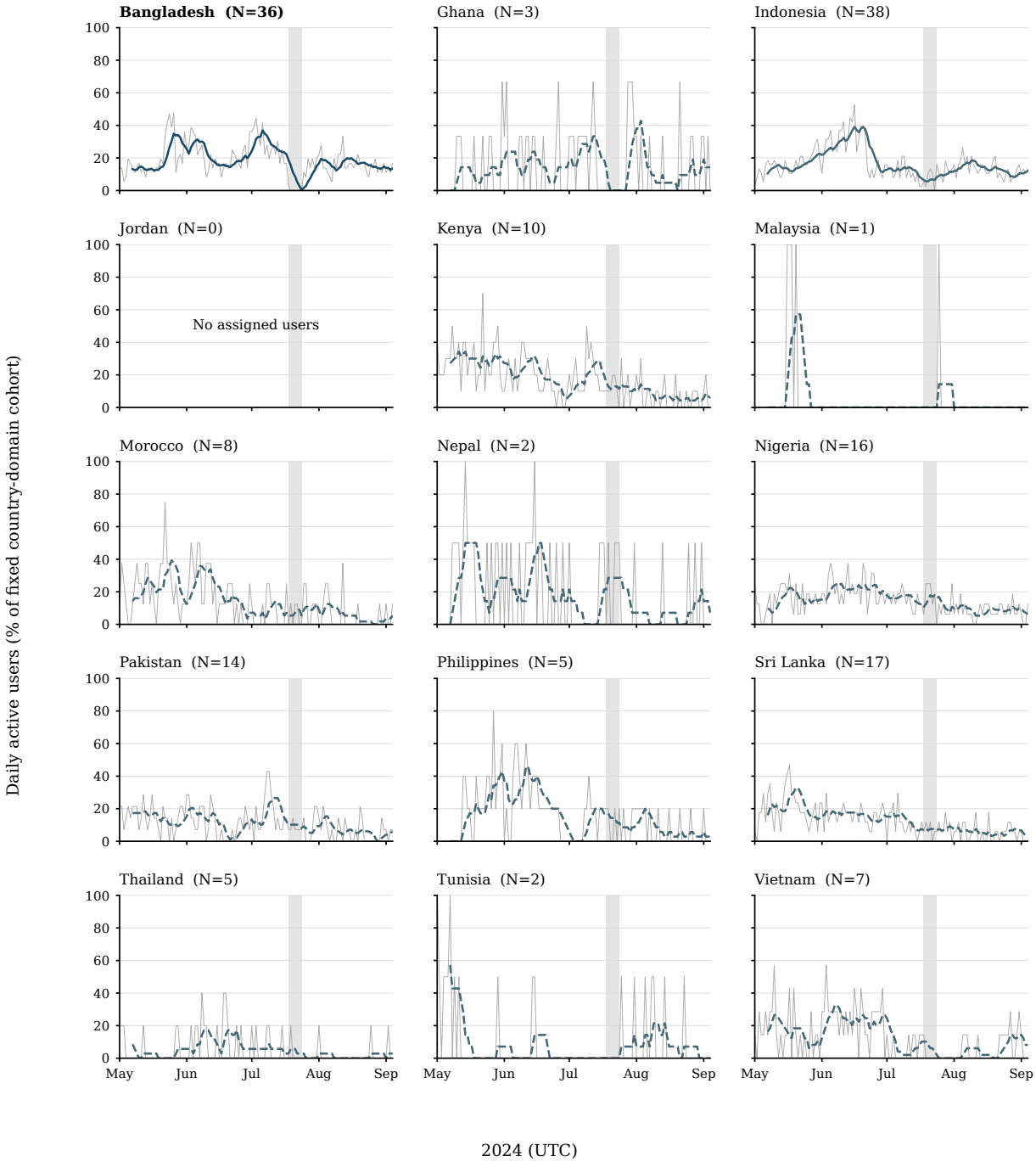
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A20: Country-by-country public activity: Professional services



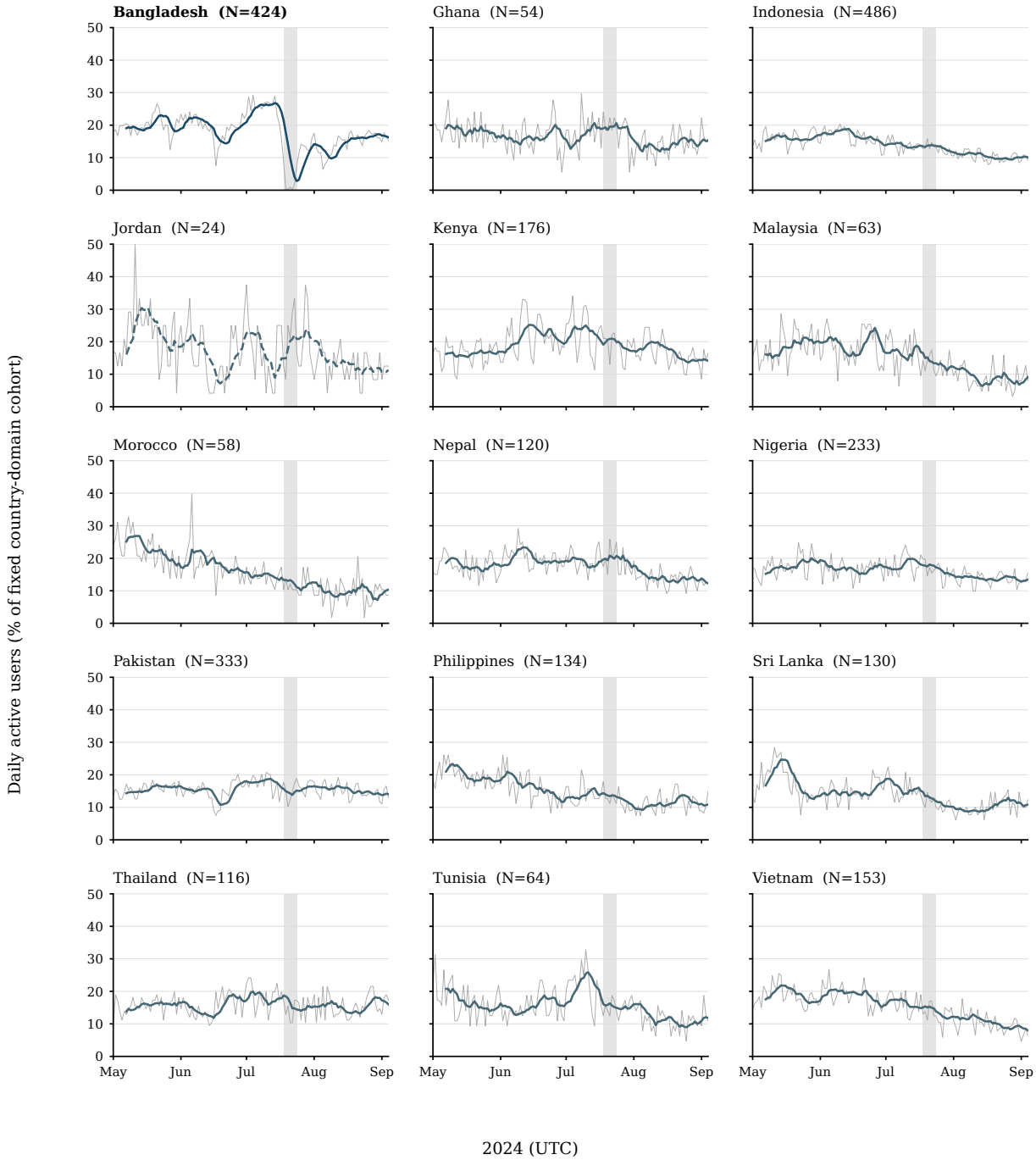
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A21: Country-by-country public activity: Administrative support



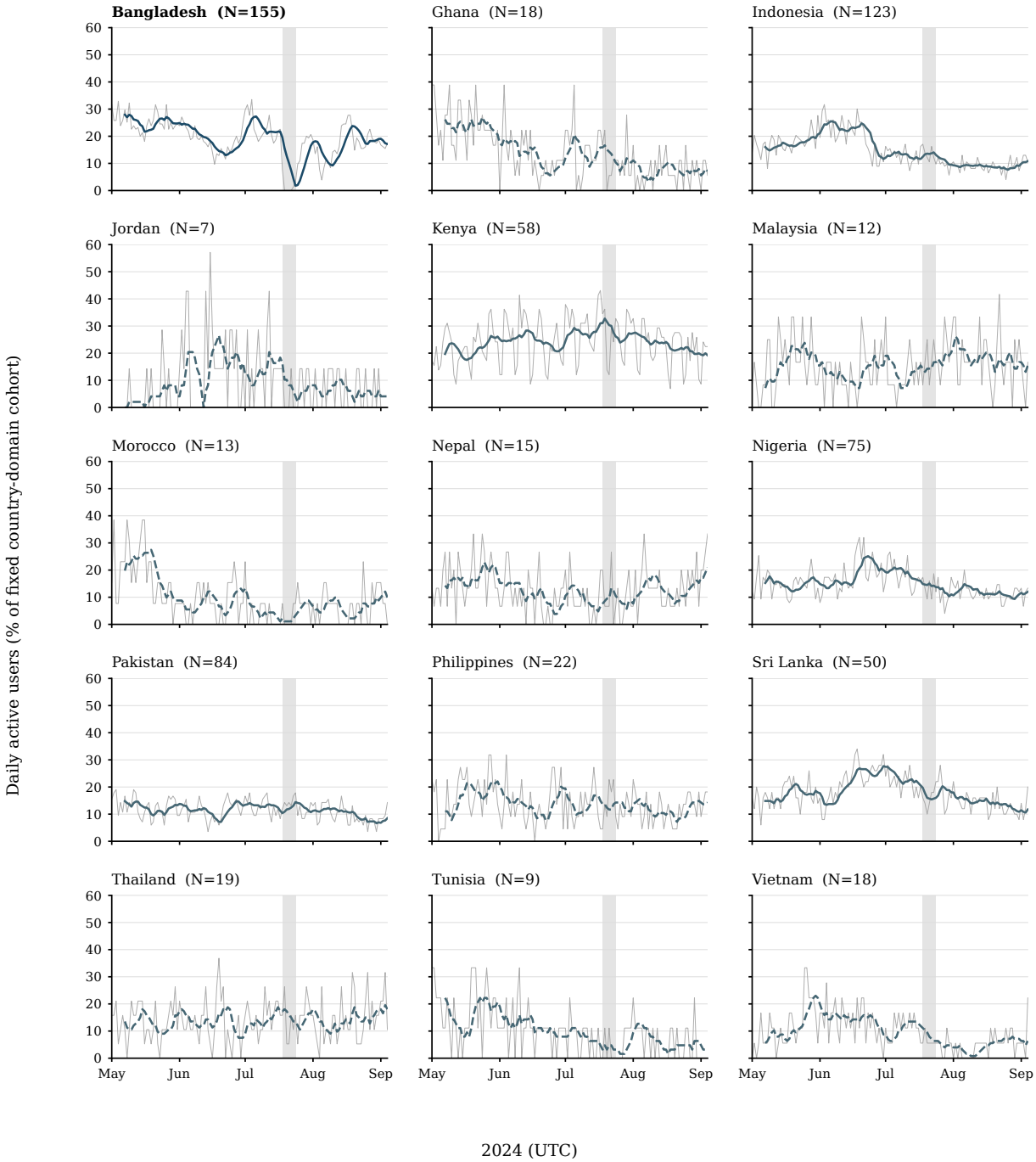
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A22: Country-by-country public activity: Education



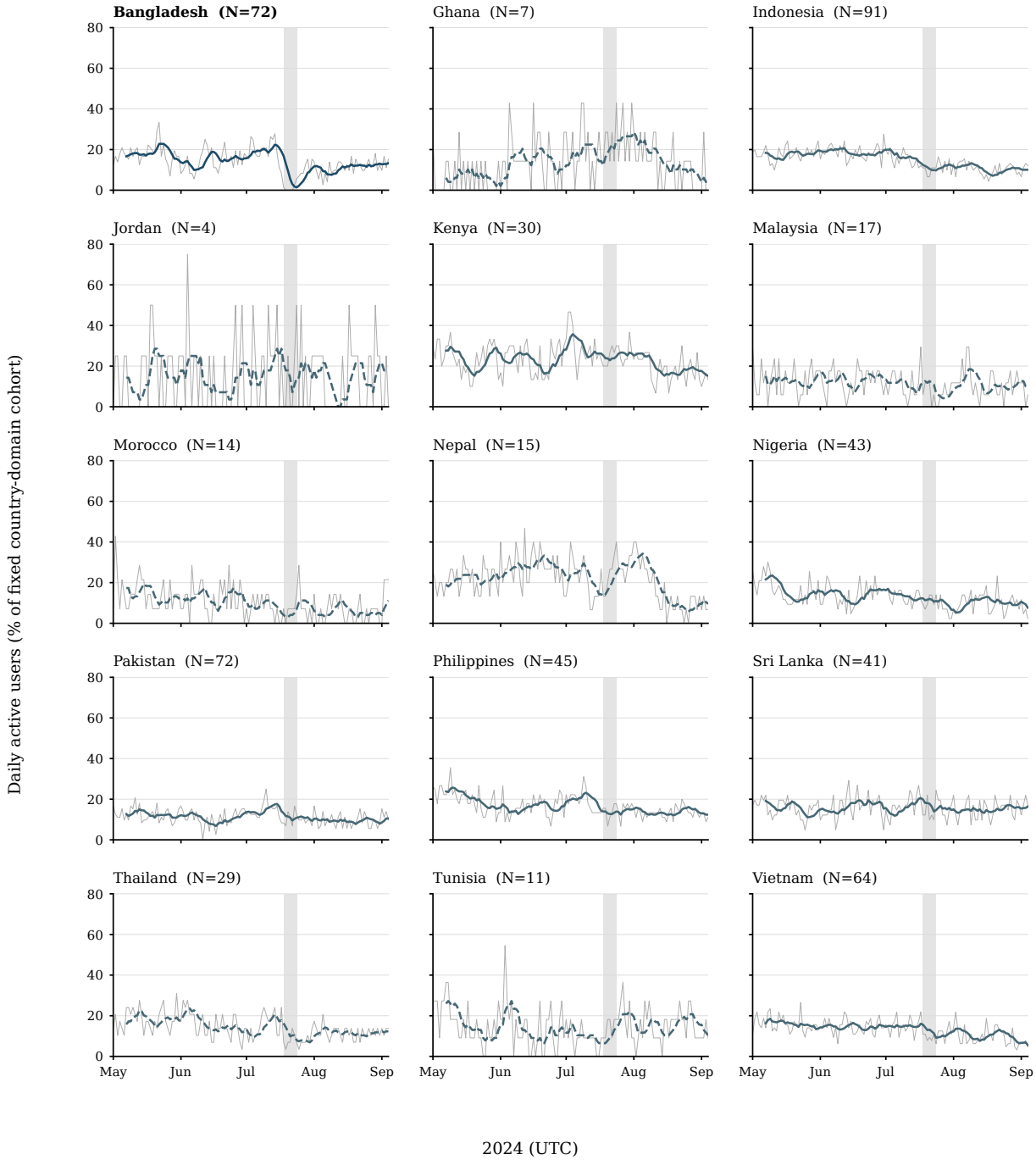
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A23: Country-by-country public activity: Health



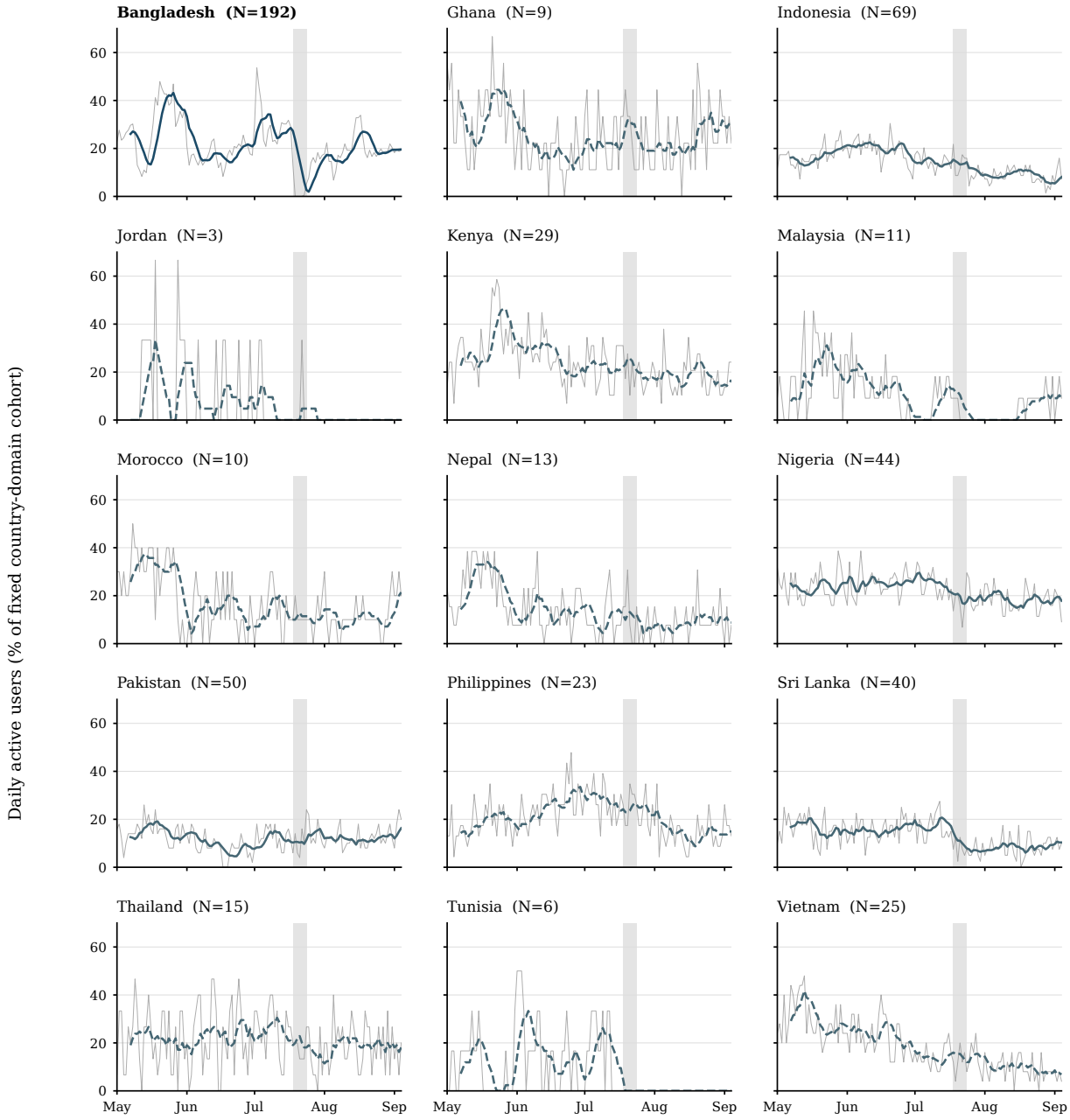
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A24: Country-by-country public activity: Arts and recreation



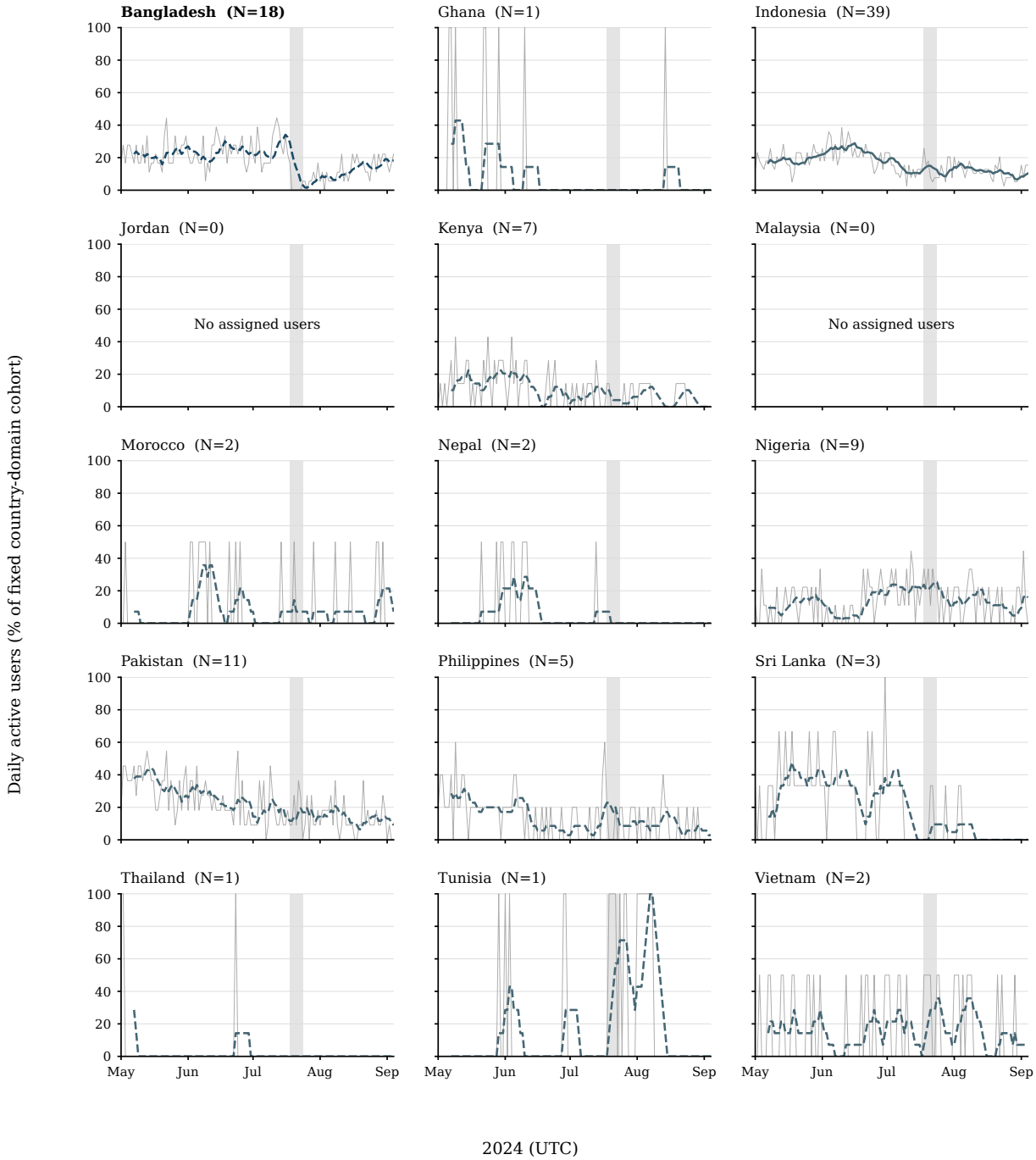
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A25: Country-by-country public activity: Accommodation and food



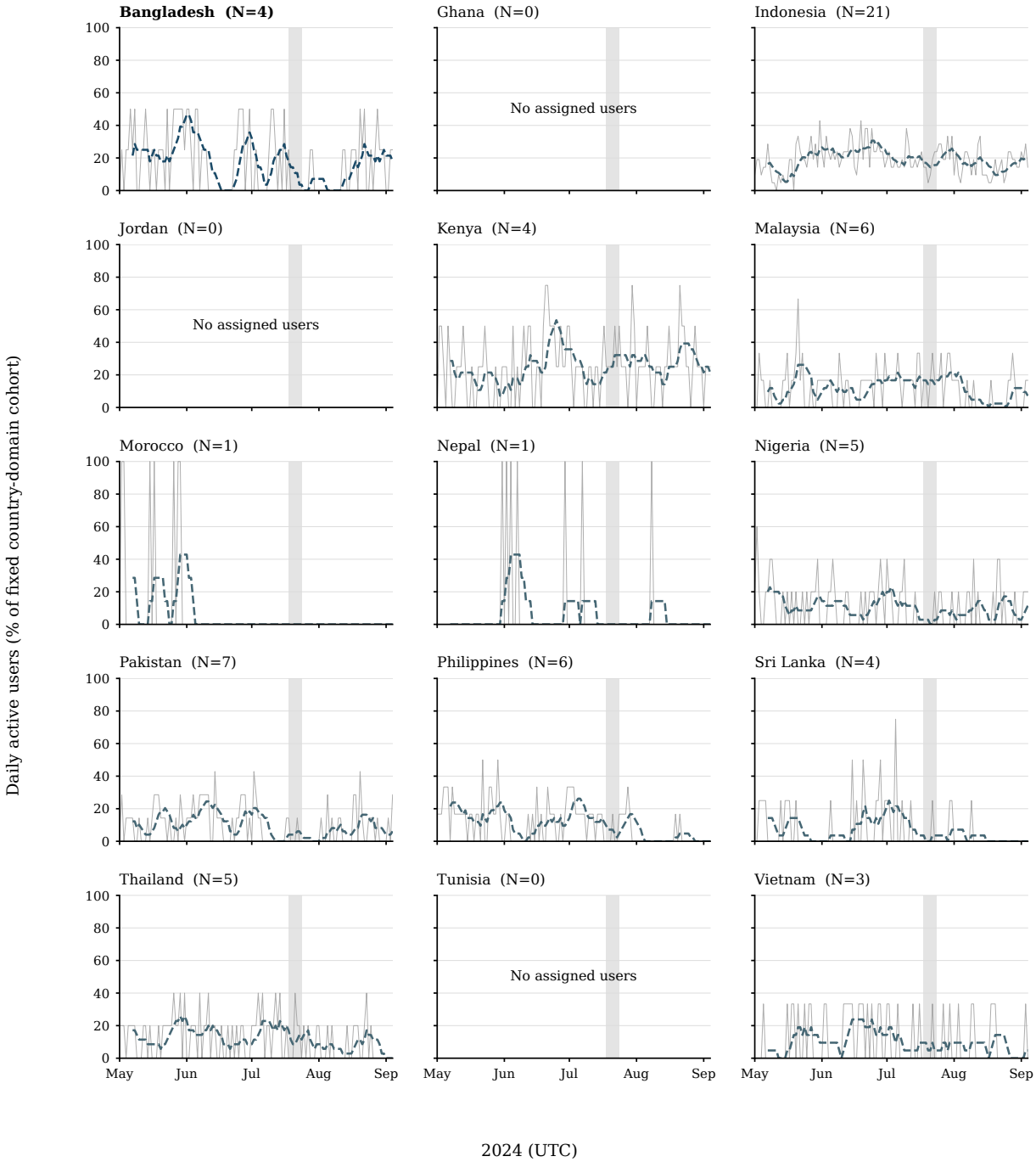
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A26: Country-by-country public activity: Other services



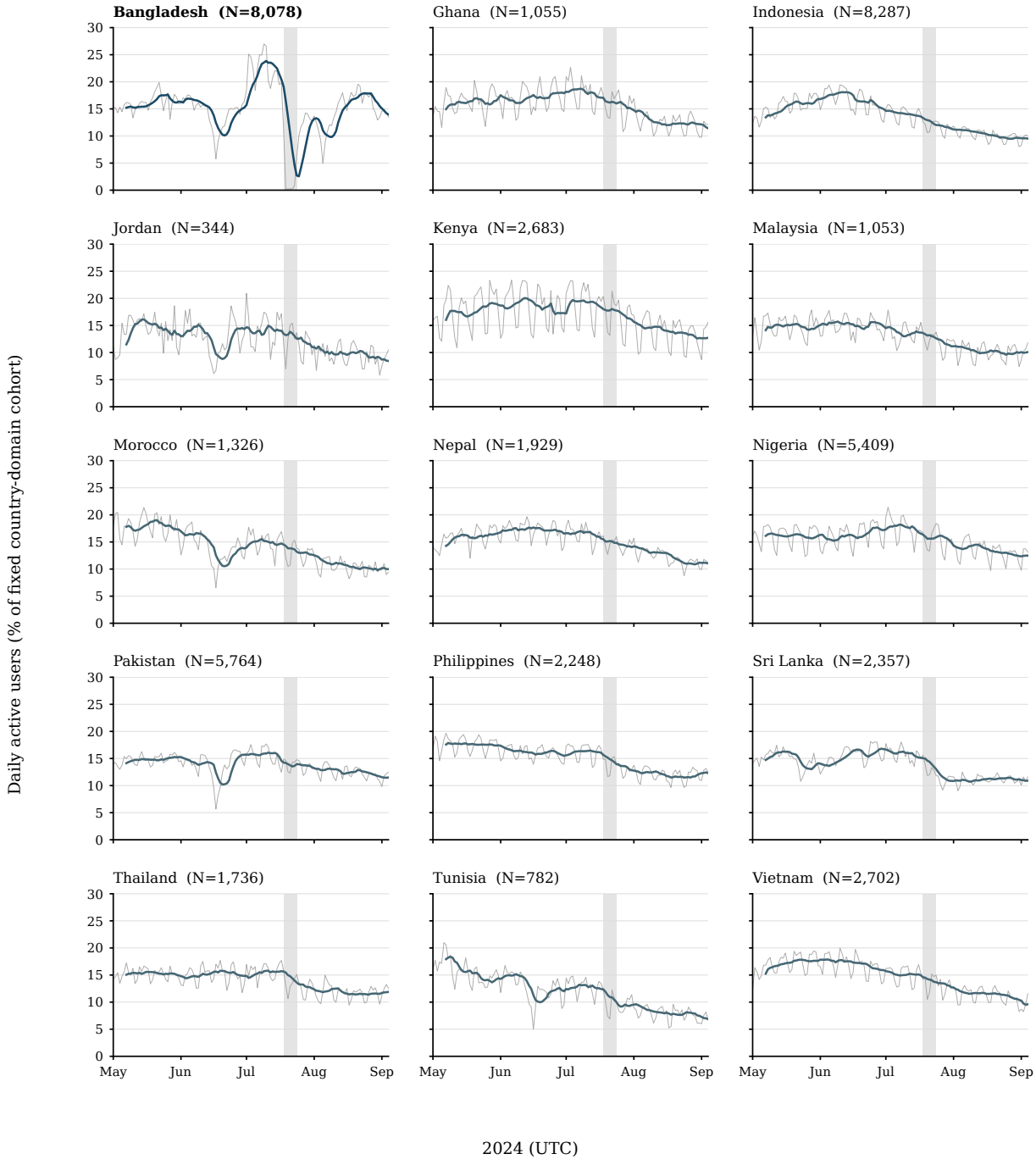
Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A27: Country-by-country public activity: Public administration



Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.

Figure A28: Country-by-country public activity: Unclassified



Notes: Fixed historical-README cohort, confidence threshold 0.80. Thin grey: daily rates; coloured: trailing seven-day mean. Dashed means indicate fewer than 30 users; empty cells have no assigned users. Scales are common across countries on this page but may differ across domains. Grey shading: July 18–23. The outcome counts each assigned user's primary activity anywhere on GitHub, not only activity in the repository used for assignment.